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Experimental & Numerical Analysis of Flow Dynamics, Flame Morphology and Emissions in a 100% Hydrogen Double-Swirl Burner under Co- & Counter-Swirl Configurations

Tesi di Laurea Magistrale in
Aeronautical Engineering - Ingegneria Aeronautica

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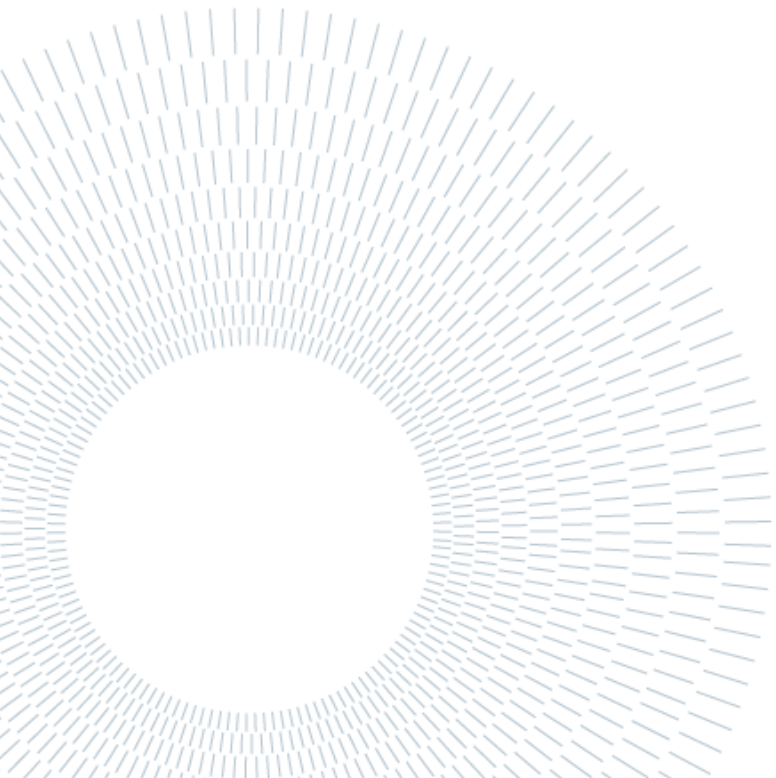


Abstract

The projected steep increase in global energy demand coupled with the need to decarbonize aviation has led to hydrogen emerging as a viable zero-carbon fuel alternative. However, hydrogen combustion presents significant technical challenges: its high diffusivity, rapid flame speed, and elevated adiabatic flame temperature complicate flame stabilization and increase nitrogen oxide emissions via the Zeldovich mechanism. This thesis investigates the aerodynamic and combustion effects of reversing the inner swirl direction within a RAPIDS-H double-swirl burner operating with 100% H_2.

The methodology couples isothermal Computational Fluid Dynamics (CFD) modeling, validated semi-qualitatively against previously acquired Particle Image Velocimetry (PIV) data for a baseline co-swirl configuration with a comparative numerical analysis of the counter-swirl flow structures. Experimentally, UV chemiluminescence imaging and exhaust gas analysis were utilized to characterize flame morphology, stability & pollutant formation.

The results demonstrate that under counter-swirl operation at higher global equivalence ratios ($\Phi_{global} = 0.45$ & 0.35), flashback consistently occurs. Isothermal CFD provides an aerodynamic interpretation consistent with these observations, suggesting that the intense interfacial shear generated by the opposing streams may disrupt the central recirculation zone (CRZ). When operating at a highly lean condition ($\Phi_{global} = 0.25$, $SR = 0.6$), the counter-swirl flame initially anchors but remains transiently unstable, ultimately flashing back after approximately four minutes of operation. Finally, exhaust gas analysis indicates that while the counter-swirl configuration offers slight reductions, overall improvements in emission indices ($EINO_x$) are marginal compared to the stable co-swirl baseline.

Keywords: *Hydrogen Combustion, Double-Swirl Burner, Computational Fluid Dynamics (CFD), Flame Stabilization, Flow Dynamics, NOx Emissions.*

Abstract in italiano

Il previsto aumento della domanda energetica globale e la necessità di decarbonizzare l'aviazione rendono l'idrogeno una valida alternativa a zero emissioni. Tuttavia, la sua combustione presenta sfide tecniche: l'elevata diffusività, la rapida velocità di fiamma e l'alta temperatura adiabatica complicano la stabilizzazione e incrementano le emissioni di NOx tramite il meccanismo di Zeldovich. Questa tesi indaga gli effetti aerodinamici e di combustione derivanti dall'inversione della direzione dello swirl interno in un bruciatore a doppio swirl RAPIDS-H alimentato al 100% a H_2.

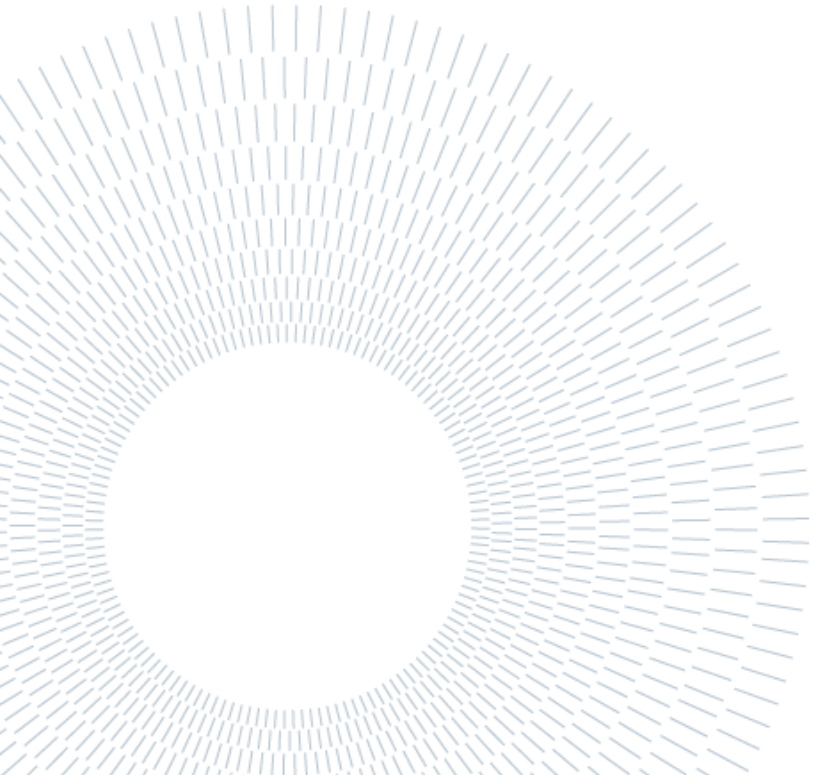
La metodologia combina simulazioni CFD isotermiche — validate in modo semi-quantitativo con dati PIV pregressi per la configurazione co-swirl di base — con un'analisi numerica comparativa delle strutture di flusso counter-swirl. Sperimentalmente, l'imaging UV e l'analisi dei gas di scarico sono stati impiegati per caratterizzare la morfologia della fiamma, la stabilità e la formazione di inquinanti.

I risultati dimostrano che, in configurazione counter-swirl con rapporti di equivalenza globali più elevati ($\Phi_{global} = 0,45$ e $0,35$), si verifica costantemente il fenomeno di flashback. La CFD isotermica fornisce un'interpretazione aerodinamica coerente con queste osservazioni, suggerendo che l'intenso sforzo di taglio generato dai flussi opposti comprometta la zona di ricircolazione centrale (CRZ). Operando in condizioni molto magre ($\Phi_{global} = 0,25$, $SR = 0,6$), la fiamma counter-swirl si ancora inizialmente ma rimane instabile, subendo un flashback dopo circa quattro minuti di operatività. Infine, le riduzioni degli indici di emissione ($EINO_x$) ottenute con la configurazione counter-swirl risultano marginali rispetto alla più stabile configurazione co-swirl.

Parole Chiave: *Combustione dell'Idrogeno, Bruciatore a Doppio Swirl, CFD, Stabilizzazione della Fiamma, Dinamica del Flusso, Emissioni di NOx*

"In Fluids and in Life it is the Turbulence that sustains the Flame, although viewed as a Perturber of Tranquility, it is essential to sustain the Fire that fuels your Forge."

~Anonymous



Contents

Abstract i

Abstract in italiano iii

Contents v

Introduction 1

Background and Motivation 1

Objectives and Scope of the Work 2

Thesis Outline 3

1 Literature review 4

1.1. Hydrogen combustion: Fundamentals & Challenges 4

1.2 Pollutant emissions resulting from Hydrogen Combustion 4

1.2.1 Nitrogen oxide formation 5

1.2.2 Hazard of NOx to environment and human health 6

1.3 Double Swirl Burner 8

1.3.1 Co-swirl combustor 13

1.3.2 Counter-swirl combustor 15

1.4 The RAPIDS-H Burner, Previous Investigations & Knowledge gap 17

2 Numerical Methodology 19

2.1 Numerical Background 19

2.1.1 The Finite Volume Method (FVM) and Mesh Topology 20

2.1.2 Governing Equations and URANS formulation 20

2.1.3 The SST k-ω Turbulence model 20

2.1.4 Near wall treatment: Kader Blending 20

2.1.5 Advanced Turbulence Modifications for Swirling flows 22

2.2 Geometry preparation 23

2.3 Mesh Scheme 25

2.4 Solver Setting 26

2.5 HPC cluster 28

2.6 Solution & Convergence monitoring 28

2.7 Mesh Independence 29

3 Experimental Methodology 31

3.1 Experimental Setup & Burner Configuration 31

3.1.1 Development and fabrication of Counter swirl injector 33

3.2 Measurement Procedures & Data Processing 35

3.2.1 Isothermal Operating Conditions & CFD Boundary setup 38

3.2.2 UV imaging and Flame Morphology 40

3.2.3 Stereoscopic Particle Image Velocimetry (Stereo-PIV) 41

3.2.4 Emission Measurements and Gas Analysis 43

4 Numerical Results and Comparison with Experimental Data 46

4.1 Co-swirler validation against Stereo-PIV data 46

4.1.1 Coordinate system & axis of rotation 46

4.2 Isothermal Flow Structures of the Co- & Counter-Swirler 50

4.3 Comparative analysis of the CRZ zones (Co- vs Counter- swirl) 54

4.3.1 Vortex breakdown mechanics and radial expansion 54

4.3.2 CFD investigation into CRZ development 55

4.3.3 Downstream evolution of velocity fields 57

4.3.4 Turbulent Kinetic Energy & Shear Dissipation 59

5 Experimental Investigation and Results 61

5.1 UV Flame Morphology and Stability Analysis 61

5.1.1 Influence of the Secondary Split Ratio (SR) 61

5.1.2 Premixing Sensitivity and Flashback limits- Test Observations 62

5.1.3 Lean Blowout and Lift-off Dynamics 64

5.2 NOx Emission Results 64

6 Conclusions, Further developments & Declarations 68

Bibliography 70

A Appendix A 73

B Appendix B 81

List of symbols 82

Acknowledgments 84

Introduction

Background and Motivation:

Global energy demand is expected to increase significantly in the coming decades. According to projections by the International Energy Agency (IEA), worldwide energy consumption may rise by approximately 30% by the year 2040 [1]. At present, the majority of this demand is still satisfied through fossil fuels such as oil, coal, and natural gas, which together account for more than 85% of the global energy supply [1]. The extensive use of these fuels has resulted in substantial emissions of greenhouse gases, with carbon dioxide (CO_2) representing roughly 77% of total anthropogenic greenhouse gas emissions [2],[5]. A large fraction of these emissions originates from the combustion of fossil fuels, which remains a dominant component of the global energy system [5]. In particular, the combustion of gaseous fuels contributes significantly to global CO_2 emissions [5]. If current trends continue, the progressive depletion of fossil resources together with the environmental consequences of their combustion will require a rapid transition toward alternative energy sources and propulsion technologies [1],[2].

Within this context, hydrogen has emerged as one of the most promising candidates for decarbonizing several industrial sectors, including aviation. When produced from renewable sources (often referred to as green hydrogen), hydrogen can provide energy without direct carbon emissions during combustion [12],[13]. This characteristic has attracted considerable interest from both academia and industry, particularly in the aerospace sector, where reducing environmental impact is becoming increasingly critical. Aviation currently contributes a significant share of greenhouse gas emissions within the transport sector, and passenger air traffic is expected to grow steadily in the coming decades [6]. Consequently, several research initiatives and technological studies have been launched to evaluate the feasibility of hydrogen-powered aircraft, including programs such as FlyZero [7], ENABLEH2 [8], and CHEETA [9]. These studies highlight hydrogen as a key element in bridging the decarbonization gap of the aviation industry and enabling future low-emission propulsion systems.

The concept of hydrogen-fueled aviation is not entirely new. Early investigations conducted in the late twentieth century already explored the possibility of using hydrogen as an aircraft fuel, highlighting both its potential advantages and the technological challenges associated with its storage and utilization [10],[11]. Subsequent studies and technological developments have further demonstrated that hydrogen-powered flight could become technologically feasible within the next decades, provided that advancements are achieved in fuel storage, propulsion technologies, and aircraft integration [11].

Despite its advantages as a carbon-free fuel, hydrogen combustion presents several technical challenges. Hydrogen flames typically exhibit higher adiabatic flame temperatures compared with conventional hydrocarbon fuels when operating at the same equivalence ratio, which can significantly increase the formation of nitrogen oxides (NOx) [13],[19]. Moreover, hydrogen is characterized by high chemical reactivity and a relatively high flame speed, which increases the likelihood of undesirable phenomena such as flashback, autoignition, and combustion instabilities [13],[14],[15]. These characteristics make the design of stable and low-emission hydrogen combustors particularly challenging and require advanced strategies for fuel injection, mixing, and flame stabilization [13].

Objectives and Scope of Work

One widely adopted solution for stabilizing flames in gas turbine combustors involves the use of swirling flows. Swirl generates a central recirculation zone (CRZ), which produces a region of low axial momentum and allows hot combustion products to recirculate toward the burner outlet. This mechanism enhances mixing between fuel and oxidizer while providing a stable region where the flame can anchor away from the injector surfaces [16]. Swirl-stabilized combustion systems therefore represent a promising approach for achieving stable hydrogen combustion while controlling pollutant formation.

In this context, several studies have investigated the use of double-swirl injector configurations for hydrogen combustion. Experimental investigations by Marragou [17] and Leroy [18] have shown that hydrogen–air flames can be effectively stabilized using coaxial double-swirl burners, where the interaction between multiple swirling streams promotes mixing and influences flame structure and pollutant formation. Their results indicate that variations in swirl intensity and injection parameters can significantly modify the flame topology and the resulting NO_x emissions [17],[18].

Building upon previous experimental studies conducted on the RAPIDS-H double swirl burner located at the Energy Department of Politecnico di Milano by Ciuffoli & Rocco [28], the present work investigates hydrogen combustion in a modified double-swirl configuration. In particular, the injector geometry is adapted to generate a counter-swirling flow field, where the inner and outer swirling streams rotate in opposite directions. This configuration is expected to modify the interaction between the flow streams, altering the formation of the central recirculation zone and the resulting flame stabilization mechanisms.

Thesis Outline

The objective of this thesis is therefore to analyze the flow fields & combustion behavior of hydrogen in both Co- and Counter-swirl double-swirl burner configurations and their comparisons. The comparison of the effects of swirl reversal (Co and Counter injectors) are done. An isothermal transient CFD [Ansys FLUENT] simulation of the whole test setup was modelled for the the Co-swirler for a specific test run condition [#4 PIV] of available experimental PIV data and the velocity plots of the CFD are compared with the relevant plots of the PIV velocimetry experimental work done previously by Ciuffoli & Rocco [28] Upon sufficient validation agreement, convinced of the accuracy and physical reliability of the CFD methodology we continue to model the same condition for the Counter-swirler *A separate and identical Counter-swirl CFD model was developed and run for the same condition to be compared*, from this we obtain an idea on the effect that reversing the swirl direction has on the flow structures *–the internal flow structures responsible for flame stabilization are investigated through computational fluid dynamics (CFD) simulations of the cold flow, enabling the identification of the main aerodynamic features governing the combustion process.*

Experimentally, particular attention is devoted to the characterization of flame morphology, stability, and pollutant emissions under partially premixed operating conditions. To perform the analysis, UV cameras were used for image-based diagnostics, and a gas analyzer was employed to measure pollutant emissions.

1. Literature Review

1.1. Hydrogen Combustion: Fundamentals & Challenges

The problem with Hydrogen combustion is that it burns with a high flame temperature in stoichiometric condition [13], this enhances the dissociation of Nitrogen molecules causing significant increase in NO_x emissions through the Zeldovich mechanism [19] the High temperature also causes mechanical damage due to the greater thermal stresses on the combustor walls and affects the integrity & service life of the combustion system [4], NO_x affects the environment, health etc. (this will be discussed in the following sub-section 1.2)

Hydrogen possesses a significantly higher laminar flame speed with respect to hydrocarbons [15] , this can improve flame stability but at the risk of flashback, especially in fully premixed configurations, the flashback can cause the flame to propagate into the premixing regions causing damage to the combustor.

The high reactivity of Hydrogen also causes susceptibility to thermoacoustic disturbances and oscillations [13], hence becoming a challenge to maintaining dynamic stability. The solution to this is the goal of the research work of the Combustion Lab- injectors & combustor development which uses swirl-based stabilization strategies & advanced staging techniques aimed at enhancing flame anchoring and suppressing undesirable combustion dynamics. The effects on combustion due to design modifications to the injectors are studied (discussed in section 1.4)

1.2 Pollutant emissions resulting from Hydrogen Combustion

The combustion of Hydrogen doesn't involve Carbon so unlike hydrocarbons we don't have the production of CO or CO_2, but it generates water vapor H_2O(g) and Nitrous oxides NO_x (NO and N_2O), the NO_x formation is the bigger source of concern here. We will understand the production routes of this undesirable product of Hydrogen Combustion in the following section.

1.2.1 Nitrogen oxide formation:

The formation of nitrogen oxides (NO_x) during hydrogen combustion is heavily dominated by the temperature & residence time, occurring through three primary chemical pathways:

Thermal NOx (Zeldovich Mechanism):

This is the main producer of NO_x in hydrogen combustion due to the exceptionally high adiabatic flame temperatures (>1800K). This mechanism is initiated by the reaction of an oxygen radical (O) on the stable N_2 molecule (detailed in Equation 1.1). The extended Zeldovich mechanism is defined by the following elementary reactions:

$$O + N_2 \leftrightarrow NO + N \quad (1.1)$$

$$N + O_2 \leftrightarrow NO + O \quad (1.2)$$

$$N + OH \leftrightarrow NO + H \quad (1.3)$$

Because hydrogen flames typically exhibit higher peak temperatures than hydrocarbon flames at identical combustion conditions like lean, rich or stoichiometric ie. Equivalence Ratios, controlling the residence time of reactants within these high-temperature (stoichiometric) zones is critical for emissions reduction. Note the OH* radical that is being produced in the process (see 1.3) will become significant later to help visualize the flame topography.

NNH Pathway:

In hydrogen-air combustion, the NNH pathway can become a significant contributor to NO formation, particularly within the flame fronts where there are high radical concentrations. The mechanism initiates with the reaction of a nitrogen molecule with an hydrogen radical (see eq 1.4), followed by oxidation. This route remains *relevant even at lower temperatures* where the thermal Zeldovich mechanism is less dominant.

$$N_2 + H \leftrightarrow NNH \quad (1.4)$$

$$NNH + O \leftrightarrow NO + NH \quad (1.5)$$

Nitrous Oxide (N_2O) Intermediate Mechanism:

Under high pressures and very lean conditions, N_2O can form as an intermediate species & subsequently decompose into NO. However, in the usual atmospheric hydrogen-air combustion which has high-temperatures, this contribution is generally lower than the thermal Zeldovich & NNH pathways.

Fuel-bound NOx:

This contribution originates from chemically bound nitrogen within the fuel. Since the hydrogen molecule, being the simplest of molecules, does not contain nitrogen making this contribution completely absent.

1.2.2 Hazard of NOx to environment and human health:

Beyond the chemical mechanisms of NO_x formation, the environmental and health impacts of the emissions due to Hydrogen Combustion require careful evaluation. Environmentally, regarding altitude-dependent effects: The greenhouse impacts of both carbon dioxide and water vapor vary with altitude [21] . Below 10 km, the impact of water vapor as a greenhouse gas is negligible, but above this threshold, the effect increases with height. While CO_2 exhibits a residence time of over 100 years independent of altitude, water vapor persists for only 3 to 4 days at mean sea level (MSL) compared to 6-12 months in the stratosphere [20]. On the other hand, a small reduction in cruise altitude by adjusting the flight paths can eliminate the warming effect caused by water vapor emissions.

Table 1
GWP regarding altitude [5].

Altitude (km)	GWP (CO_2)	GWP (H_2O)	GWP (NO_x)
0	1	0.00	−7.1
1	1	0.00	−7.1
2	1	0.00	−7.1
3	1	0.00	−4.3
4	1	0.00	−1.5
5	1	0.00	6.5
6	1	0.00	14.5
7	1	0.00	37.5
8	1	0.00	60.5
9	1	0.00	64.7
10	1	0.24	68.9
11	1	0.34	57.7
12	1	0.43	46.5
13	1	0.53	25.6
14	1	0.62	4.6
15	1	0.72	0.6

Fig 1.1 GWP regarding altitude [20]

In addition to the altitude considerations, nitrogen oxides (NOx) pose severe hazards to both human health and ecosystems. NOx reacts with volatile organic compounds in the presence of sunlight to generate ozone at ground level altitude, thereby exacerbating photochemical smog and causing respiratory ailments in humans due to it. Direct exposure to NO_x is linked to airway inflammation, reduced lung function, the aggravation of asthma and chronic respiratory diseases- these become concerns to public health.

Environmental consequences are equally critical: NO_x oxidizes into nitric acid (HNO_3) , causing acid rain that acidifies ecosystems and affects water quality, while nitrogen deposition promotes excessive algal growth that suffocates aquatic systems. Furthermore, NO_x interacts with ammonia and sulfur oxides to form fine particulate matter, diminishing visibility and further endangering public health. On a broader atmospheric scale, certain nitrogen oxides like nitrous oxide (N_2O), though unregulated as standard NO_x, act as potent greenhouse gases and contribute significantly to stratospheric ozone depletion.

1.3 Double Swirl Burner

To mitigate and handle the critical issues associated with hydrogen combustion; like flashback and elevated NO_x emissions, a commonly adopted strategy is a delayed injection of the fuel as late as possible within the combustion chamber. [13] [17]

Anchoring of diffusion flames directly to burner components creates severe operational issues. Local overheating occurs because hydrogen flames reach very high temperatures, leading to thermal damage of injector hardware. Furthermore, anchored flames increase flashback risk as the flame propagates upstream into premixing passages when local velocities drop below the local laminar flame speed of hydrogen [15]. Depending on specific burner geometry, near-wall stoichiometric or slightly rich hot spots in these anchored regions can also drastically promote thermal NO_x formation. [17] [31]

To achieve aerodynamic stabilization, swirling air flows are used to generate a Central Recirculation Zone (CRZ) [16]. This low-pressure region along the combustor axis promotes the backflow of hot gases and radicals, anchoring the flame away from burner walls and allowing stable reignition if the flame extinguishes. [16] [23]

Coaxial double-swirl injectors provide flexibility over single-swirl systems. For specific geometries, altering the outer swirl intensity and momentum-flux ratio produces distinct flame topologies [18] [29]. In studies utilizing the EM2C burner geometry, researchers identified five flame topologies ranging from attached M-shaped flames (driven by strong outer swirl and forming a widened base attached to injector walls) to highly lifted V-shaped configurations (where higher momentum flux overrides recirculation).

To non-intrusively analyze these flame topologies, optical diagnostics frequently rely on OH* chemiluminescence [18] [25]. During high-temperature hydrogen-oxygen reactions, excited hydroxyl radicals (OH*) emit photons in the ultraviolet band. OH* chemiluminescence serves as an indicator of the reaction and heat-release regions under defined conditions, though it is not universally proportional to local heat release. Furthermore, Abel inversion techniques used to process these images assume approximate axisymmetry and cannot correct for flame asymmetry [25].

Consequently, aerodynamic features such as the presence, structure, or collapse of the CRZ must be formally identified from the velocity field rather than from OH* emission alone.

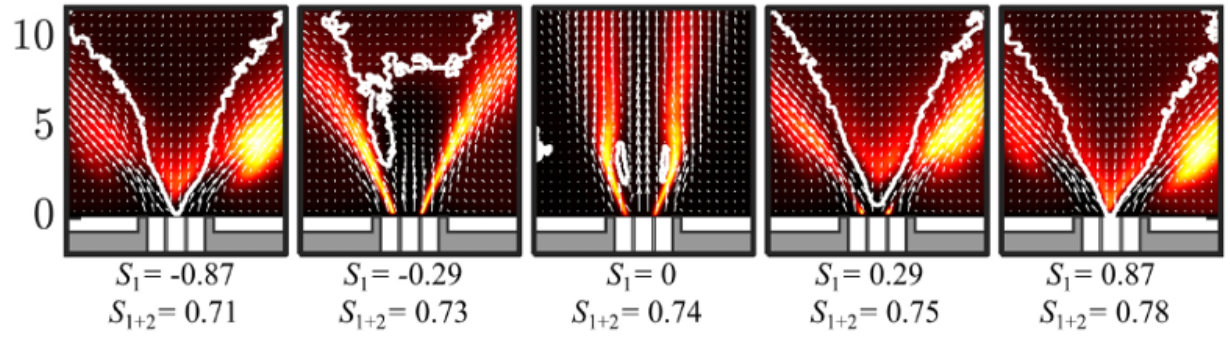


Fig 1.2 from [22] OH-PLIF/PIV measurements for flames at S 2 = 0 . 85 , J = 0 . 75 and S 1 (D 2 dataset) varying from S 1 = −0.87 to 0.87. Mean OH-PLIF signal is shown in color scale. The average velocity field is superimposed with white arrows. The recirculation zone is delineated by the thick white contours.

Introducing partial premixing (rich core and lean annulus) shifts operation toward a Rich-Quench-Lean (RQL) mode. Lowering the central equivalence ratio Φ causes a secondary premixed flame front to form upstream of the main diffusion flame, shortening flame length and nearly halving EINOx emissions [Fig.1.3 and 1.4 ref [27]]. However, parametric studies and Large Eddy Simulations (LES) show that traditional jet-flame scaling laws fail under strong swirl or equal jet velocities (u1/u2), as premixed flame fronts alter shear-layer strain distribution and disrupt classical turbulence-strain correlations. [25]

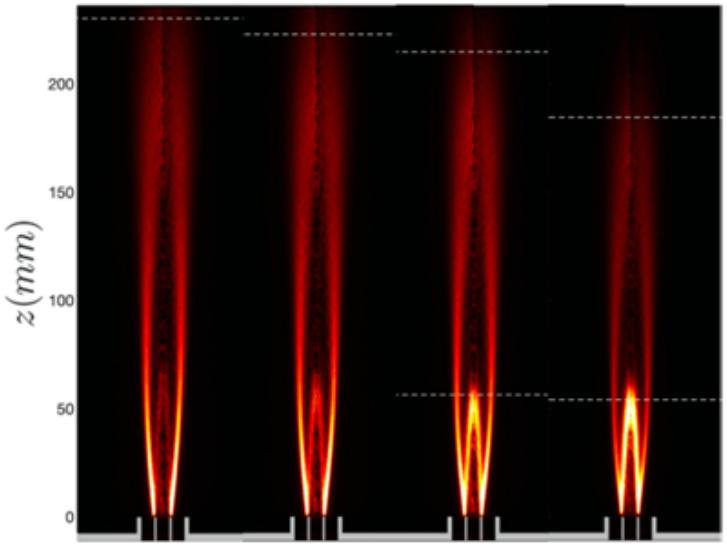


Fig 1.3 Abel-inverted OH* chemiluminescence images at various values of Φ1 with d1=6mm,Φtot=0.4, Pth=10kW. From left to right: Φ1 = 3.5, 3, 2.5 & 2 [27]

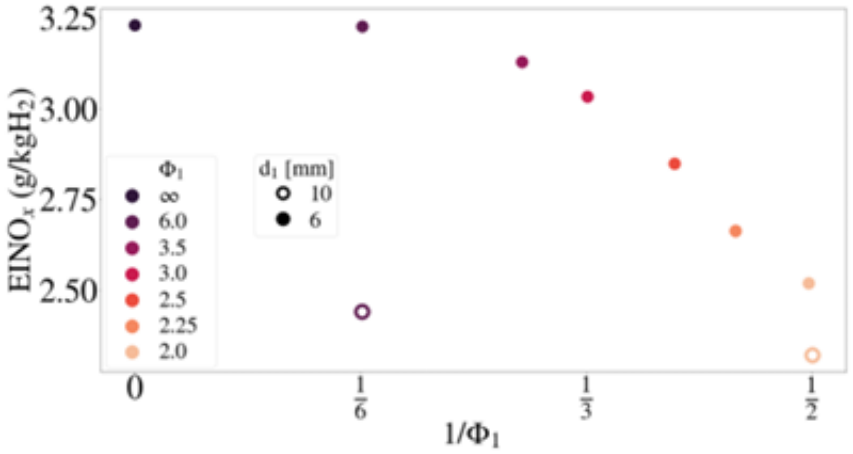


Fig 1.4 EINOx for Various air staging conditions with d1=6mm, Φtot=0.4 and Pth=10kW [27]

Flame Topologies and Momentum Effects According to research on double-swirl configurations, altering the outer swirl intensity S2 and the momentum-flux ratio (J) produces five distinct flame topologies, ranging from attached M-shaped flames to highly lifted V-shaped configurations (schematically shown in Fig. 1.5 and visibly captured in Fig. 1.7 ref [18]). The momentum-flux ratio essentially governs the jet penetration speed, which directly dictates the lift-off phenomena. The transition

across these topologies is governed by the balance between the jet momentum and the stabilizing CRZ:

- Type I (Attached M-shaped flame): Driven by a strong outer swirl (S2 = 1), the flame base widens and attaches to the injector walls. The stable CRZ keeps the flame safely anchored in an M-shape with two visible wings.
- Type II & III (Compact lifted flame): With lower momentum and swirl (S2 = 0.75 - 0.8), the flame becomes slimmer and less spread. Its root detaches slightly from the injector exit, indicating a reduction in anchoring and a transition toward a V-shape.
- Type IV (Highly lifted V-shaped flame): A higher momentum flux pushes the flame tip further downstream, establishing a clear V-shaped structure due to strong jet penetration overriding the recirculation.
- Type V (Very lifted narrow flame): Weak swirl and low momentum cause the flame to detach far downstream. This topology is highly vulnerable to blow-off and exhibits reduced stability.

The structural variations of these flame anchoring regimes are further validated.
To non-intrusively analyze these highly complex flame topologies, optical diagnostics rely heavily on OH* chemiluminescence (Fig.1.3 & 1.7) [25]. During high-temperature hydrogen-oxygen reactions, excited hydroxyl radicals (OH*) are continuously generated. As these very unstable radicals rapidly fall back to their lower energy state, they emit photons mostly close to the ultraviolet band. Because the intensity of the OH* that gives us this UV emission. Because this UV emission serves as a reliable indicator of the approximate reaction zone, OH* Imaging allows researchers to mark flame front boundaries, penetration lengths, and stabilization regimes without disturbing the flow field [25] [27]. This provides direct visualization of the flame structures and the transition from stable, compact CRZ anchoring to extended, lifted flames prone to blow-off .

Influence of Thermal Power and Equivalence Ratio on Emissions: This section evaluates key operational parameters of double swirl hydrogen burners, specifically analyzing how input thermal power and global equivalence ratio impact NO_x production. It highlights literature demonstrating that NO_x emissions consistently decrease at higher thermal power outputs and leaner operating conditions across both anchored and lifted flame regimes. [31]

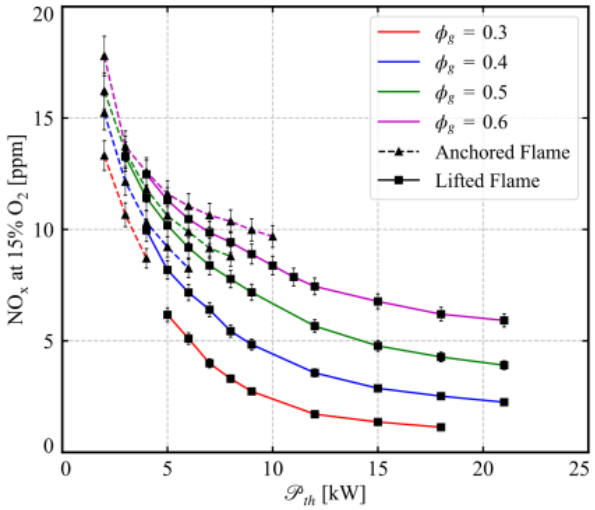


Fig 1.5 NOx emissions as a function of the input thermal power Pth for different global equivalence ratios Φg when air is injected at ambient temperature Tair,in = Tamb. Adapted from [31].

A general decrease in NO_x emissions can be noticed with the increase of thermal Power (Pth); a reduction in NO_x emissions can also be noticed as Φg reduces and tends towards a leaner operation-[31]

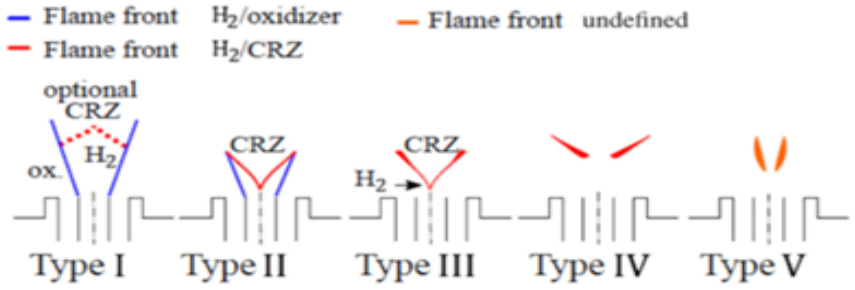


Fig 1.6 Flame Structure types observed on the dual-swirl coaxial injector. [18]

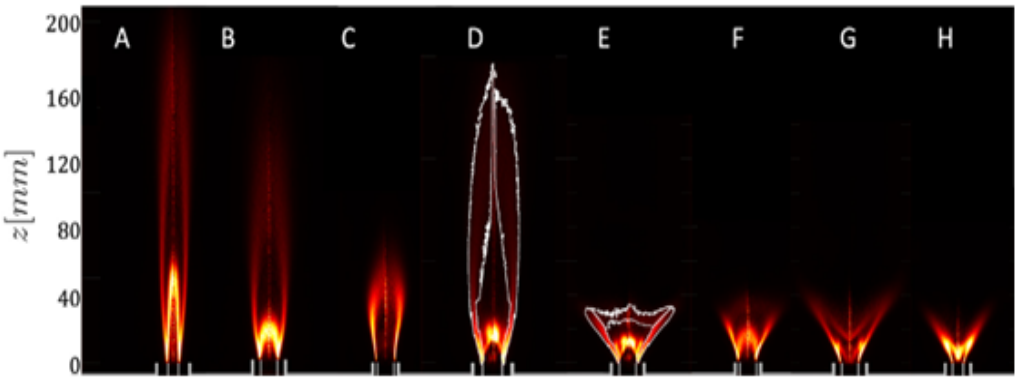


Fig 1.7 Inverse Abel transform of OH* chemiluminescence images of eight flames for different injection conditions, injector geometry and combustion chambers [25]

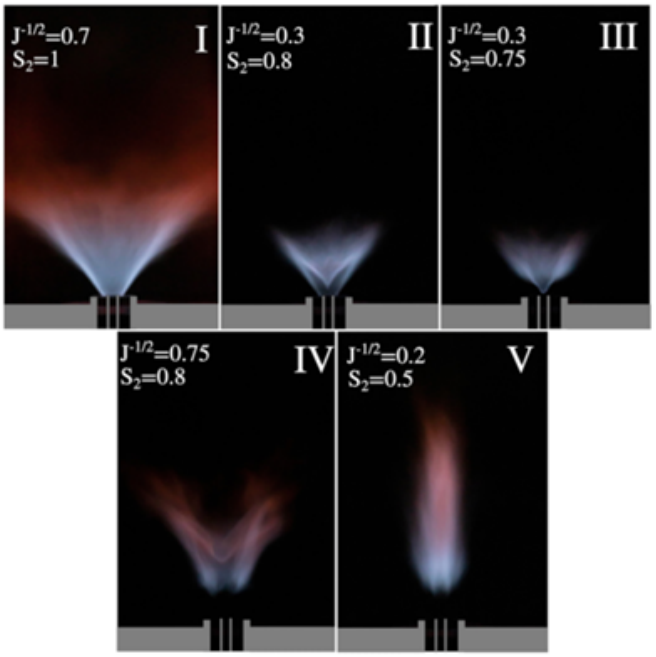


Fig 1.8 Short exposure time visible range image of the different flame types. Pth =10kW, S1=0.87. [18]

1.3.1 Co-swirl combustor

In a co-swirl configuration, the inner fuel jet and the outer air stream rotate in the same direction, this aligns and adds up their angular momentum. As established in foundational swirl aerodynamic literature [23], when the global Swirl Number exceeds a certain critical threshold, the phenomenon of vortex breakdown occurs. This breakdown induces a severe adverse (reverse) axial pressure gradient, forcing the core flow to decelerate, stagnate, and reverse upon itself, thereby establishing a unified Central Recirculation Zone (CRZ). [16] [23]

Recent numerical and experimental studies on 100% hydrogen dual-swirl combustors (such as the HYLON and DFDS burners seen in Fig 1.9 & 1.10) demonstrate that this co-rotating configuration greatly influences flame topology. [17] [24]. The strong, continuous backflow of hot combustion products (radicals) provided by the CRZ aerodynamically anchors the highly reactive hydrogen flame safely above the injector lips [24]. This stabilization mechanism typically produces a distinct attached M-shaped or lifted V-shaped flame brush Fig 1.11, where the primary reaction layer evolves along the boundary between the fresh reactants and the recirculating hot gases. [24] [29]

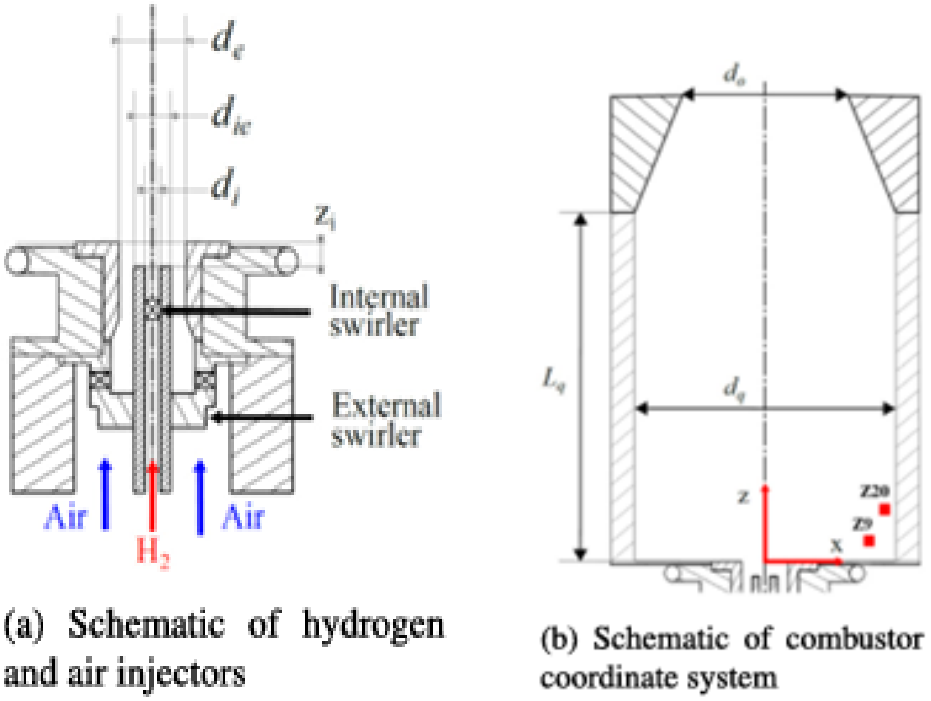


Fig 1.9 Hylon burner schematic and coordinate system [24]

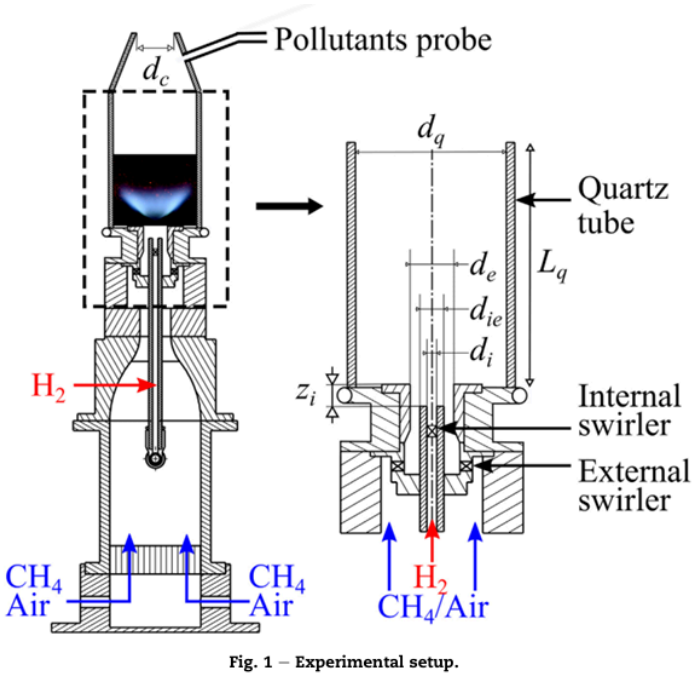


Fig 1.10 DFDS burner [17]

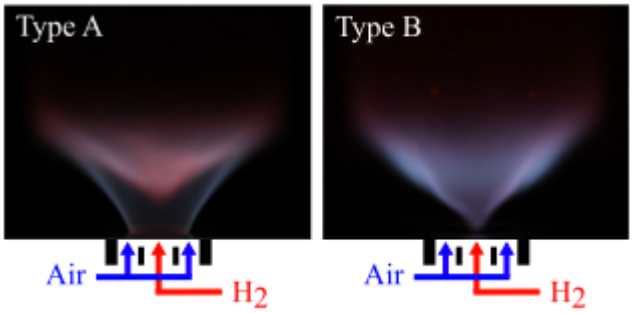


Fig 1.11 Hydrogen air flame archetypes stabilized with the HYLON burner. Left: Type A anchored flame. Right: Type B lifted flame. [29]

1.3.2 Counter-swirl combustor

Counter-swirl combustor architectures introduce opposing rotation between the inner and outer streams, generating intensified shear layers and enhanced mixing compared to co-swirling setups. [22]

The operational viability of a dual-swirl configuration is heavily influenced by the momentum ratio (J) between the inner and outer streams. For the specific geometry and confinement of the RAPIDS-H burner, the following behaviors are hypothesized based on swirling jet literature of Marragou [17], the momentum ratio can be defined by Eq (1.6):

$$J = \frac{\rho_e u_e^2}{\rho_i u_i^2} \quad (1.6)$$

Where subscripts e means external and i means internal.

For the coaxial dual-swirl burner investigated by Marragou et al. [17], decreasing (J) corresponds to increasing the momentum of the central stream relative to the annular stream. The authors show that, for their burner geometry, increasing the central hydrogen injection velocity decreases (J) and causes the central jet to penetrate further into the annular flow. At sufficiently low (J), the central jet pushes the central recirculation zone (CRZ) downstream and can destabilize it [17]. A sufficiently high internal swirl can promote aerodynamic flame stabilization above the injector; however, high internal swirl alone is not sufficient to maintain a lifted flame when (J) becomes too low. [17]

The effect of (J) is therefore coupled to the internal swirl level and injector geometry. In particular, Marragou et al. [17] demonstrate that a small recess of the inner fuel lance increases the range of operating conditions over which aerodynamically stabilized, lifted flames can be maintained by modifying the effective momentum ratio at the burner outlet. [17]

The momentum flux of a fluid jet is proportional to $\rho U^2 A$. To achieve the same thermal power output as conventional hydrocarbon fuels, hydrogen's very low density implies that it must be injected at significantly higher velocities [12]. Because velocity is squared in the momentum flux equation, this faster injection velocity greatly increases the momentum of the hydrogen stream. In a dual-swirl configuration, this causes the inner fuel stream to travel faster, potentially shifting the operating condition toward an inner-dominated momentum ratio ($J < 1$). Depending on the specific burner geometry, this momentum shift can shorten the flame length and push the stabilization point downstream (lift-off), altering wall thermal loads but increasing the risk of blow-out. [17] [26]

Counter-swirl configurations can be leveraged for advantageous fuel stratification (staging a fuel-rich inner core with a highly lean outer flow). The intense shear layer accelerates localized mixing. As demonstrated by Leroy [27] this specific stratification strategy distributes heat release and reduces the residence time of reactants in peak adiabatic temperature zones, effectively suppressing thermal NO_x formation in hydrogen-air systems, these setups typically sacrifice axial symmetry. Depending on

specific geometry, they may feature two dominant recirculation regions and exhibit extreme sensitivity to variations in Split, Equivalence and Momentum ratios. [22]

1.4 The RAPIDS-H Burner, Previous Investigations & Knowledge gap:

The present thesis builds upon a series of progressive experimental campaigns conducted on the RAPIDS-H double-swirl burner at the Combustion and Optical Diagnostics Laboratory of Politecnico di Milano. Foundational studies by Pirci [37] and Syed [38] characterized the flame morphology and stability limits of unconfined natural gas and hydrogen blends. Recently, the architecture was upgraded to accommodate 100% hydrogen combustion in a confined, partially premixed configuration. As detailed by Cozzi [36], the baseline co-rotating configuration (Model A injector) successfully established a robust, symmetric Central Recirculation Zone (CRZ) that mitigated flashback and stabilized the highly reactive H_2 flame. In parallel, the isothermal aerodynamic flow field of this co-swirl configuration was meticulously mapped using Stereoscopic Particle Image Velocimetry (Stereo-PIV) by Bezzi and Sala [35].

This thesis explicitly reuses the Stereo-PIV dataset acquired by Bezzi and Sala [35], alongside the OH* chemiluminescence and emissions datasets established by Ciuffoli and Rocco [28], to serve as the baseline for the Model A co-swirler.

The operational baseline for the present research relies on recent experimental investigations conducted on the RAPIDS-H double-swirl burner by Ciuffoli and Rocco [28]. Utilizing Stereo-PIV and OH* chemiluminescence imaging, their work thoroughly characterized the isothermal flow field and reacting flame morphology of the co-swirler (Model A injector) Fig 1.12. They observed that the co-swirl setup provides excellent flame-holding capabilities and effectively mitigates flashback due to the massive, highly symmetric structure of its CRZ. However, because both fluid streams rotate in unison, the interfacial shear layer generated between the inner hydrogen core and the outer air annulus is relatively (to the counter-swirl) milder.

The Knowledge Gap that creates the need for our genuinely new contribution of this work is the numerical simulation and experimental testing of a modified counter-swirl inner injector and compared with equivalent conditions of the model A co-swirl inner injector.

The objective is to analyze and compare the flow structures, flame morphology, and NO_x emissions between the co- and counter-swirl configurations to determine if the theoretical benefits of emissions of the counter-swirl staging outweigh the aerodynamic penalties (instability) in a 100% hydrogen environment.

This established baseline forms the comparative foundation for the present thesis. While the co-swirler inherently prioritizes ultimate aerodynamic stability through a unified CRZ, the present study investigates the counter-swirler to intentionally disrupt this unified rotation. Reversing the inner swirl is expected to drastically amplify the turbulent shear mixing at the interface of the two streams, a mechanism that directly alters both the flame's morphological stability and the localized thermal NOx formation routes.

Fig 1.12 Model A co-swirl Injector [28]

2. Numerical Methodology

2.1 Numerical Background

Computational Fluid Dynamics (CFD) works on the basis of the numerical resolution of the fundamental governing equations of fluid mechanics to simulate the fluid dynamics phenomena.

In this study, the commercial software ANSYS Fluent is utilized to solve these equations across a discretized computational domain (Mesh), with careful modelling and tweaks to the solver to enable a detailed and an accurate as possible reconstruction of the highly swirling, transient flow fields characteristic of the RAPIDS-H burner's swirl flow structures.

2.1.1 The Finite Volume Method (FVM) and Mesh Topology

ANSYS Fluent like any other CFD software employs the Finite Volume Method (FVM) to transform the continuous partial differential equations of fluid flow into a system of discrete algebraic equations. The fluid domain is divided into a finite number of discrete control volumes (cells), where the conservation of mass, momentum & energy is enforced over each individual volume.

To optimize computational efficiency while maintaining high spatial accuracy, a polyhedral mesh topology is utilized. This advanced meshing strategy discretizes the fluid domain using polyhedrons rather than standard tetrahedral elements. Because polyhedral cells possess significantly more adjacent faces, they provide superior spatial gradient approximations across the control volumes, effectively minimizing false numerical diffusion, which is particularly important for our case- highly swirling flows. Additionally, this topology drastically lowers the overall cell count required to achieve grid independence compared to equivalent tetrahedral grids. Near the solid boundaries, the automatic mesher algorithm created *structured prism inflation layers* to accurately resolve the viscous sublayer- *this assists with the Boundary Layering operation*, smoothly transitioning into the purely polyhedral cells within the bulk flow. This architecture allows for superior resolution of the high-shear mixing zones (sharp gradients) and the Precessing Vortex Core (PVC) while balancing our memory limitation of the High-Performance Computing (HPC) cluster.

2.1.2 Governing Equations and the URANS Formulation

The isothermal flow field of the burner is governed by the conservation of mass and momentum, described by the Navier-Stokes equations. The general form of the mass conservation (continuity) equation is given by:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = S_m \quad (2.1)$$

where ρ is the fluid density, $\vec{v}$ is the velocity vector, and S_m accounts for any mass added from user-defined sources. The conservation of momentum is described by:

$$\frac{\partial}{\partial t}(\rho\vec{v}) + \nabla \cdot (\rho\vec{v}\,\vec{v}) = -\nabla p + \nabla \cdot (\bar{\bar{\tau}}) - \nabla \cdot (\rho\,\overline{\vec{v}'\vec{v}'}) + \rho\vec{g} + \vec{F} \quad (2.2)$$

where p is the static pressure, $(\bar{\bar{\tau}})$ is the mean viscous stress tensor, $\rho\,\overline{\vec{v}'\vec{v}'}$ represents the unknown Reynolds stress tensor that requires the turbulence model equations for closure $\rho\vec{g}$ is the gravitational body force, and $\vec{F}$ represents external body forces.

2.1.3 The SST k-ω Turbulence Model

Directly resolving all temporal & spatial scales of turbulence is computationally infeasible. Therefore, the Unsteady Reynolds-Averaged Navier-Stokes (URANS) approach is applied, which averages the governing equations but introduces a closure problem via the Reynolds stress tensor.

To achieve closure, the Shear-Stress Transport (SST) k-ω model is employed. The SST model acts as a hybrid, utilizing the robust k-ω formulation in the near-wall region and smoothly switching to the k-ϵ formulation in the free stream to avoid the sensitivity to inlet turbulence. A notable feature of the SST model is its limitation of the eddy-viscosity μ_t to account for the proper transport of turbulent shear stress:

$$\mu_t = \frac{\rho k}{\omega} \frac{1}{\max\left[\frac{1}{\alpha^*}, \frac{SF_2}{a_1\omega}\right]} \quad (2.3)$$

where S is the strain rate magnitude, F_2 is a blending function, and a_1 is a model constant.

2.1.4 Near-Wall Treatment: Kader Blending

Resolving the boundary layer is critical for capturing the wall-induced shear stresses that sustain swirling flows. ANSYS Fluent utilizes the *Kader blending wall functions* to seamlessly bridge the **laminar sublayer** and the **logarithmic law** of the wall regions.

When the mesh is fine enough to resolve the laminar sublayer, the wall shear stress is determined via the ***laminar stress-strain*** relationship:

$$\frac{\bar{U}}{u_\tau} = \frac{\rho u_\tau y}{\mu} \tag{2.4}$$

Conversely, in the logarithmic region, the ***logarithmic law-of-the-wall*** is employed:

$$\frac{\bar{u}}{u_\tau} = \frac{1}{\kappa}\ln(E\,\frac{\rho u_\tau y}{\mu}) \tag{2.5}$$

where κ is the von Kármán constant and E = 9.793 and the non-dimensional wall distance is $y^+ = \frac{y u_\tau}{\nu}$. If the first near-wall grid node falls within the transitional buffer region. This will be critical for us to resolve the y^+ threshold problems as parts of our combustor see such high velocity gradients (outer-swirl bottle zone see Fig 2.1) that and meshing them to have a very fine boundary layering to capture the viscous effects which will cause the total cell number to increase exponentially. Due to the complex geometry of the double-swirl burner and the particular fluid physics causing intense velocity gradients, creating a strictly wall-resolved mesh (uniformly $y^+ \sim 1$) across the entire domain was computationally and practically (due to mesh collapse) infeasible. While the bulk of the combustor mesh targeted a low y^+ value, the intense velocity gradients within the outer-swirler bottle generated unavoidable localized peaks in the y^+ distribution, reaching values between 20 and 45 (as shown in Fig 2.1). As a consequence, Kader blending was strictly necessary. This formulation mathematically blends the sublayer and log-law equations to ensure an accurate, continuous velocity profile when the first grid node falls within the transitional buffer region. This provides a necessary trade-off: it relaxes the stringent near-wall requirements of the standard k-ω model, keeping the overall cell count computationally affordable while still resolving viscous effects with sufficient accuracy to capture the flow physics.

$$y^+ = \frac{y u_\tau}{\nu} \tag{2.6}$$

$$\text{Where, } u_{\tau} = \sqrt{\frac{\tau_w}{\rho}} \tag{2.6.1}$$

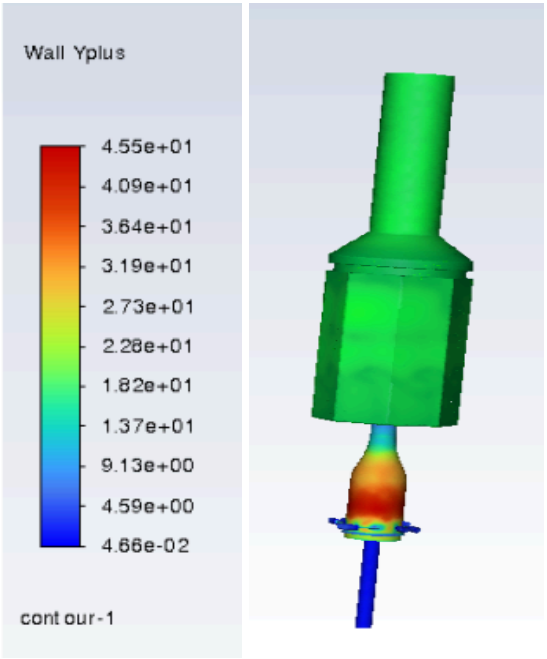


Fig 2.1 Wall Y^{+}volume contour- the intense velocity gradients in the outer-swirl bottle zone cause the $Y^{+}$to visibly peak- a BL refinement fix will add millions of cells.

2.1.5 Advanced Turbulence Modifications for Swirling Flows

Standard eddy-viscosity models fundamentally assume isotropic turbulence and are generally ineffective in sensing streamline curvature, system rotation & corner flows. To properly capture the highly anisotropic & asymmetric dynamics of the double-swirl burner, the following mathematical corrections are applied for the most accurate modelling of the phenomena. All the following are adapted from Ansys Fluent- Theory Guide [30]. The Shear-Stress Transport (SST) k-ω model was utilized in conjunction with the following specific mathematical corrections enabled within the ANSYS Fluent solver:

- Curvature Correction: To account for the strong streamline curvature of the swirling jets, a curvature correction multiplier (f_k) is applied directly to the turbulence production term (P_k). This sensitizes the model to rotation, preventing artificial dampening of the angular momentum required to sustain the Central Recirculation Zone (CRZ):

$$P_k \rightarrow P_k \cdot f_r \tag{2.7}$$

- Corner Flow Correction: Complex injector geometries often induce secondary flow complexity. To prevent the over-prediction of corner separation zones, a quadratic non-linear algebraic extension is applied to the modeled Reynolds stress tensor. This artificially drives momentum into the corners, mimicking the anisotropic behavior of true differential Reynolds stress models without the massive computational requirement, essentially it dynamically modifies the turbulent viscosity to accurately preserve swirling streamlines and vortex boundaries.
- Production Limiters (Kato-Launder): Standard two-equation models suffer from an excessive, unphysical generation of turbulent kinetic energy (k) (TKE overprediction that we will discuss when comparing the PIV and CFD velocity profiles) near stagnation points. Because the flow field near a vortex breakdown stagnation point is nearly irrotational, the Kato-Launder formulation modifies the turbulence production term (G_k) to scale with both the strain rate magnitude (S) and the vorticity magnitude (ω):

$$G_k = \mu_t S\,\Omega \tag{2.8}$$

This limiter suppresses unphysical turbulent bursts while preserving shear layer performance and prevents the unphysical overprediction of turbulent kinetic energy (k) in the irrotational stagnation zones generated near the injector rims.

2.2 Geometry Preparation

The computational fluid domain (VolumeExtract function- see Fig 2.2) was extracted from the three-dimensional CAD model of the RAPIDS-H burner using ANSYS SpaceClaim- The conventional Watertight geometry workflow was chosen, and the domain was broken down strategically into Zones to give us versatility during the meshing phase, allowing localized surface & volumetric sizing controls to be applied independently to each fluid stream without forcing a uniformly dense grid across the entire burner. This is crucial to maintain physical accuracy and capture the fluid dynamic phenomena.

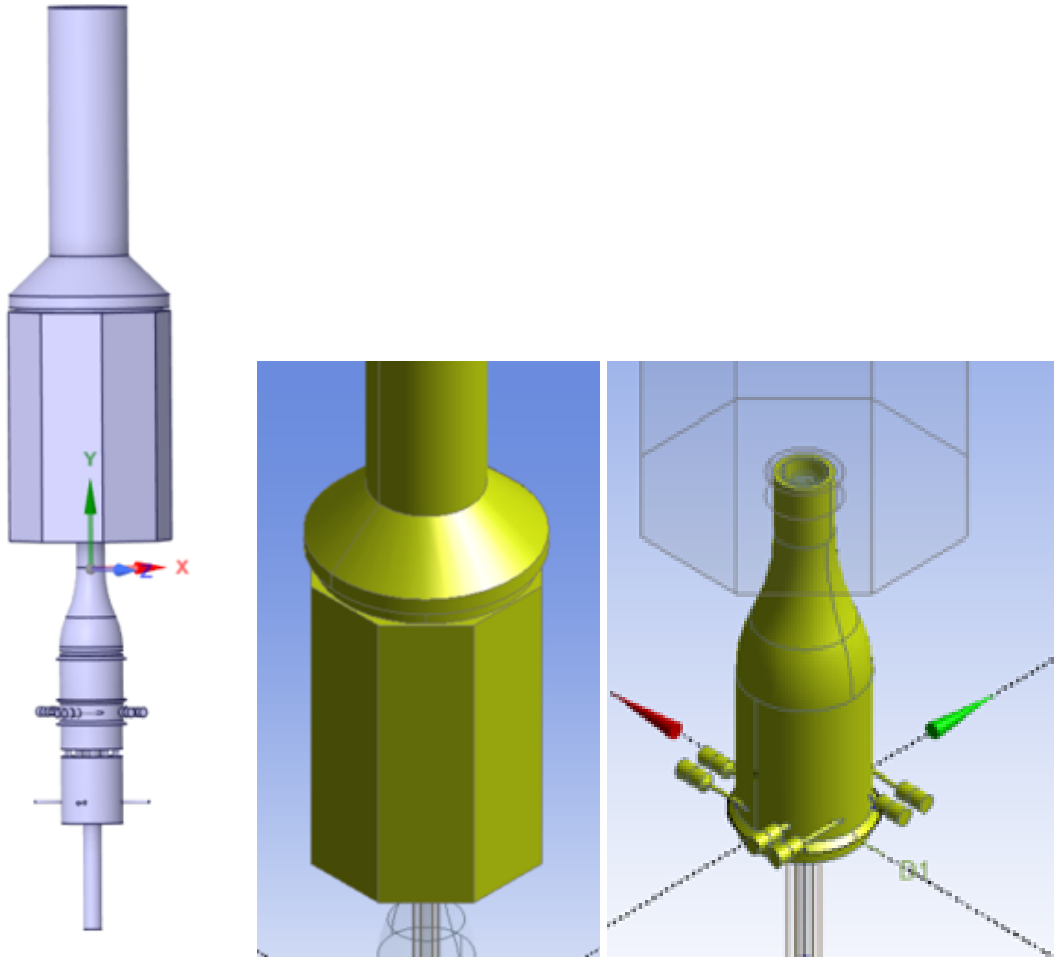


Fig 2.2 Full Volume Extract (left). Sample pictures of Zone separations; Combustor zone (center) and Outer-swirler zone (right) can be seen illustrated in the picture.

Due to the intricate injection patterns, narrow coaxial passages, and the massive scale (relatively) of the downstream combustion chamber, simulating the entire physical volume uniformly was computationally infeasible. To simplify the domain without sacrificing aerodynamic fidelity, a Body of Influence (BOI) strategy was implemented to achieve local refinement. A virtual cylinder *(168 mm in diameter & 170 mm in height)*- Fig 2.3a was modeled originating from the combustor base to act as the BOI.

This allowed the mesh refinement to be strictly constrained to the critical near-field mixing regions and the Central Recirculation Zone (CRZ), while the downstream far-field was aggressively coarsened to minimize the computational footprint (Fig 2.3b).

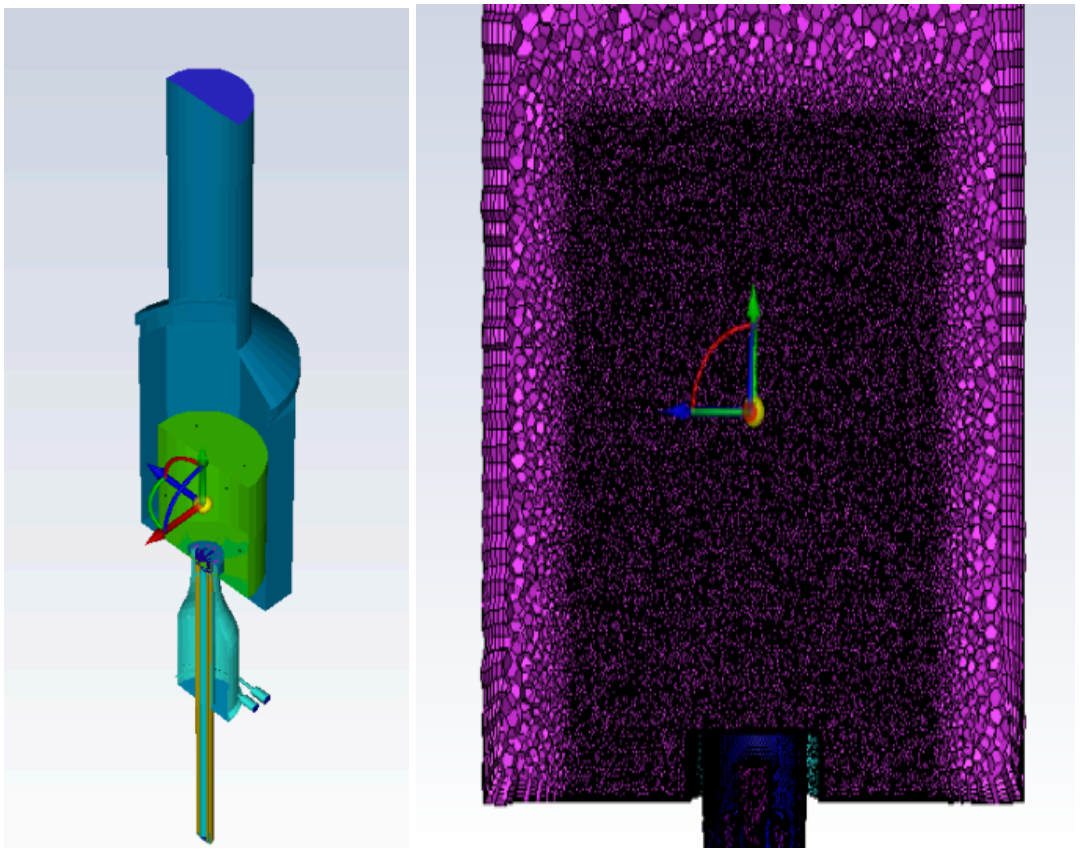


Fig 2.3 a) BOI illustrated in green b) local refinement for the BOI cylinder, the downstream coarsening of the combustor mesh is also visible

2.3 Mesh Scheme

To accurately capture the high shear gradients and the Precessing Vortex Core (PVC) with minimal numerical diffusion, a **3D Polyhedra mesh topology** was utilized. The final volumetric meshes comprised approximately 2.8 million cells for the co-swirler and 2.2 million cells for the counter-swirler.
Resolving the boundary layer was critical to preserving swirl momentum, as artificial numerical dampening at the walls would destroy the angular momentum required to trigger vortex breakdown. The first cell height was assigned such that to target a Y^+ value close to 1 and under 5.

Because the complex double-swirl geometry generates extreme velocity variations across the domains, yet extreme refinements to boundary layers to meet the y+ ~1 demand in all zones caused the cell number to explode and also collapsed the mesh as the mesher could not smoothly transition between zones of very different refinement scales, therefore a global boundary layer treatment of 5 inflation layers with a 1.2 growth rate was applied.

While this resulted in localized y+ deviations (e.g., peaking near 20 in the high-velocity outer-swirler) in certain locations as discussed before, the SST k-ω

model's automatic wall treatment (Kader wall blending) smoothly bridged the sublayer ($y^+ \sim 1$ under 5) and log-law regions (y^+: $30 - 300$):, fully protecting global mass conservation and macro-scale recirculation dynamics.

The inlet & outlet assignments, computations for the inlet velocity for the inlet boundary patches from the experimental flowrate data and meshing setup (BOI refinement, Face & Body size, surface mesh, fluid domain definition, share topology, boundary layering and Volume mesh Generation are available in the **Appendix A**.

2.4 Solver Setting

The numerical analysis was executed in ANSYS Fluent utilizing a two-phase solver strategy to balance physical accuracy with computational efficiency:

- Steady-State Initialization: A pressure-based coupled solver was initially run for 3,000 iterations. This rapidly established the fully developed velocity profiles and adverse pressure gradients of the CRZ, saving us precious computational time steps that would have otherwise been lost to the flow and CRZ establishment, This also aids in efficiently flushing out initialization noise before engaging the transient solver.
- Transient Phase: The simulation transitioned to the Pressure-Based Coupled Solver, utilizing a bounded second-order implicit transient formulation. Unlike segregated algorithms that solve the mass and momentum equations one after the other, the coupled solver resolves them simultaneously. For highly swirling flows characterized by severe adverse pressure gradients and intense pressure-velocity coupling, this simultaneous resolution significantly enhances numerical robustness (stability and accuracy). While it demands a higher memory footprint per iteration, the coupled approach prevents divergence during the critical transition to transient operation and drastically reduces the number of inner iterations required to achieve convergence within each time step The transition from a steady to a transient solver inherently introduces a numerical shock to the flow field, as the time-derivative terms in the governing equations are activated instantaneously. To prevent this shock from destabilizing the pressure-velocity coupling and diverging the solution, a conservative time-stepping "ease-in" and successive time step optimization strategy was employed.

Initially, an extremely small time step (Δt = 0.00001 s) was prescribed, with a strict maximum of 25 internal iterations per step to ensure absolute stability and sufficient

convergence. Once the residual trends confirmed that the initial numerical transients had safely dissipated (typically after 50–100 time steps), the step size was incrementally increased to 0.0001 s. Upon verifying that the Cell Convective Courant Number (CFL) (see Eqn 2.9) provided sufficient headroom, the final operational time step was set to 0.00025 s. This optimized value maintained a highly conservative global CFL of approximately 0.11, peaking at only 1.35 within the highest-velocity regions of the outer swirler, thereby significantly accelerating the computation toward the targeted physical wall-clock time without sacrificing temporal fidelity, computational & physical accuracy.

$$CFL = \frac{u\,\Delta t}{\Delta x} \tag{2.9}$$

Where Δt is the time step size, Δx is the cell size, and u is the velocity of a fluid particle through the cell in question. While a CFL of ~1 is considered ideal the reduction of the time step greatly increases the storage demand, while an increment of the cell size can cause loss of refinement, hence loss of accuracy in capturing the flow physics. So the time step size and local refinement must be chosen carefully, keeping in mind the node storage and accurately capturing the flow physics.

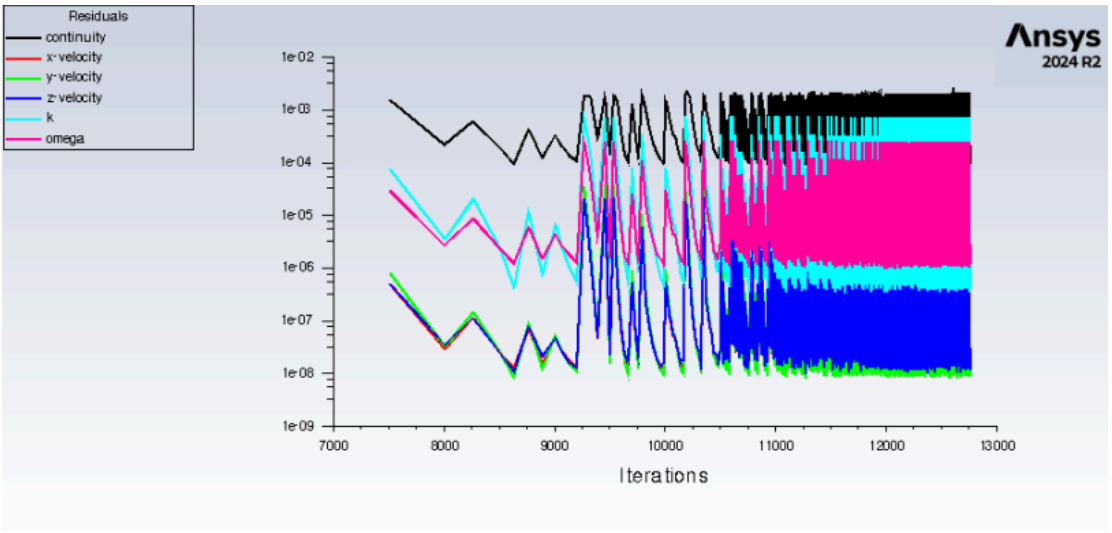


Fig 2.4 Convergence pattern of the transient run seen in residual plots of the 6 parameters of the k-ω SST solver.

The success of this optimization scheme is visibly captured in the residual convergence history (Fig. 2.4). In a transient CFD plot, each "sawtooth" represents the residual spike at the initialization of a new time step, followed by its decay over the internal sub-iterations. The distinct transition across three zones of varying sawtooth density visually maps the solver's stabilization. As the flow dynamics stabilize, the

iterations required to close each time step decrease, yielding the densely packed, high-frequency sawtooth pattern observed in the latter stages of the plot.

2.5 HPC Cluster:

The computational work was conducted utilizing a High-Performance Computing (HPC) cluster architecture associated with the Department of Energy. A primary technical limitation governing the numerical methodology was a 15 GB storage assigned for a computational node. This hardware constraint heavily dictated the meshing strategy; by utilizing a polyhedral meshing scheme coupled with a Body of Influence (BOI) local refinement strategy, the total file storage per run was successfully compressed to approximately 4 GB for the 3 million-cell co-swirl model and 4 GB for the 2.2 million-cell counter-swirl model. Each case was computed separately by partitioning the domain across 12 of the 16 available cores per node, resulting in an efficient load distribution of approximately 180,000 to 250,000 cells per core. This optimized configuration allowed the transient simulations to successfully compute roughly 0.3 seconds of physical flow time with Data sampling for time statistics enabled, storing the data at a 0.002s interval to extract mean velocity fields instead of an instantaneous time stamp, this however costs storage space, so care was needed taken to model without exhausting the node storage limits.

2.6 Solution & Convergence Monitoring:

Strict convergence criteria were enforced to ensure the high-fidelity validation against experimental PIV data. Residual targets were set to 10^{-06} for Cartesian velocity components (x, y, z) to accurately resolve directional shear layers, and 10^{-05} for turbulent kinetic energy (k) and specific dissipation rate (ω) to stabilize mixing fluctuations.

Due to the intense pressure-velocity coupling inherent in highly swirling flows, the continuity residual naturally plateaued at 10^{-04}. To demonstrate sufficient numerical stability despite this plateau, a global boundary mass flux balance was evaluated. The net mass flow rate imbalance between all inlets and outlets was calculated to be exceptionally low: yielding a global percentage mass error on the order of 0.0002%, satisfying the rigorous < 0.2% industry threshold for proving sufficient continuity resolution. [32]. The final closing residuals of the solution for both models are mentioned in Appendix A section A.3.

2.7 Mesh Independence:

To verify that the simulated flow structures were independent of spatial discretization, grid sensitivity studies were conducted. Multiple grids were generated with uniform refinement factors (e.g. coarse, medium, and fine cell counts). Steady-state runs were executed and radial lines were probed at specific axial heights to extract the axial velocity profiles. The extracted velocity profiles for the medium and fine meshes exhibited near-identical overlay patterns, proving that further increases in cell density yielded no significant change to the macroscopic flow physics.

The cell numbers for the coarse mesh ~440k, medium mesh ~830k and the fine mesh (due to the aggressive refinement of the BOI) ~2.2mil.

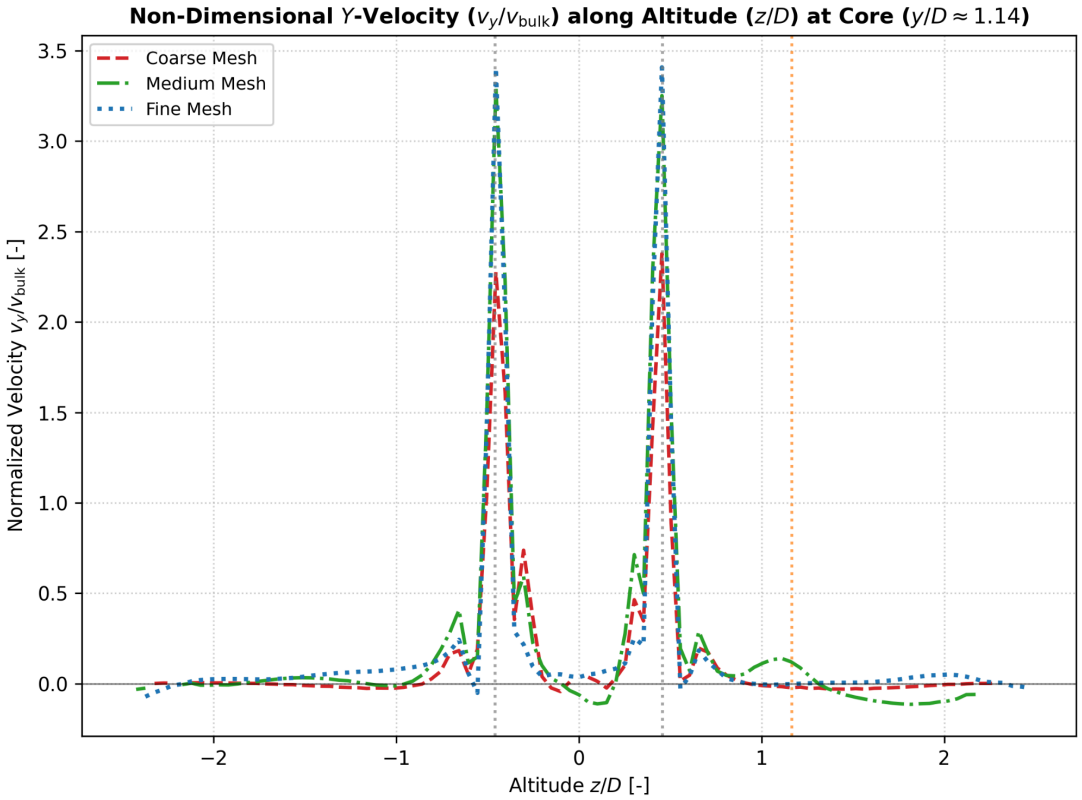


Fig 2.5 Mesh independence- sensitivity study by overlaying axial velocity plots for different mesh conditions over a certain position along the y-axis.

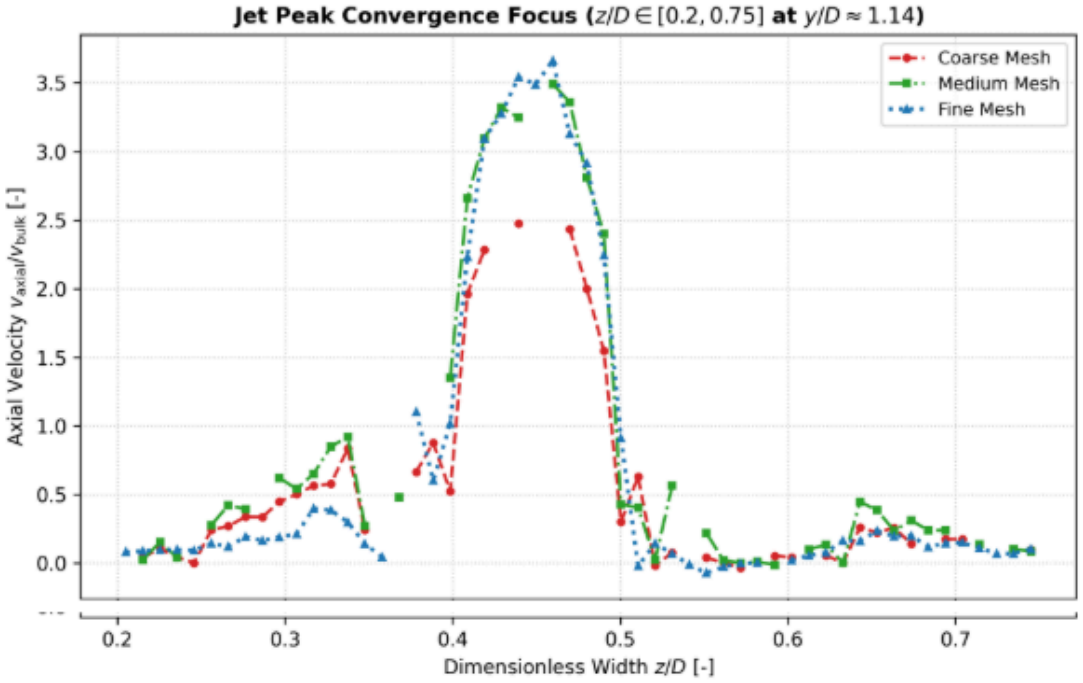


Fig 2.6 For a greater clarity the right axial velocity peak is isolated and plotted to better notice the sensitivity and convergence towards independence.

3.Experimental Methodology

3.1 Experimental Setup & Burner Configuration

The experimental activities were conducted at atmospheric pressure utilizing the RAPIDS-H coaxial double-swirl burner, specifically adapted for 100% hydrogen combustion. The test rig consists of a swirl generator base, an octagonal combustion chamber equipped with four steel walls and four quartz windows for optical diagnostics, and a converging exhaust hood. (see Fig 3.1)

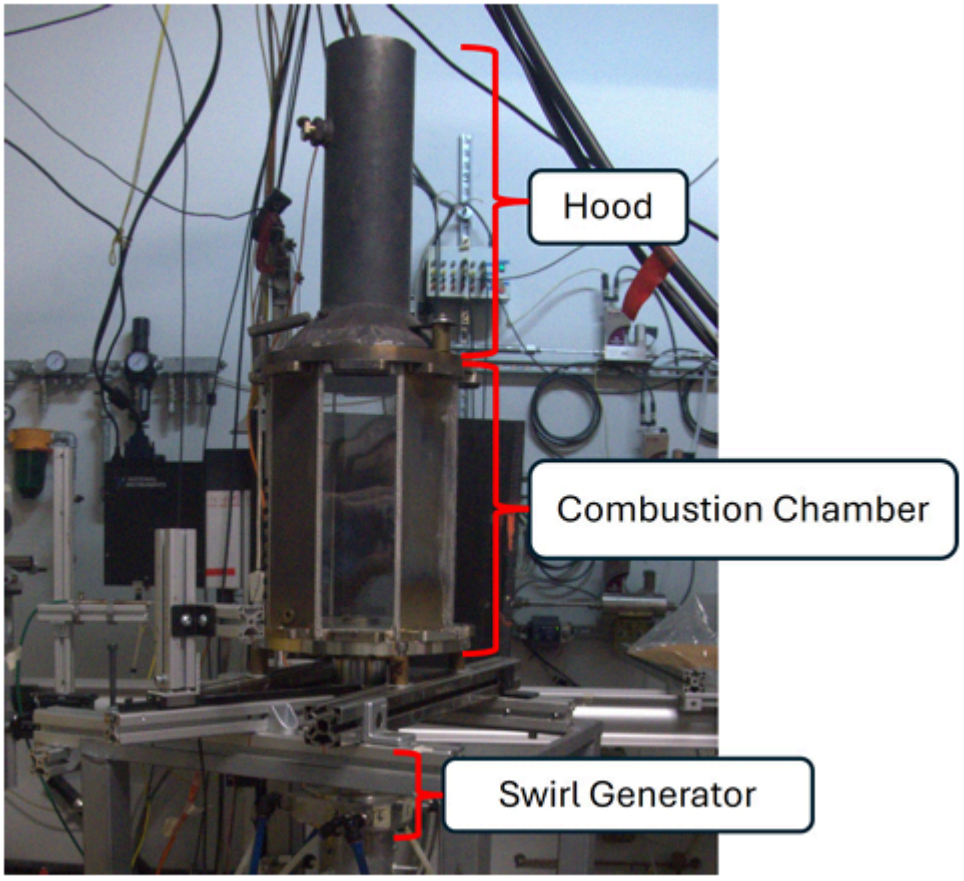


Fig 3.1 Combustor setup adapted from Ciuffoli & Rocco Thesis [28]

The injection system is divided into two primary streams. The injector (refer Model A Fig 1.12) utilizes 18 tangential holes (arranged in a 3 × 6 configuration) to impart swirl to the premixing air, while pure hydrogen is injected axially through 12 discrete holes

located in the injector crown (seen from the top view of the injector shown in Fig 3.4). The secondary outer air is routed through a divided plenum that splits the flow into axial and tangential components (see Fig 3.3), allowing precise control over the outer swirl intensity (Split Ratio, SR). To execute the comparative study, a custom counter-swirl inner injector was designed and fabricated and physically swapped with the baseline co-swirler to reverse the inner flow rotation.

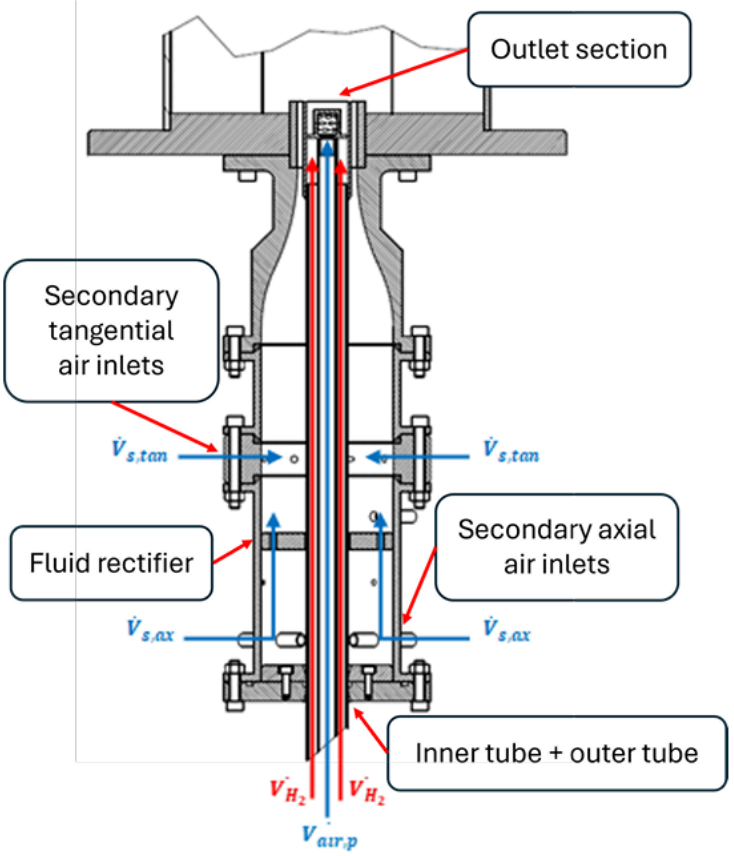


Fig 3.2 Double Swirl implementation, adapted from Ciuffoli & Rocco [28]

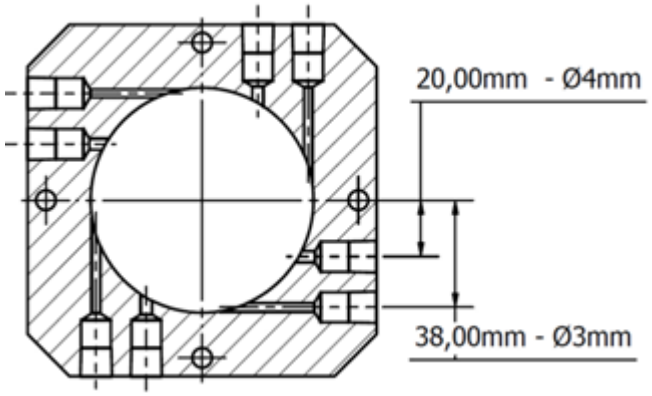


Fig 3.3 Outer swirl is created via air injected in a tangential pattern as detailed above

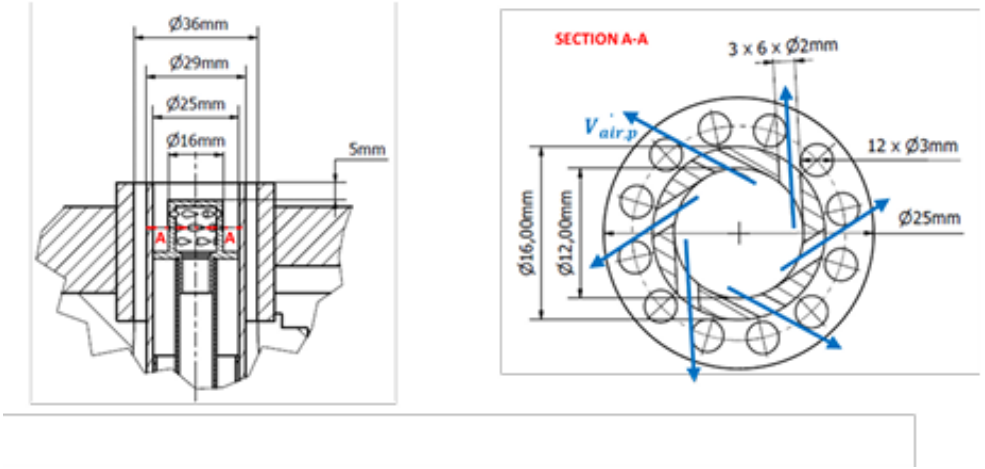


Fig 3.4 Inner swirl is created via air injection into a plenum that has tangential holes drilled into it, imparting a tangential component to the velocity as detailed above.

The nominal thermal power P_{th} was maintained at a constant 12 kW across all configurations, mathematically defined as:

$$P_{th}[KW] = \frac{\dot{V}_{H_2} \cdot LHV_{H_2}}{60} \quad (3.1)$$

where $\dot{V}_{H_2}$ represents the fuel volumetric flow rate and expressed in Normal liters per minute (Nl/min), and LHV_{H_2} is the Lower Heating Value of hydrogen (10.78 kJ/Nl). Both parameters are referenced to standard conditions ($0\,^{\circ}C$ & 1 atm).

3.1.1 Development & Fabrication of Counter-swirl injector:

Based on the core design of the injector Model A. A redesign was done (using AutoCAD) to reverse the direction of the swirl and the new injector was fabricated (Fig 3.5a).

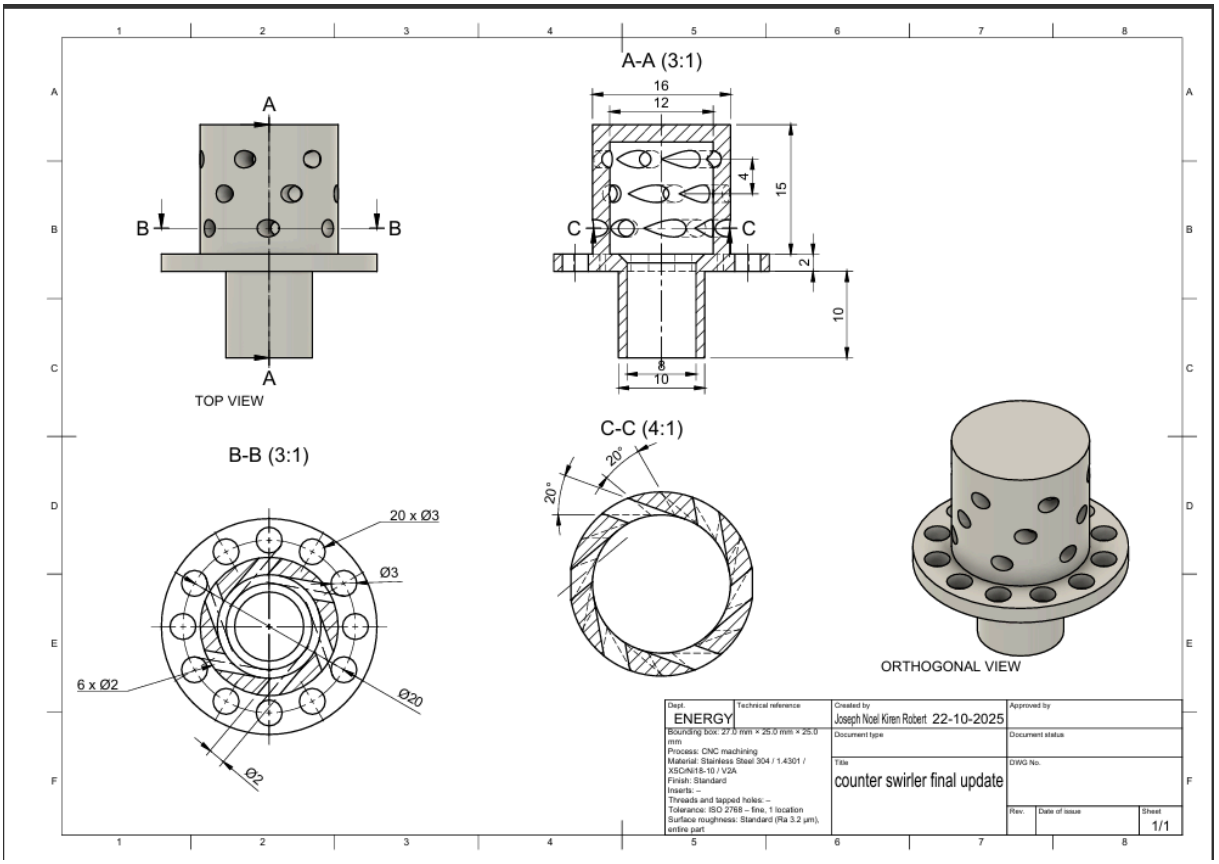


Fig 3.5a Fabrication details of the newly designed Counter Swirl Injector

Fabrication Details:

Bounding box: 27.0 mm × 25.0 mm × 25.0 mm

Process: CNC machining

Material: Stainless Steel 304 / 1.4301 / X5CrNi18-10 / V2A

Finish: Standard

Inserts: –

Threads and tapped holes: –

Tolerance: ISO 2768 – fine, 1 location

Surface roughness: Standard (Ra 3.2 µm), entire part

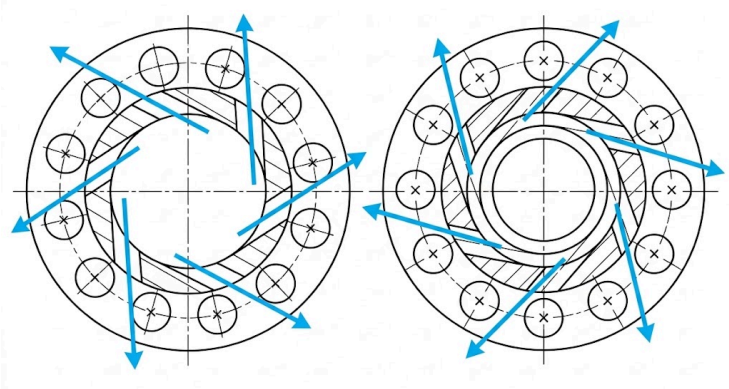


Fig 3.5b Side by side comparison of the A-A section of the Co and Counter swirl injectors

3.2 Measurement Procedures & Data Processing

The fluid delivery network was managed using Bronkhorst thermal mass flow meters integrated with a LabVIEW control system, enabling stable, steady-state regulation of the incoming air and hydrogen streams. Temperature monitoring was conducted utilizing type N and type S thermocouples positioned strategically along the chamber walls and near the recirculation zones to ensure structural safety and establish thermal equilibrium before data acquisition.

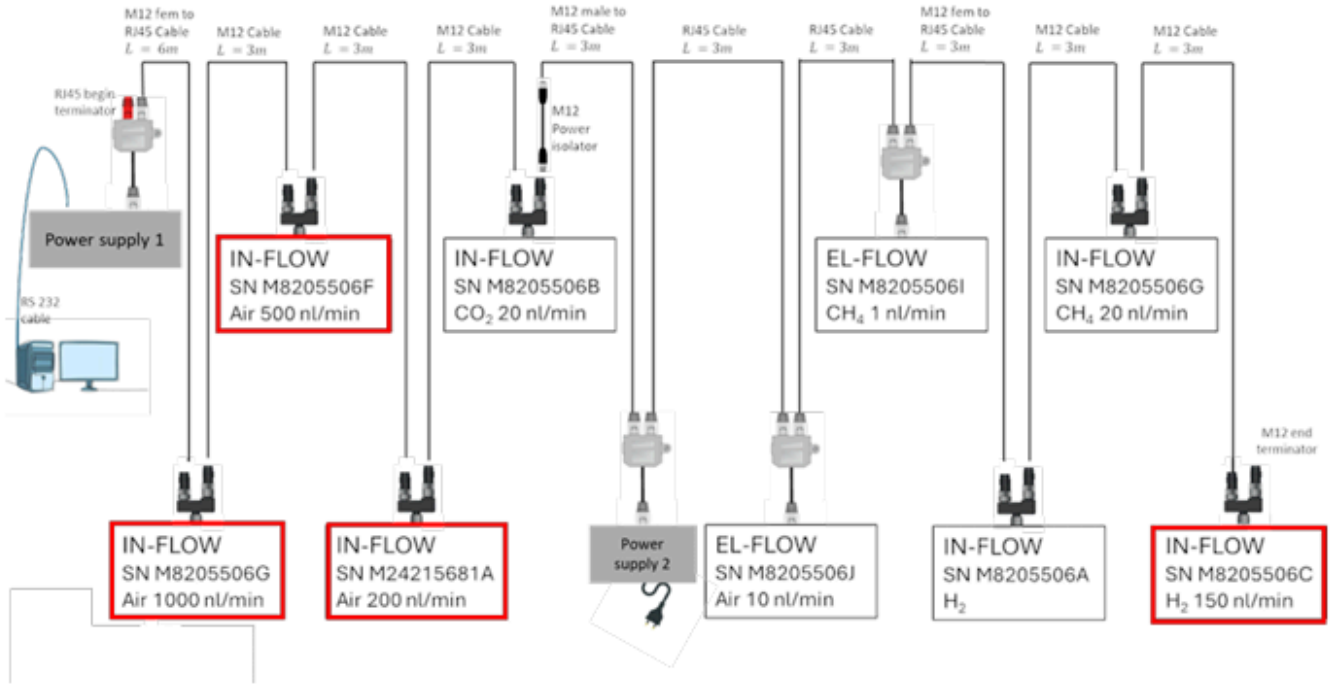


Fig 3.6 LabVIEW logics for the control of the flowmeters adapted from Ciuffoli & Rocco [28]

Throughout the testing matrix, the system systematically varied the premixed equivalence ratio, the global equivalence ratio, and the secondary air split ratio. These governing operational parameters are defined as follows:

Secondary Split Ratio (SR):

$$SR = \frac{\dot{V}_{as,tg}}{\dot{V}_{as,tg} + \dot{V}_{as,ax}} \tag{3.2}$$

where $\dot{V}_{as,tg}$ and $\dot{V}_{as,ax}$ denote the tangential and axial volumetric flow rates of the secondary air, respectively.

Global Equivalence Ratio (Φ_{global}):

$$\Phi_{global} = \frac{\dot{V}_{H_2}/\dot{V}_{a,tot}}{(F/A)_{st}} \tag{3.3}$$

Premix Equivalence Ratio (Φ_{premix}):

$$\Phi_{premix} = \frac{\dot{V}_{H_2}/\dot{V}_{ap}}{(F/A)_{st}} \tag{3.4}$$

where $\dot{V}_{H_2}$ is the fuel volumetric flow rate, $\dot{V}_{a,tot}$ is the total air flow rate, $\dot{V}_{ap}$ is the premixing air flow rate, all flow rates measured in Nl/min, and $(F/A)_{st}$ represents the stoichiometric fuel-to-air ratio, it is evaluated on a consistent volumetric (molar) basis under identical standard conditions, taking a value of approximately 0.42 (or 1/2.38) for hydrogen-air combustion.

To characterize the aerodynamic rotational intensity of the flow, the swirl number is defined as the ratio of the axial flux of angular momentum (G_θ) to the axial flux of axial momentum (G_x), normalized by the characteristic burner radius (R):

$$S_n = \frac{G_\theta}{RG_x} \tag{3.5}$$

Following the rigorous formulations evaluated by Vignat et al [34] for swirling flows in polar-cylindrical coordinates, the exact momentum fluxes in an axisymmetric confined flow must include both mean and fluctuating velocity components, alongside the static pressure field. Accurate evaluation of S_n from experimental or numerical velocity fields requires careful treatment of the static pressure term $p - p_{ref}$. Near the plane where the different inflows are dumped into the combustor, neglecting the pressure term does not violate continuity; rather, it directly affects the correct evaluation of the swirl number. Consequently, utilizing boundary layer approximations for pressure is mathematically valid only deeper and more downstream into the flow field where the shear layers are fully established.

When extracting the swirl intensity close to the injector outlet, or when defining the fixed hardware capabilities of the burner (like a swirling capability fixed by geometry alterable only by flow rate), the simplified geometrical swirl number provides a very reliable baseline approximation before the flow is fully established.

The geometrical swirl number for the secondary air stream ($S_{g,s}$), which is tangentially introduced around the central injector, is evaluated as:

$$S_{g,s} = \frac{\pi R_{0,s} D}{2A_{T,s}} SR_s^2 \tag{3.6}$$

where $R_{0,s}$= 38 mm is the radial distance from the axis of each tangential inlet to the swirler centerline, D = 36 mm is the outlet diameter, $A_{T,s}$ represents the total cross-sectional area of the tangential air passages, and SR_s is the secondary split ratio.

Similarly, the geometrical swirl number magnitude for the primary premixed stream $|S_{g,s}|$ is formulated as:

$$|S_{g,s}| = \frac{R_{o,p} D_{crown}}{18 D_{tan}^2} SR_p^2 \tag{3.7}$$

In this formulation, $R_{o,p}$is the radial distance between the centerline of each premixed tangential channel and the injector axis. It is calculated as $R_{o,p} = 0.5(D_{inj} - D_{tan} - 2t)$, where D_{inj}denotes the injector outer diameter, $D_{tan}$is the tangential channel diameter, and t is the wall thickness of the injector.

Important Note: Because the geometric formulas utilize squared split ratios, they only give positive magnitudes. To mathematically differentiate the flow regimes and reflect opposing angular momentum, a directional sign convention is applied: a positive sign is assigned to the rotation of the secondary outer stream (clockwise). Consequently, the primary inner swirl number $S_{g,p}$ is strictly positive for the co-swirl configuration and strictly negative for the counter-swirl configuration.

The premixed split ratio (SR_p) governing this inner stream is defined by:

$$SR_p = \frac{\dot{V}_{air,p}}{\dot{V}_{air,p} + \dot{V}_{H_2,eq}} \tag{3.8}$$

where $\dot{V}_{air,p}$represents the premixed air volumetric flow rate, determined from the specified local equivalence ratio. The term $\dot{V}_{H_2,eq}$represents the equivalent air volumetric flow rate required to replicate the exact momentum flux of the hydrogen

stream during isothermal, non-reacting PIV and calibration tests to ensure safe burner operation. This is explicitly derived by equating the momentum flux of the hydrogen jet to that of an equivalent air jet passing through the same cross-sectional area (A):

$$\rho_{H_2} u_{H_2}{}^2 A = \rho_{air} u_{air}{}^2 A \tag{3.9}$$

Solving for the equivalent air velocity yields $u_{air} = u_{H_2}\sqrt{\rho_{H_2}/\rho_{air}}$. Multiplying both sides by the flow area defines the equivalent volumetric flow rate formulation:

$$\dot{V}_{H_2,eq} = \dot{V}_{H_2}\sqrt{\frac{\rho_{H_2}}{\rho_{air}}} \tag{3.10}$$

where the densities ρ_{H_2} and ρ_{air} are evaluated at identical standard reference conditions.

3.2.1 Isothermal Operating Conditions and CFD Boundary Setup

A single reference operating condition was selected to conduct the isothermal Stereoscopic Particle Image Velocimetry (Stereo-PIV) measurements and the corresponding Computational Fluid Dynamics (CFD) numerical validation. The baseline condition corresponds to a nominal thermal power of P_{th}=12.00 kW, a global equivalence ratio Φ_{global}=0.45, a premixed equivalence ratio Φ_{premix}=5.00, and a secondary split ratio SR=0.60. To safely conduct the isothermal non-reacting tests and numerical simulations, the actual hydrogen volumetric flow rate of 66.8 Nl/min was substituted with a momentum-equivalent air flow rate of 17.6 Nl/min. This equivalent air flow rate ensures the preservation of the inner jet momentum flux, which is the primary aerodynamic driver of the central recirculation zone. To accurately define the inlet boundary conditions for the CFD domain, the standard volumetric flow rates (Nl/min) were converted to bulk velocities (m/s) at the respective inlet patches, assuming an ambient reference temperature of $20^{\circ}C$. The calculated inlet velocities applied to the secondary axial annulus (outer diameter 80 mm, inner diameter 25 mm), the 8 secondary tangential injectors (11.257 mm diameter), the primary premix pipe (10 mm diameter), and the primary equivalent-air annulus (outer diameter 23 mm, inner diameter 12 mm) are summarized in Table 3.1, the computations for the same are available in **Appendix A.**

Table 3.1: Isothermal Operating Conditions and CFD Inlet Boundary Velocities

Parameter	Value
Global Parameters	-
Nominal Thermal Power P_{th}	12.00 kW
Global Equivalence Ratio Φ_{global}	0.45
Premixed Equivalence Ratio Φ_{premix}	5.00
Secondary Split Ratio (SR)	0.60
Volumetric Flow Rates	-
Secondary Axial Air $\dot{V}_{axial}$	129.30 Nl/min
Secondary Tangential Air ($\dot{V}_{tangential}$)	193.80 Nl/min
Premix Air $\dot{V}_{premix}$	31.95 Nl/min
Equivalent Air Sub. $H_2(\dot{V}_{af})$	17.60 Nl/min

CFD Boundary Patch Velocities	-
Secondary Axial Inlet	0.510 m/s
Secondary Tangential Inlet	4.352 m/s
Premix Air Inlet	7.278 m/s
Air sub H_2 Inlet	1.041 m/s

3.2.2 UV Imaging and Flame Morphology

Because hydrogen flames lack carbonaceous soot and emit predominantly in the ultraviolet spectrum, standard visible imaging is insufficient for topological analysis. Flame morphology and structural stability were characterized by capturing the chemiluminescence of excited hydroxyl radicals (OH*), which peak at a wavelength of approximately 300 nm. The optical diagnostic setup employed specialized UV-sensitive camera equipped with narrow bandpass optical filters to isolate the OH* emissions while blocking background radiation. To capture these specific emissions, the optical diagnostic setup employed an *fxo487MCX 8.1 MP camera (SVS-Vistek)*. This sensor features a 2840 × 2840 pixel array with a 2.74 × 2.74 μm pixel size, optimized for high relative quantum efficiency in the 200–400 nm range. The camera was fitted with an *Azure Photonics 75 mm F/3.8 ultraviolet lens* (covering a 200–1000 nm spectral range) and coupled with a *MIDOPT Near-UV Bandpass Filter*. This filter provides a specific transmission bandwidth of 315–365 nm, successfully isolating the OH* chemiluminescence while blocking background visible and infrared radiation. The multiple captured two-dimensional flame images were post-processed and time-averaged to define the distinct flame boundaries, penetration length, and the

presence or collapse of the Central Recirculation Zone (CRZ) under varying swirl conditions.

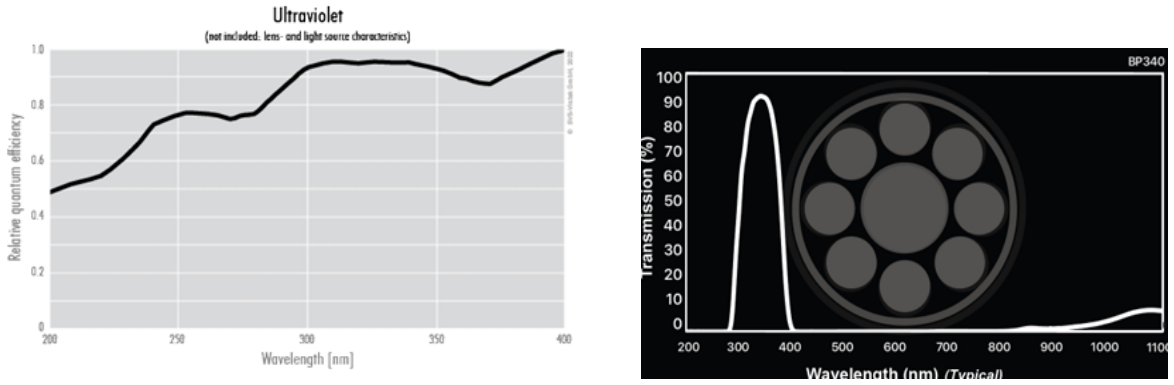


Fig 3.7 fxo487MCX Quantum Efficiency (left) & Bandwidth MIDOPT BP340 (right)

3.2.3 Stereoscopic Particle Image Velocimetry (Stereo-PIV)

To characterize the aerodynamic flow field and provide high-fidelity validation data for the numerical models, Stereoscopic Particle Image Velocimetry (Stereo-PIV) was utilized under isothermal conditions. The optical system mapped the highly three-dimensional velocity vectors within the combustion chamber, capturing the boundaries of the Central Recirculation Zone (CRZ) and the intricate shear layers generated by the interacting swirling streams. The PIV work in this project is referenced from the thesis work of Bezzi & Sala [35] . The velocity profiles extracted from their Velocimetry work serve as the foundational experimental baseline for the numerical validations presented in **Chapter 4** of this thesis.

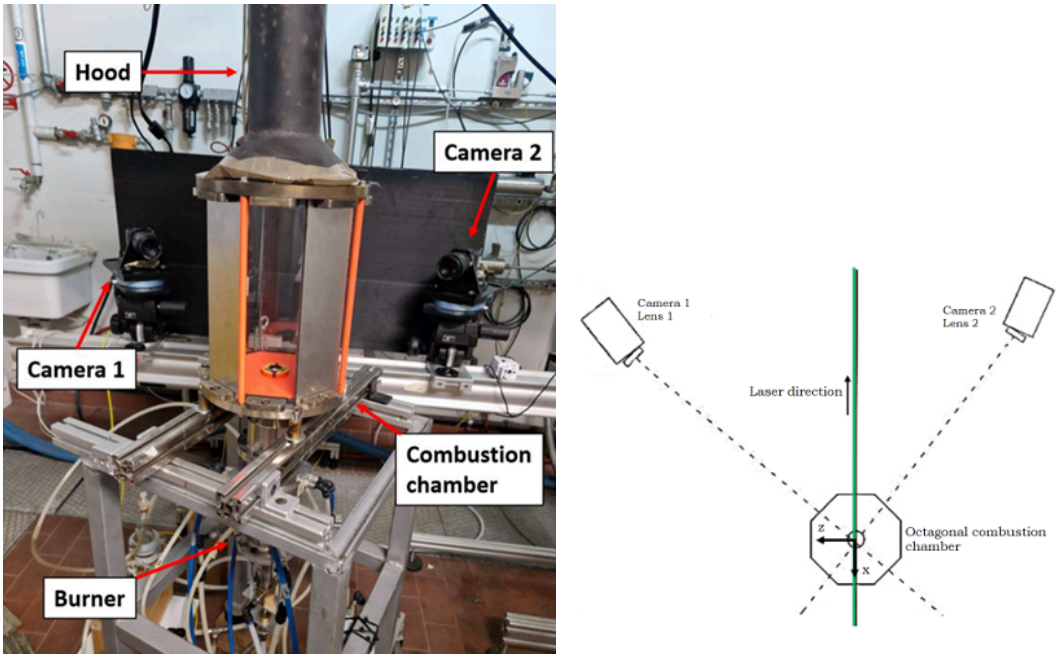


Fig 3.8 Stereo PIV setup detailing for the combustor, adapted from **Bezzi & Sala** [35]

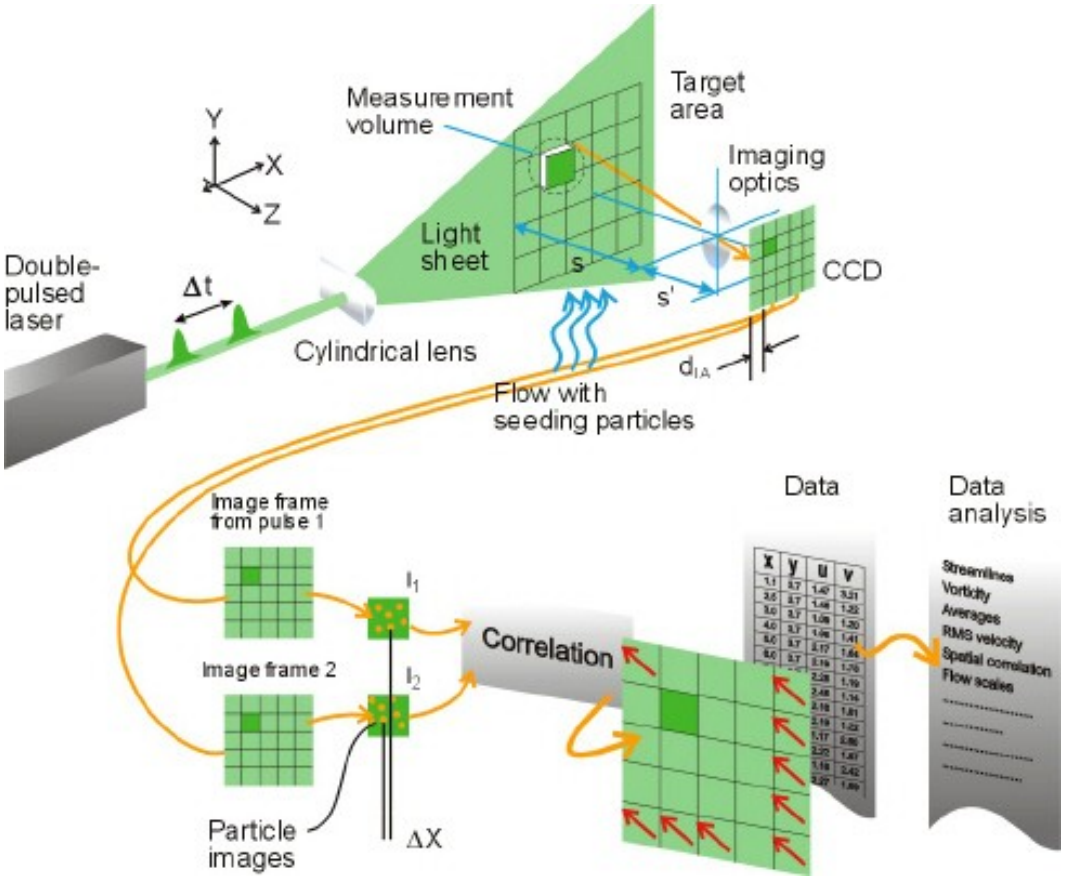


Fig 3.9 Stereo PIV working adapted from Dantec site [33]

The setup and workflow of the Experiment as referenced in the thesis work of ***Bezzi & Sala*** [35] from whose setup the PIV data extraction followed: The isothermal validation dataset was acquired using Stereoscopic Particle Image Velocimetry (Stereo-PIV). This optical diagnostic technique extracts three velocity components on a two-dimensional measurement plane (2D3C) aligned with the combustor's longitudinal symmetry axis. Images were captured using two digital cameras arranged in a *Scheimpflug configuration*, with the flow seeded by *atomized Vaseline oil droplets* and illuminated by a *double-pulsed Nd:YAG laser sheet*. Vector fields were processed via an adaptive multi-pass cross-correlation algorithm, starting with a 64 × 64 pixel interrogation window and refining to a final 32 × 32 pixel size with a 50% overlap. To ensure statistical convergence of the turbulent structures, the mean velocity profiles were calculated by averaging 1000 instantaneous image pairs. The *maximum statistical uncertainty* at a 95% confidence level was evaluated at 4.5% for the *axial velocity*, 3.2% for the *tangential velocity*, and 3.3% for the *radial velocity*. [35]

3.2.4 Emission Measurements and Gas Analysis:

Pollutant formation was evaluated through exhaust gas sampling to quantify the overall NO_x emissions generated by the high adiabatic flame temperature of hydrogen. A sampling probe was positioned centrally within the exhaust hood to extract the combustion products. Measurements were conducted using continuous gas analyzers (MRU MGAprime and MRU Optima7), which explicitly measure the volumetric molar fractions of total NO , NO_2 & O_2 in parts per million (ppm). These instruments measure total NO_x on a dry basis and do not experimentally isolate specific formation mechanisms such as thermal NO_x. Prior to the testing campaigns, the analyzers were calibrated using certified span gases to ensure measurement accuracy.

Because the raw data is measured on a nominally dry basis that may still contain residual moisture, a correction is first applied to the measured oxygen concentration based on the ambient water vapor pressure. To ensure consistent comparability of emission levels across varying excess air conditions and dilution levels, the dry NO_x measurements are normalized to a reference oxygen concentration of 15% O2: a standard metric for aeronautical and gas turbine applications. The correction is performed as follows:

$$NO_{x,ref} = NO_{x,measured} \frac{[21-O_{2,ref}]}{[21-O_{2,measured}]} \quad (3.11)$$

where $O_{2,ref}$= 15 and $O_{2,measured}$ is the actual measured oxygen percentage in the exhaust gases.

To evaluate the absolute mass of pollutant generated per unit mass of fuel consumed, the volumetric data is mathematically converted into the Emission Index ($EINO_x$), expressed in g/kg H2. This parameter provides a robust framework for comparing environmental performance independent of the specific combustor geometry. The $EINO_x$ is calculated using the wet-basis NO_xmole fraction:

$$EINO_x = X_{NO_{x,wet}} \cdot M_{NO_2} \frac{\dot{m}_{tot,\Phi}}{\overline{M_\Phi} \cdot \dot{m}_{H_2}} \quad (3.12)$$

Where M_{NO_2} is the molar mass of NO_2(46.01 g/mol) evaluated as the reference mass for NO_x emissions, $\dot{m}_{H_2}$ is the hydrogen mass flow rate, and M_Φ is the total mass flow rate of the reactants.

The measured dry NO_x concentration is converted to a wet basis by accounting for the water vapor produced during combustion:

$$X_{NO_{x,dry}} = \frac{NO_x\,[ppm]}{10^6} \quad (3.13)$$

$$X_{NO_{x,wet}} = X_{NO_{x,dry}} (1 - X_{H_2O,\Phi}) \quad (3.13)$$

The molar fraction of water vapor in the combustion products $X_{H_2O,\Phi}$ and the average molar mass of the exhaust gas mixture $\overline{M_\Phi}$ are determined from the global equivalence ratio Φ assuming complete combustion:

$$X_{H_2O,\Phi} = \frac{\Phi}{\Phi+(0.5-\frac{\Phi}{2})\ +1.88} \quad (3.14)$$

$$\overline{M}_{\Phi} = \frac{\Phi \cdot M_{H_2O} + (0.5 - \frac{\Phi}{2}) \cdot M_{O_2} + 1.88 \cdot M_{N_2}}{\Phi + (0.5 - \frac{\Phi}{2}) + 1.88} \tag{3.15}$$

The dominant source of uncertainty in the $EINO_x$ calculation arises from the intrinsic accuracy of the gas analyzer ($\pm$ 2 ppm, or approximately 5%). Combined with the 0.6% maximum uncertainty associated with the thermal mass flow meters, the overall propagated uncertainty for the reported $EINO_x$ values is estimated using the root-sum-square method to be approximately $\pm$ 5. 1%.

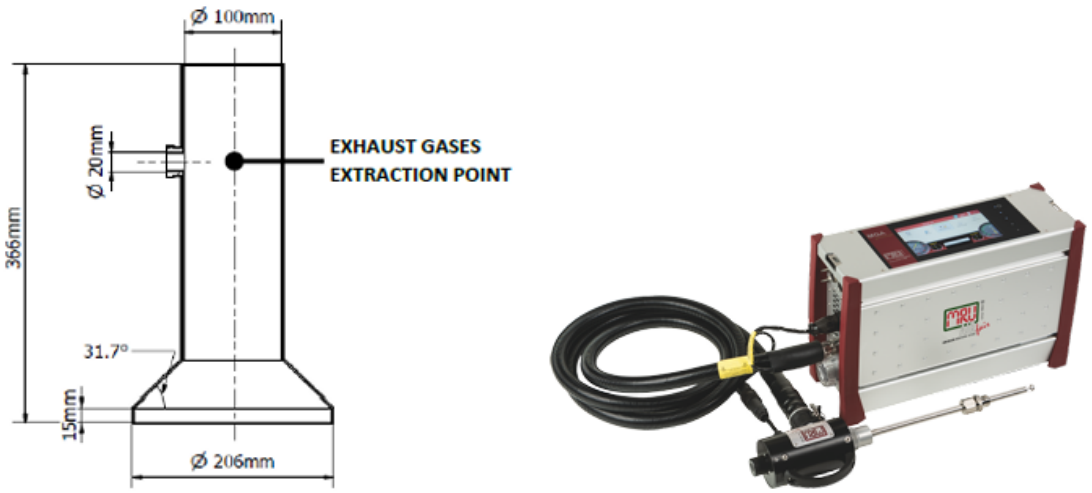


Fig 3.9 On the left the hood on top of the octagonal combustor body which hosts the port to support the probe of the MRU MGA prime Gas Analyser that can be seen on the right.

4.Numerical Results and Comparison with Experimental Data

4.1 Co-Swirler Validation Against Stereo-PIV Data

The numerical validation of the co-swirl configuration compares the time-averaged URANS simulation against the isothermal Stereo-PIV dataset. To achieve statistically converged mean velocity profiles, the simulation was executed in a transient mode with flow data sampled every 0.002 s over a total physical sampling period of 0.25 s.

4.1.1 Coordinate System and Axis of Rotation

In this study, the primary axis of rotation for the RAPIDS-H burner is aligned with the vertical Y-axis. Consequently, longitudinal flow profiles are evaluated along planes parallel to the Y-axis (the YZ plane), while downstream cross-sectional profiles are evaluated on the XZ plane.

Cartesian Projection Conventions: Throughout this analysis, velocities on all visualized planes are reported strictly as Cartesian components (V_x, V_y, V_z) rather than mathematically transformed cylindrical components. Because a swirling flow is nominally axisymmetric, projecting its vectors onto a 2D Cartesian grid naturally results in a sign inversion across the central origin.

Longitudinal Plane Conventions (Cartesian Projections on YZ plane): During the analysis of the longitudinal centerplane, V_y dictates the axial flow, the out-of-plane velocity V_x captures the swirl intensity, and the transverse velocity V_z centrifugal expansion moving radially outward registers as a positive velocity ($+V_z$) on the right side of the axis, and a negative velocity ($-V_z$) on the left side of the axis. Similarly, a uniform out-of-plane swirl (relating to V_x) exhibits opposing signs on either side of the origin. These projected fields are explicitly labeled as "Projected Transverse Velocity" or "Projected Out-of-plane Velocity."

Cross-Sectional Plane Conventions (Cylin*drical Components)*: On the downstream horizontal planes, the swirling flow is captured via its Cartesian in-plane

components (V_x & V_z), which manifest as distinct crescent-shaped contours of opposing signs. This Cartesian visualization is intentionally retained because the distinct inversion of these lobes at the core provides immediate, physical visual evidence of the counter-rotating inner stream.

Note on True Cylindrical Transformations: While the Cartesian projections on the XZ and YZ planes are retained in the main body of this chapter to provide immediate, intuitive visualization of the counter-rotating flow structures and structural inversions, a rigorous mathematical transformation to true cylindrical coordinates has also been performed. Readers seeking the purely axisymmetric representations of the true tangential (V_θ) and true radial (V_r) velocity fields for the transverse cross-sections are directed to **Appendix A.5.**

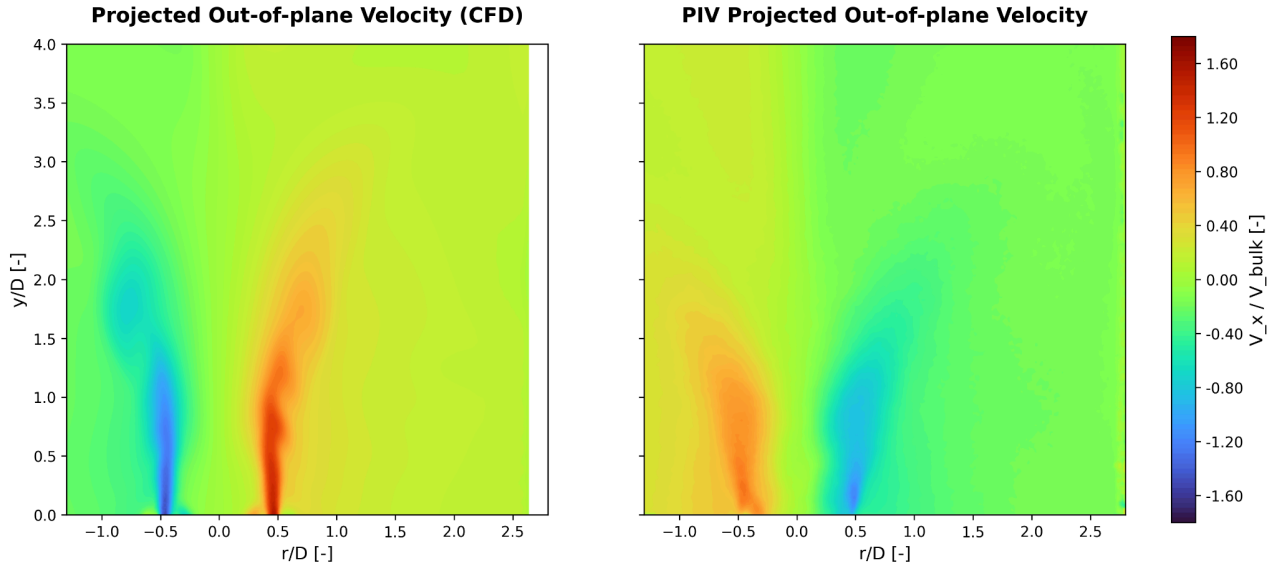


Fig 4.1: Out-of-plane Velocity *(Tangential) projec*tion contours CFD (left) & PIV (right)

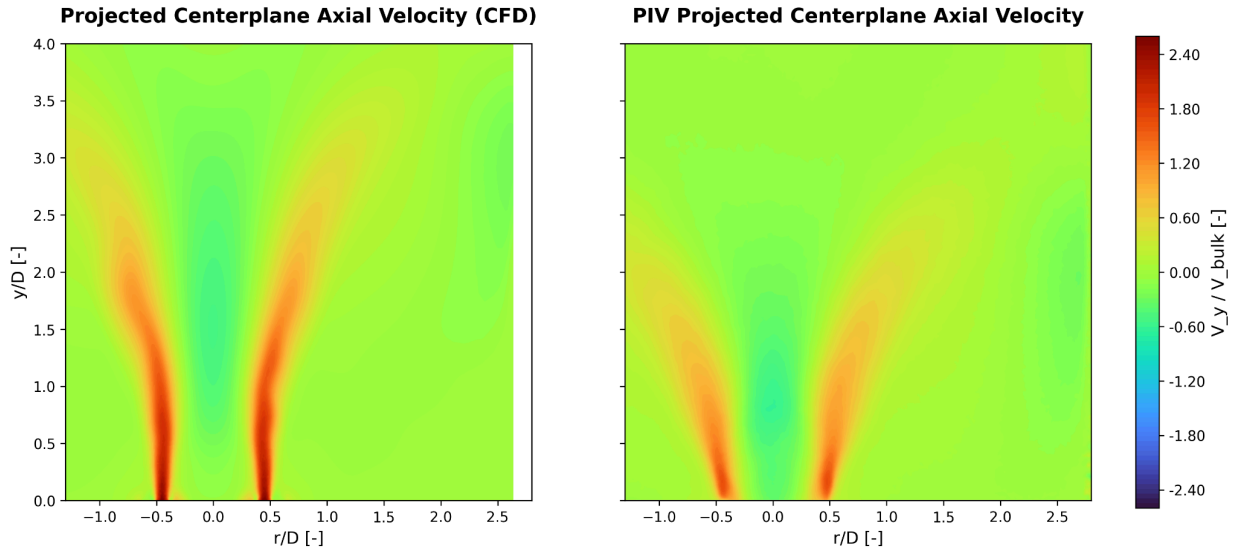


Fig 4.2: Axial (Y) Velocity Contour CFD(left) vs PIV(right)

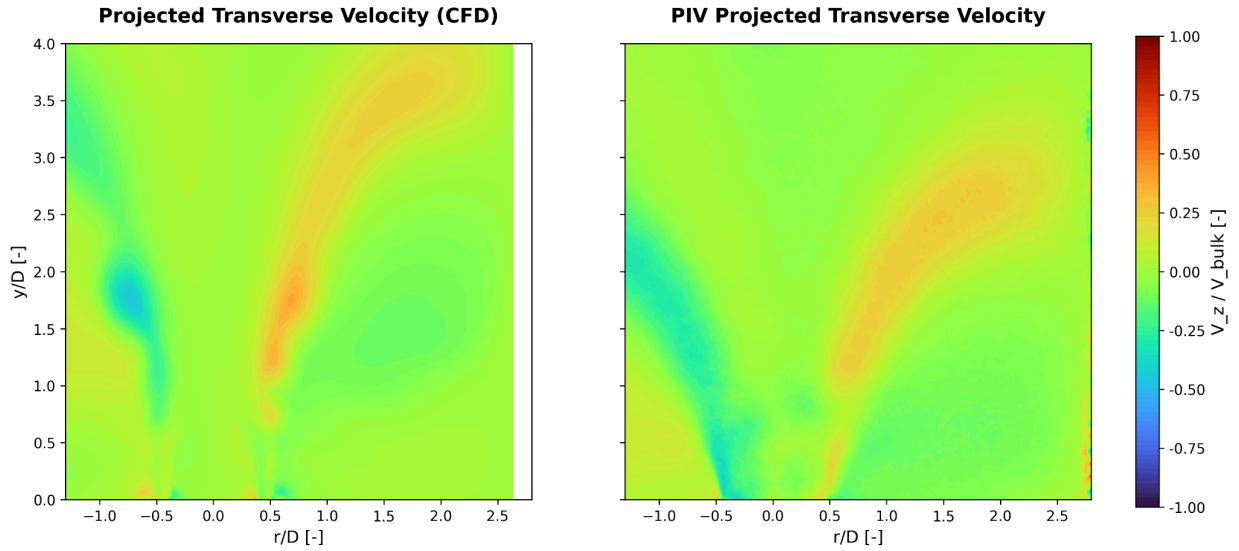


Fig 4.3: Transverse Velocity *(Radial)* projection contours CFD(left) vs PIV(right)

Spatial Velocity Profiles The 1D velocity profiles for the axial, transverse, and out-of-plane components were extracted at sequential transverse planes (y/D = 1.0, 1.5, 2.0, 2.5) to map the downstream aerodynamic evolution. The CFD provides a semi-quantitative validation, capturing the fundamental macroscopic flow topology, including the central recirculation region and the divergent trajectory of the swirling jet which spreads out.

CFD Post-processing Note:

- The CFD contours of velocity are depicted within a particular frame z(-0.0432, 0.0972)m & y(0.041, 0.144)m coordinate to be in agreement with the PIV view window.
- An iso-clip was created with these dimensions and the relevant parameters extracted as .csv file data which was mapped & plotted using Python.

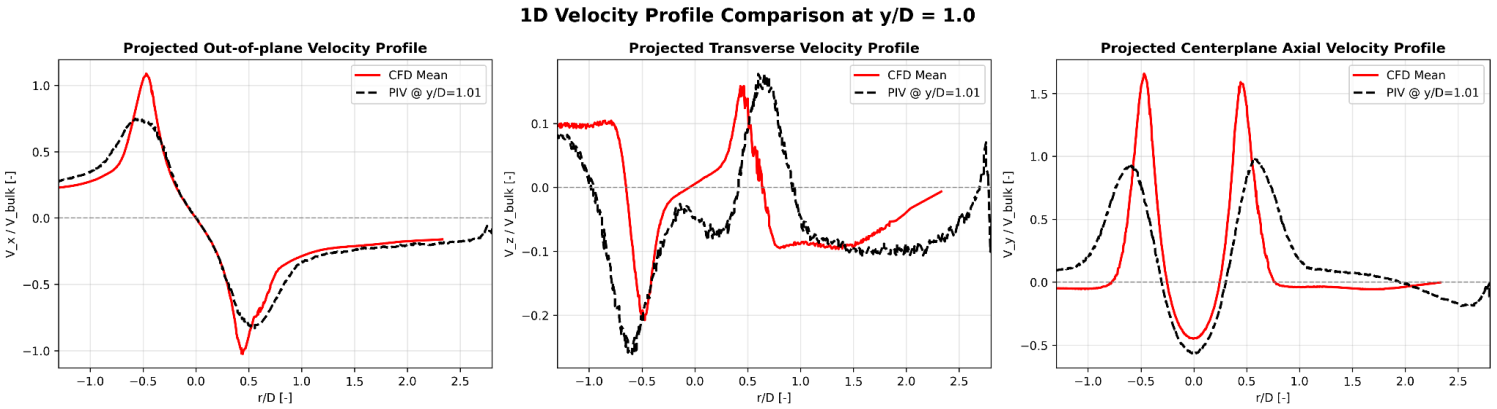


Fig 4.4 plots the 3 CFD & PIV overlaid velocity plots of the Co-swirl combustor @ y/D=1

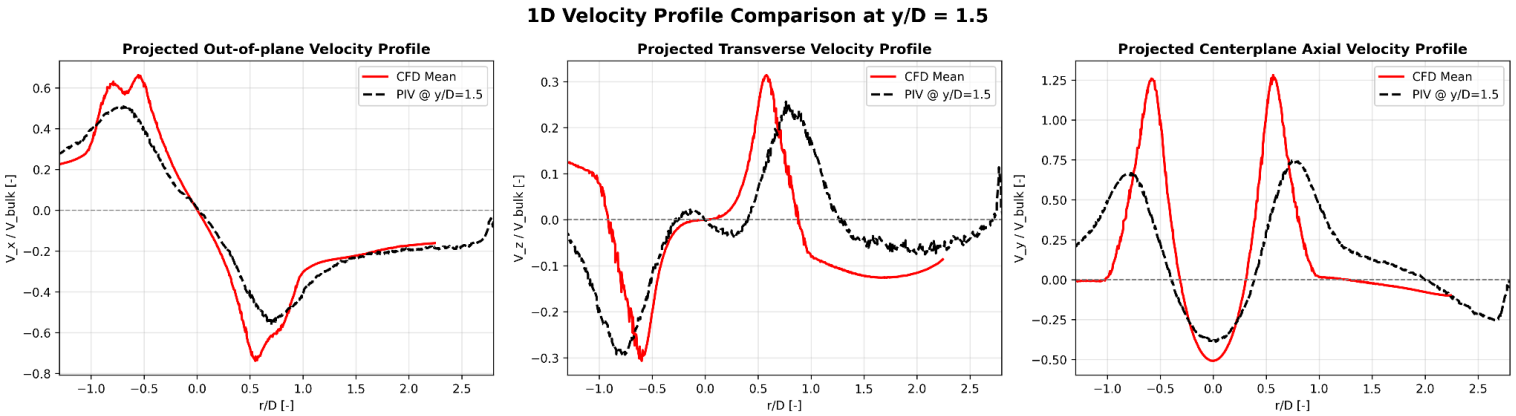


Fig 4.5 plots the 3 CFD & PIV overlaid velocity plots of the Co-swirl combustor @ y/D= 1.5

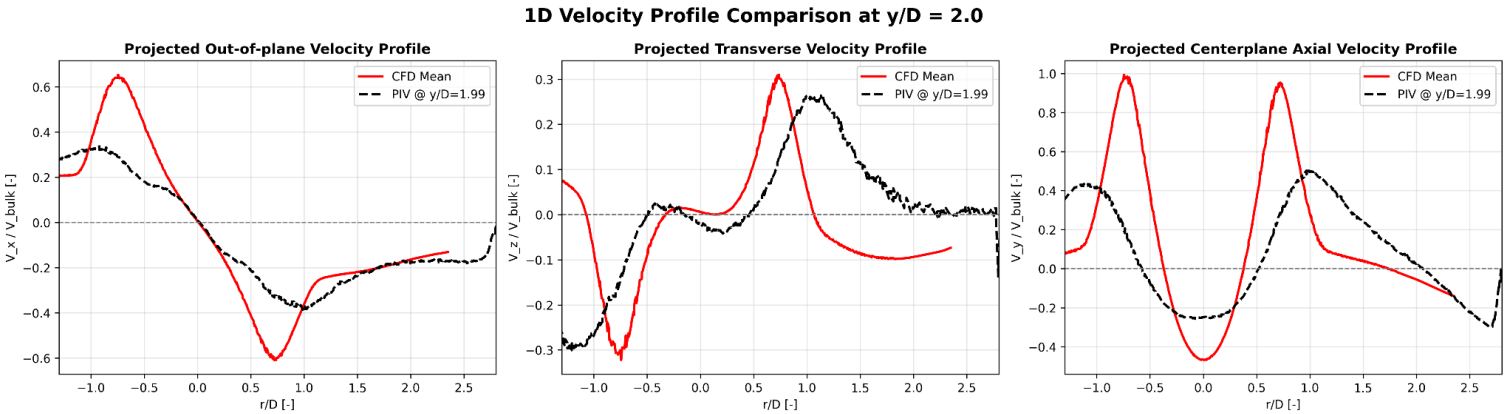


Fig 4.6 plots the 3 CFD & PIV overlaid velocity plots of the Co-swirl combustor @ y/D=2

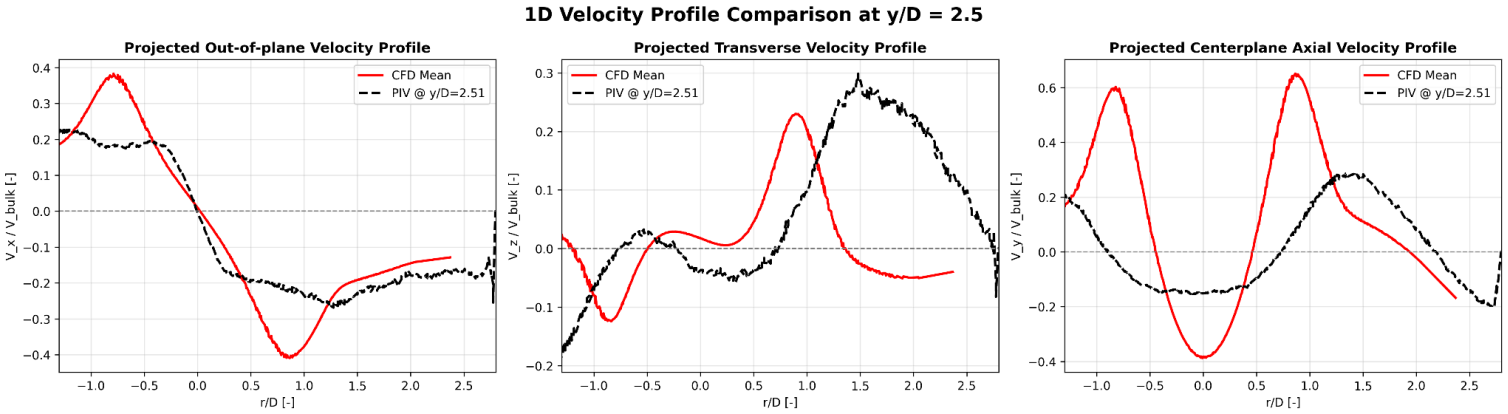


Fig 4.7 plots the 3 CFD & PIV overlaid velocity plots of the Co-swirl combustor @ y/D=2.5

Turbulence Diffusion Discrepancies: While the global flow structures align, differences emerge in the shear layer resolution. The numerical CFD predictions consistently exhibit sharp, narrow velocity peaks, whereas the experimental PIV profiles are broader and more smeared across the radial coordinate. This discrepancy may be attributed to inherent limitations of the URANS SST k-ω model, which typically underpredicts turbulent diffusion and lateral mixing in highly swirling flows. Furthermore, this numerical sharpness contrasts with the experimental spatial averaging inherently caused by the finite 1.5 mm thickness of the PIV laser sheet, which smooths out sharp velocity gradients. However, without a higher-fidelity simulation (e.g., LES) or quantitative error metrics, these differences cannot be exclusively attributed to modelling limitations and optical averaging alone.

4.2 Isothermal Flow Structures of the Counter-Swirler

The counter-swirl isothermal flow field was evaluated numerically to identify the transient flow development and the specific aerodynamic shifts induced by the hardware modification. At the inlet plane (y = 0.041 m), the time-averaged velocity profiles explicitly confirm the intended geometric effect: the outer out-of-plane velocity peaks perfectly match the co-swirl baseline, while the inner out-of-plane peaks are distinctly reversed (see Fig 4.8b), accurately reflecting the swapped inner swirler.

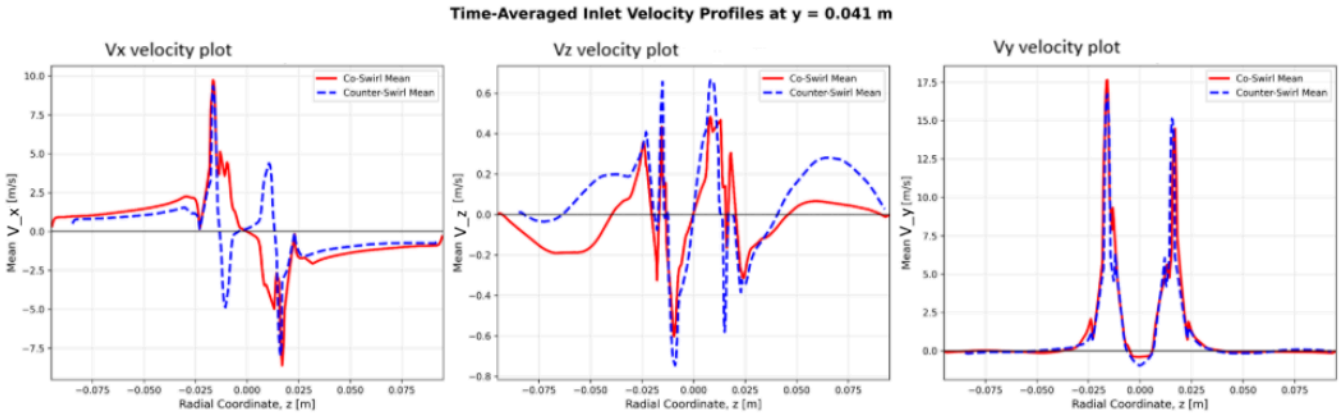


Fig 4.8a plots the 3 CFD Co & Counter swirl overlaid dimensional velocity plots at the injector plane at y=0.041 m

Fig 4.8b the plot Counter swirl injector (blue) in the region of the inner swirl is reversed for the Projected Out-of-plane component Vx *(Tangential component)*

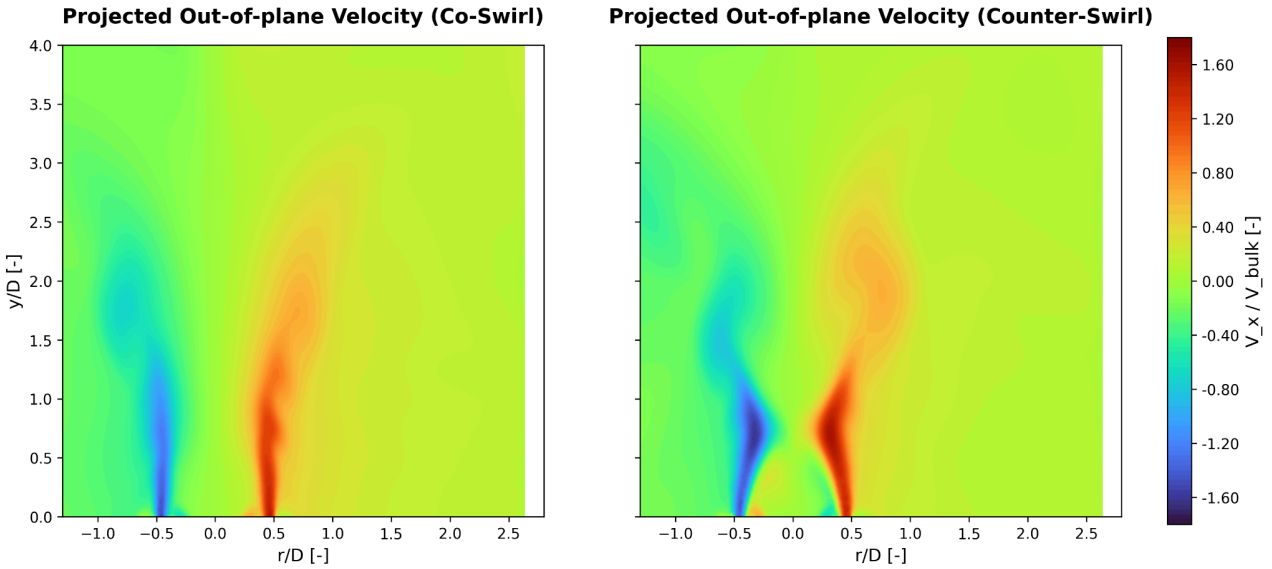


Fig 4.9 CFD Out-of-plane Velocity *(Tangential)* projection contours Co (left) & Counter (right)

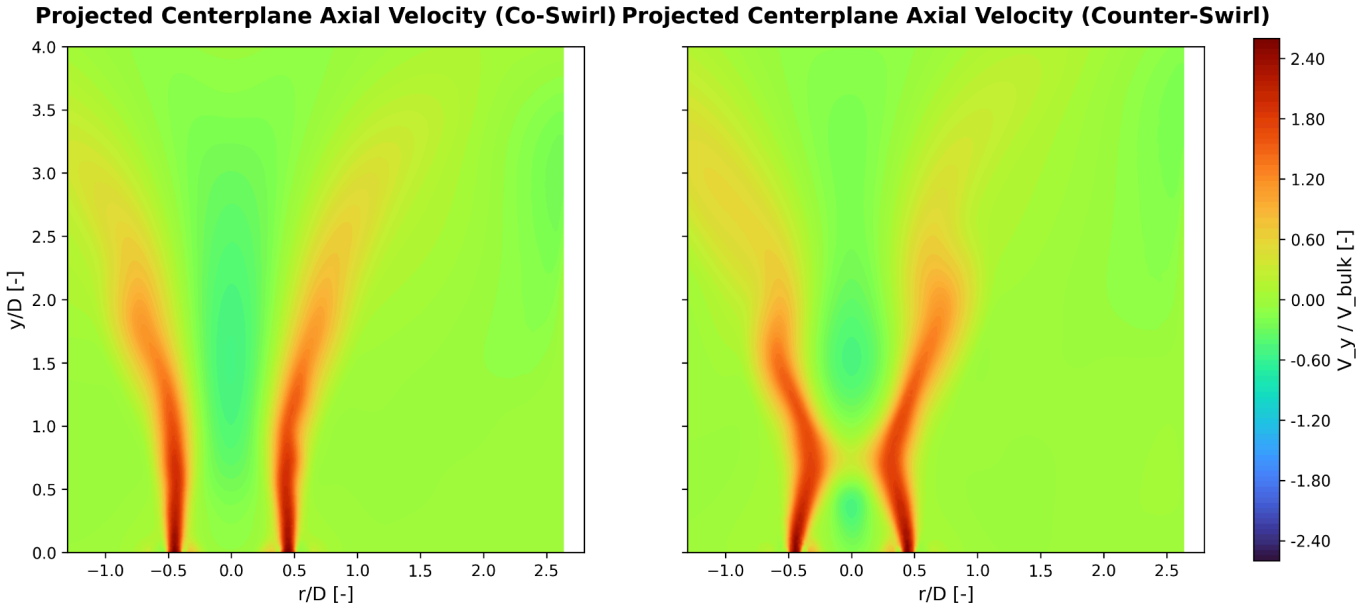


Fig 4.10 CFD Axial (Y) Velocity Contour Co(left) vs Counter(right)

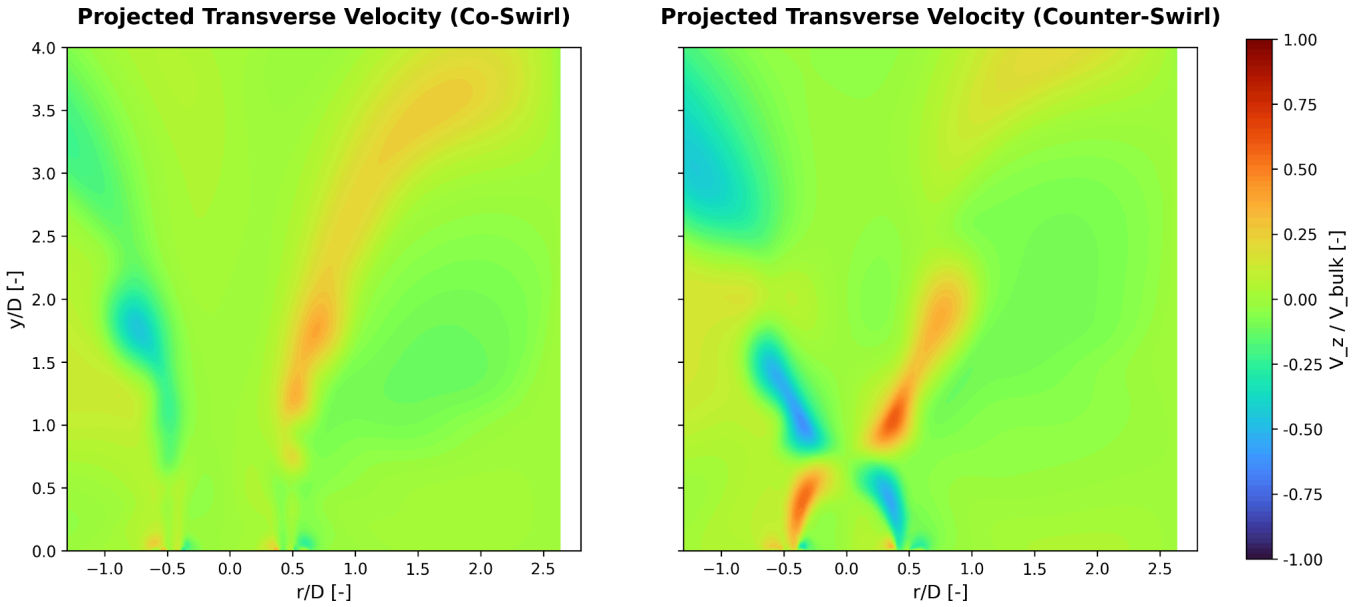


Fig 4.11 CFD Transverse Velocity *(Radial)* projection contours Co(left) vs Counter(right)

As the transient flow develops downstream, this reversal fundamentally alters the jet trajectory. The transverse and axial velocity contours in the counter-swirl configuration show a significant narrowing of the flow field and a disruption of the typical divergent swirling jet structure compared to the co-swirl baseline.

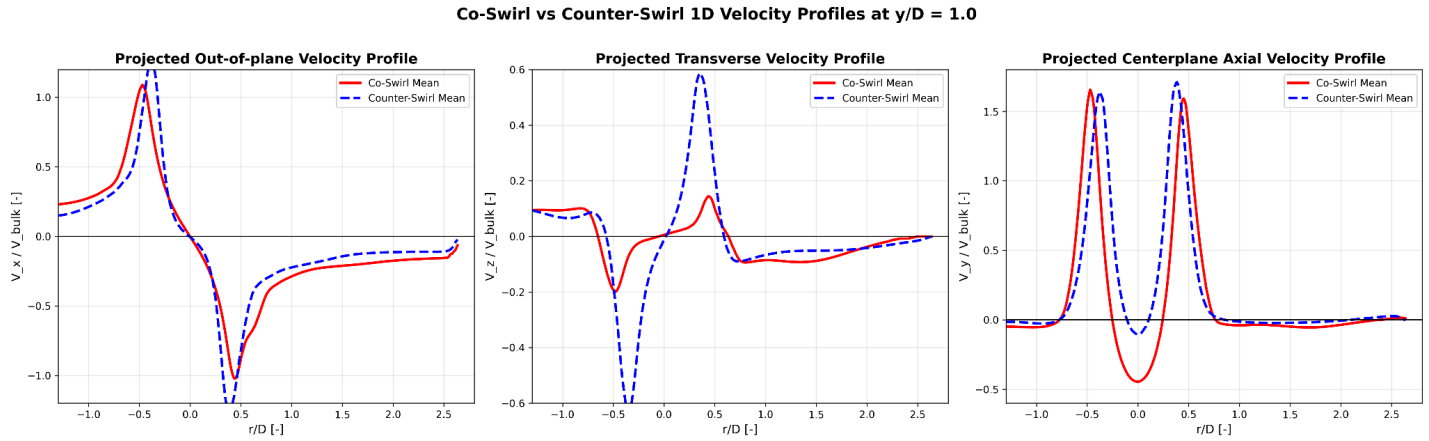


Fig 4.12 plots the 3 CFD Co- & Counter swirl overlaid velocity plots at y/D= 1.0

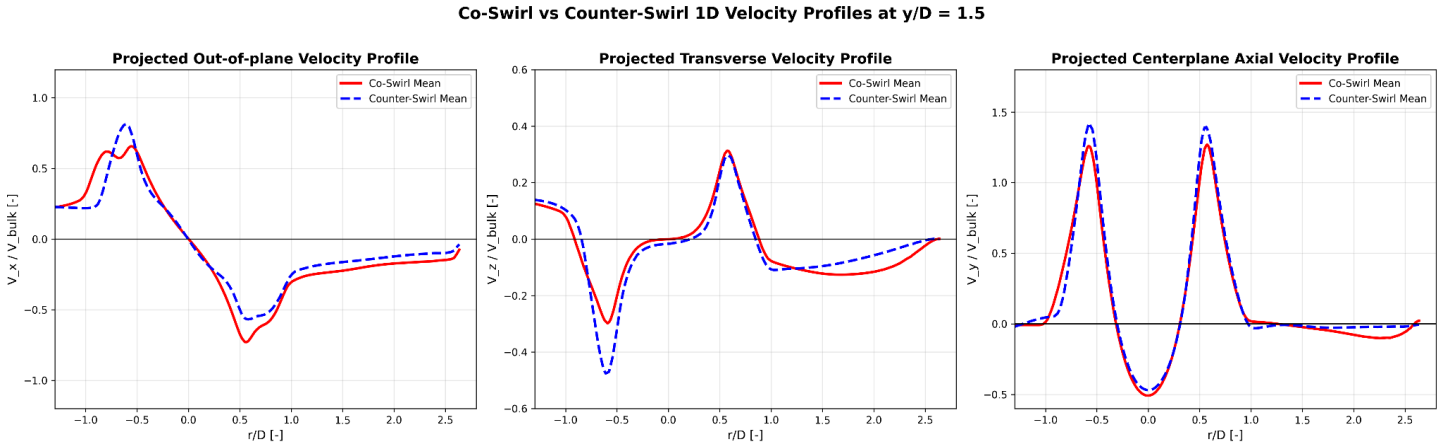


Fig 4.13 plots the 3 CFD Co- & Counter swirl overlaid velocity plots at y/D= 1.5

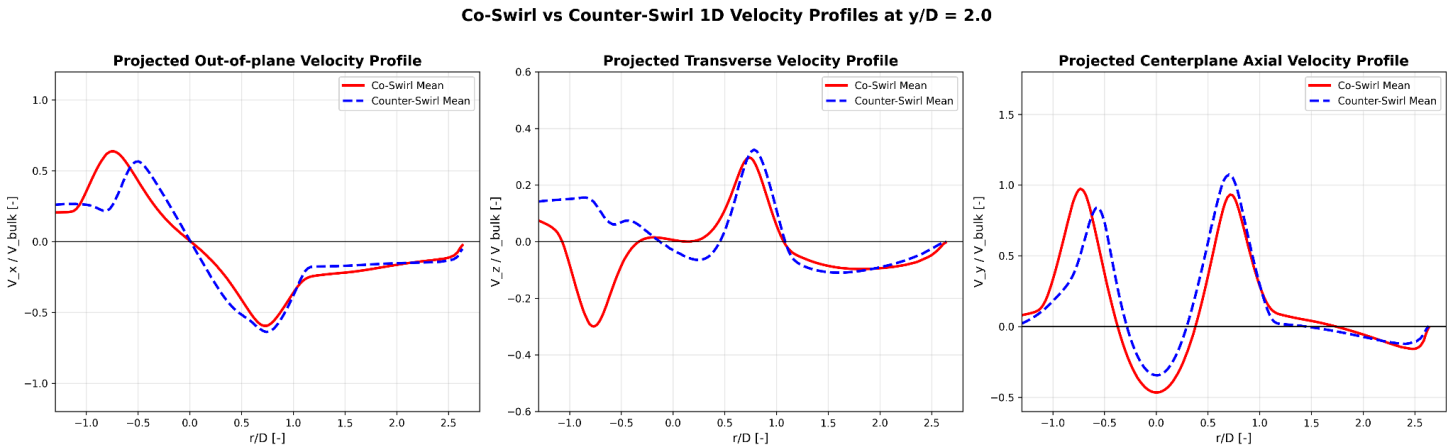


Fig 4.14 plots the 3 CFD Co- & Counter swirl overlaid velocity plots at y/D= 2.0

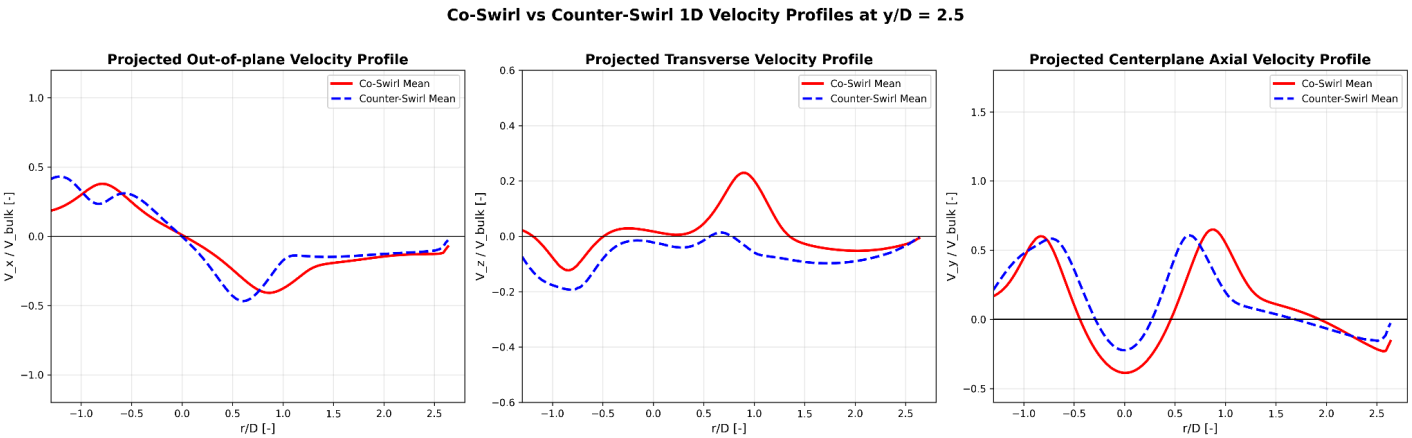


Fig 4.15 plots the 3 CFD Co- & Counter swirl overlaid velocity plots at y/D= 2.5

4.3 Comparative Analysis of the Central Recirculation Zones (Co- vs. Counter-Swirl)

The aerodynamic stability of the double-swirl burner relies fundamentally on the establishment of a robust Central Toroidal Recirculation Zone (CTRZ). To understand the reduced flame stability of the counter-swirler, a comparative isothermal numerical analysis of the co-rotating and counter-rotating configurations was conducted under identical test conditions ($\Phi_{global} = 0.45$, $\Phi_{premix} = 5$, SR=0.6). Because the CFD is isothermal and is not performed at all the same reacting operating conditions as the flashback experiments, the following analysis provides an aerodynamic interpretation consistent with the experimental observations, rather than demonstrating a direct mechanism.

4.3.1 Vortex Breakdown Mechanics and Radial Expansion

IIn the co-rotating configuration, the inner and outer swirling jets rotate in the same direction, collectively augmenting the global angular momentum flux. The resulting centrifugal forces drive the fluid mass to migrate radially outward, creating a strong adverse pressure gradient along the burner axis [40]. Once the swirl number exceeds the critical threshold, the axial flow decelerates, stagnates and reverses, thereby establishing a wide and open conical form of vortex breakdown. This forms the well-defined M-shaped axial velocity profile and the visible reverse-flow pocket captured in the computational data. This CTRZ provides the aerodynamic anchoring

necessary to stabilize the highly reactive hydrogen flame safely away from the injector hardware.

Conversely, reversing the inner swirl direction drastically alters the internal fluid dynamics and the resulting vortex breakdown mechanism. In the counter-swirler, an intense shear layer develops at the interface between the opposing inner and outer rotational streams. This severe interfacial shear results in a combination of opposing angular-momentum fluxes, enhanced turbulent transport, and dissipation. As demonstrated by Soni et al. [39], the opposing momentum of the counter-swirl airstream counteracts the inner jet's momentum, strictly limiting the radial dispersion of the flow compared to the co-swirl configuration. Because the centrifugal expansion is suppressed, the flow fails to induce a strong adverse axial pressure gradient. Without this gradient, the counter-swirl configuration fails to sustain a robust vortex breakdown, leading to an aerodynamic interpretation of the CTRZ collapse.

4.3.2 CFD investigation into the CRZ development

To comprehensively capture the complex spatial evolution of these flow structures—specifically the birth, peak radial expansion, and subsequent decay of the CTRZ—the aerodynamic flow field is analyzed across 6 sequential transverse planes. These planes are normalized by the burner outlet diameter (y/D=0.25 through 2.00) to map the aerodynamic lifecycle of the swirling jets:

- Near-field planes (y/D=0.25 to 0.75): Selected to isolate the initial momentum flux, the intense interfacial shear between streams, and the exact aerodynamic stagnation point where flow reversal is triggered.
- Intermediate planes (y/D=1.00 to 1.50): Selected to capture the fully developed vortex core, identifying the maximum radial spread and the peak magnitude of the reverse-flow bubble.
- Far-field planes (y/D=2.00): Selected to document the natural viscous dissipation of the angular momentum and the eventual closure of the recirculation bubble.

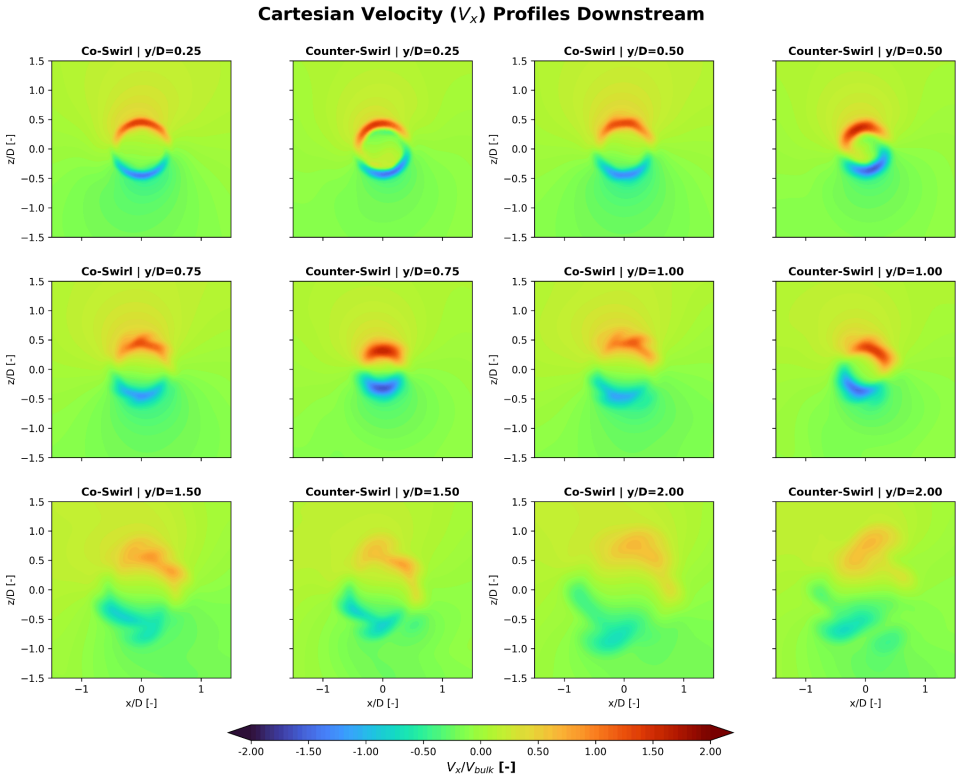


Fig 4.16 Evolution of Cartesian Velocity (V_x) downstream of the CRZ

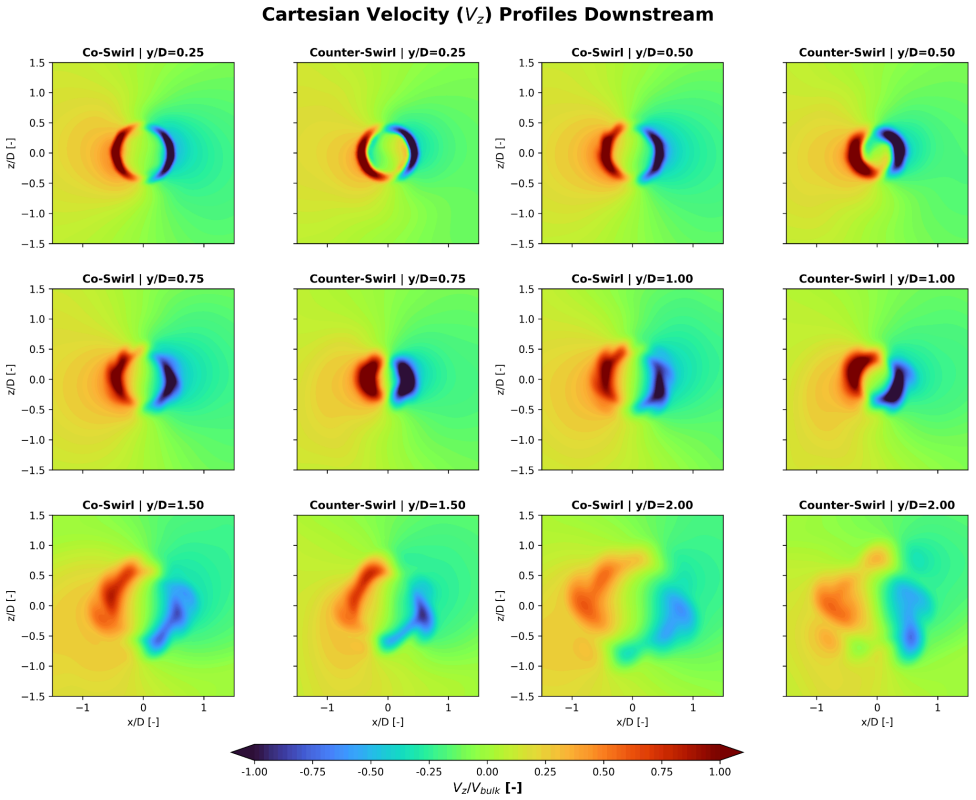


Fig 4.17 Evolution of Cartesian Velocity (V_z) downstream of the CRZ

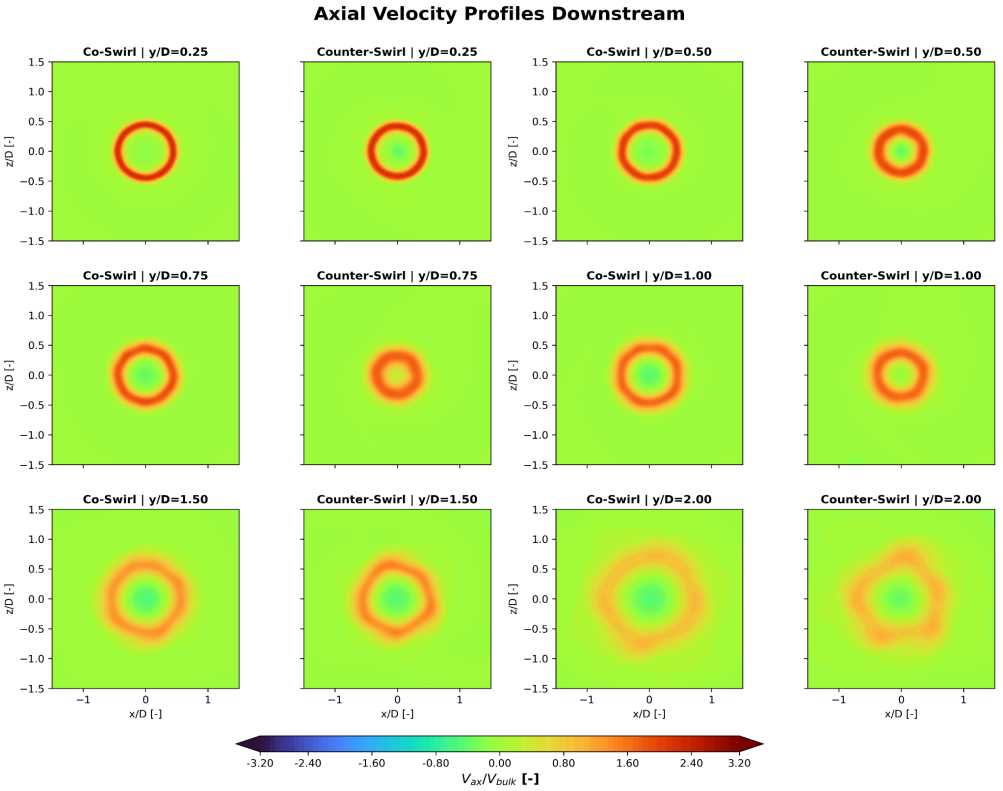


Fig 4.18 Evolution of Axial Velocity (V_y) downstream of the CRZ

4.3.3 Downstream Evolution of the Velocity Fields

The numerical extraction of the velocity components across these sequential planes reveals the exact point of aerodynamic bifurcation between the two configurations:

- Dump Plane and Initial Shear (y/D=0.25-0.50):

Immediately downstream of the injector, both configurations exhibit a distinct, high-velocity annular jet surrounding a primary central core. The Cartesian V_x contours (Fig. 4.16) provide direct visual confirmation of the intended geometric modifications. At y/D=0.25, the co-swirler displays a unified rotational field where both the inner and outer streams rotate in harmony. In contrast, the counter-swirler exhibits a distinct velocity inversion at its core. While its outer secondary stream matches the co-swirler's rotation, the inner primary stream rotates in the exact opposite direction: visually evidenced by the inverted polarity of the V_x core in Fig. 4.16 and confirmed by the 1D overlay plot at the injector plane (Fig 4.8b). This opposing rotation induces severe interfacial shear. By y/D=0.50, both the co-swirler and counter-swirler cores begin to exhibit distinct reverse flow (negative axial

velocity, Fig. 4.18), indicating the successful onset of aerodynamic stagnation. However, the counter-swirler struggles to establish a strong adverse pressure gradient due to the opposing velocity vectors colliding and generating massive turbulent dissipation.

- Flow Bifurcation and CTRZ Core (y/D=0.75-1.5):

This region marks the critical structural divergence. In the co-swirler, the wide-open vortex breakdown fully establishes. By y/D=1.00, the axial profile displays a massive, deep central reverse-flow region surrounded by an expanding annular jet characterized by high outward transverse velocities (Fig. 4.17). The co-swirl CTRZ reaches its maximum radial width and structural integrity at y/D=1.50. Conversely, the strong interfacial shear in the counter-swirler reduces the angular momentum. Instead of expanding radially, the counter-swirler exhibits a suppressed outward movement and hence does not spread out as much as the co-swirler, as seen in the V_z contours (Fig. 4.17). While a recirculation bubble forms, its reverse-flow magnitude is severely weakened, and beyond y/D=1.00, the annular structure collapses inward into a highly localized, forward-moving central core (Fig. 4.18).

- Far-Field Dissipation (y/D=2.00):

The co-swirl CTRZ gradually weakens and begins to close beyond y/D=2.00 as the radial pressure gradients equalize and viscous dissipation dominates. The counter-swirler, having formed only a narrow and weak recirculation bubble, transitions prematurely out of the vortex breakdown regime. It continues to diffuse as a standard forward-moving turbulent jet, with its central axial velocity peak slowly spreading into the far-field.

- Aerodynamic Implications for Combustion Stability:

Without sufficient angular momentum to sustain a strong adverse axial pressure gradient, the vortex breakdown mechanism appears visibly weakened. The numerical 1D transverse profiles and contour slices of the counter-swirler indicate a faster decay of the CTRZ under these specific operational parameters. The flow structure transitions from an aerodynamically stabilized, wide M-shape into a narrower, jet-like configuration characterized by localized, high-velocity forward flow along the centerline. This narrowing and weakening of the recirculation bubble reduces the protective, wide backflow of hot combustion products. Consequently,

this compromised aerodynamic anchoring provides an aerodynamic interpretation for the reduced flame stability and increased flashback tendency observed during the counter-swirl experimental campaign.

4.3.4 Turbulent Kinetic Energy and Shear Dissipation

To further understand the aerodynamic divergence and the mechanism behind the CTRZ collapse in the counter-swirl configuration, a comparison of the Turbulent Kinetic Energy (TKE) fields is conducted. TKE serves as a primary indicator of shear layer intensity, velocity fluctuations & momentum dissipation within the flow field.

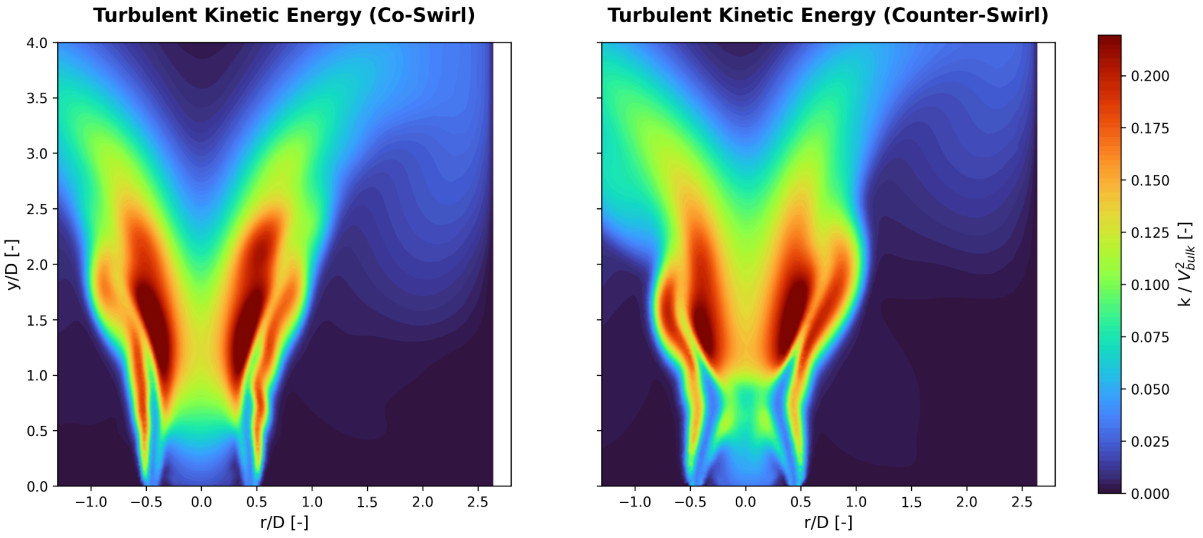
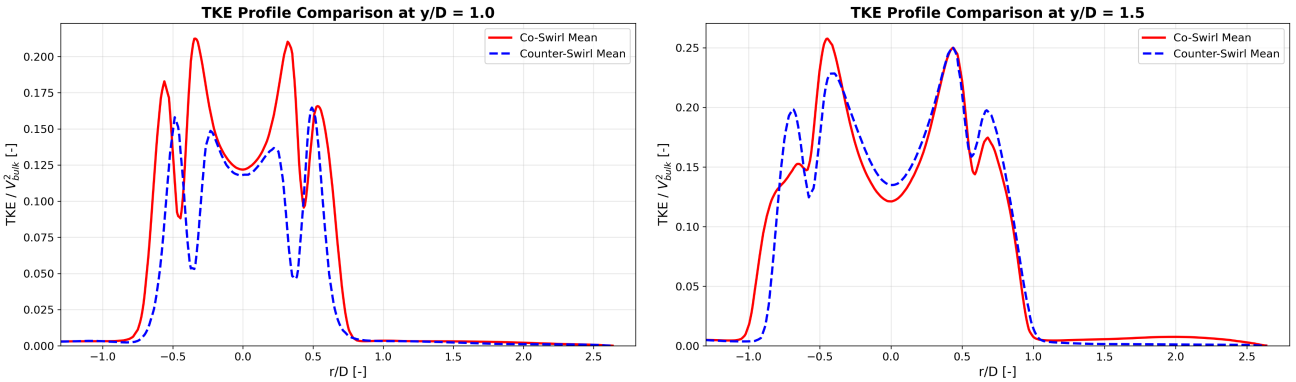


Fig 4.19: 2D contours of non-dimensional Turbulent Kinetic Energy (k / V^2_{bulk}) for Co-Swirl (left) and Counter-Swirl (right) on the longitudinal centerplane.



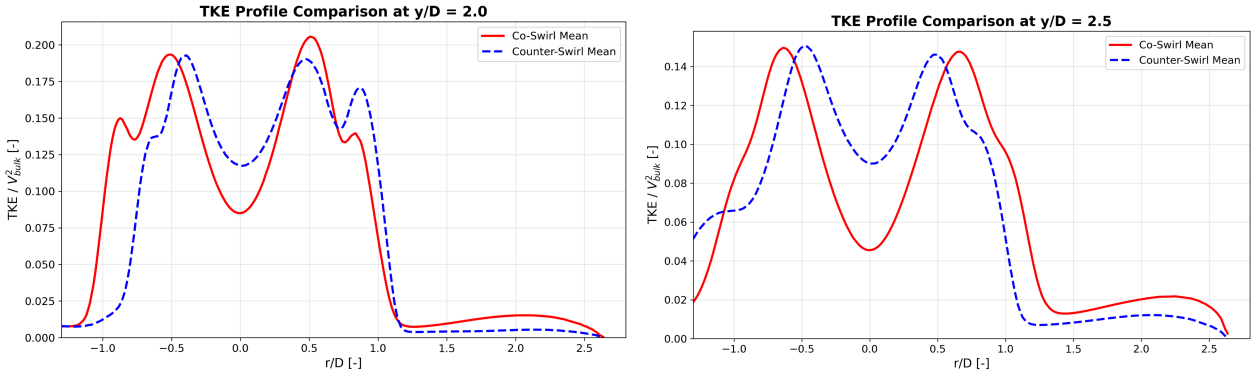


Fig 4.20: Overlaid 1D TKE profiles comparing Co-Swirl and Counter-Swirl at sequential downstream planes (y/D = 1.0, 1.5, 2.0, 2.5).

In the co-rotating configuration, the uniform rotation of the primary and secondary streams allows the flow to preserve its angular momentum as it travels downstream, as a result, the TKE generation is sustained and distributed along the expanding outer shear layers, visually tracking the outward growth of the recirculation zone. As shown in the 1D profiles, the Co-Swirl configuration produces highly energetic, coherent TKE peaks that migrate radially outward from $r/D \approx 0.5$ at $y/D = 1.0$ to $r/D \approx 0.8$ at $y/D = 2.0$.

Conversely, the opposing rotational streams in the counter-swirl configuration generate a region of extreme, localized shear immediately at the interface between the inner and outer jets at the dump plane. This violent velocity gradient acts as a massive, premature sink for angular momentum. The resulting intense turbulent dissipation rapidly extracts kinetic energy from the mean swirling flow early in the domain. As a result, by $y/D = 1.0$, the Counter-Swirl TKE peaks are noticeably suppressed compared to the Co-Swirl baseline (peaking at roughly 0.16 versus 0.21 of the Co-Swirl), and the turbulent field remains pinched near the core axis rather than expanding outward. This TKE visualization provides direct physical evidence for why the counter-swirler is starved of the momentum required to sustain a strong adverse axial pressure gradient, resulting in a weaker CRZ and premature vortex breakdown.

5.Experimental Investigation and Results

5.1 UV Flame Morphology and Stability Analysis

The morphological analysis of the UV images (*OH* chemiluminescence*) acquired provides direct optical evidence of the flame anchoring mechanisms, the structural integrity of the Central Recirculation Zone (CRZ), and the operational limits of both swirl configurations. Based on the experimental test matrix, the visual data is categorized into three primary aerodynamic phenomena: the influence of the secondary split ratio (SR), the impact of primary premixing Φ_{premix}on flashback risk and lean operational limits. For all the experimental campaign observations specified below check **Appendix B**.

5.1.1 Influence of the Secondary Split Ratio (SR)

To isolate the effect of momentum distribution on flame topology, the non-premixed conditions ($\Phi_{premix} = \infty$) at a nominal global equivalence ratio of $\Phi_{global} = 0.45$ are compared across SR = 0.9, 0.6, and 0.4.

- High Split Ratio (SR = 0.9): In both the co-swirl and counter-swirl configurations, allocating 90% of the air to the outer tangential channels generates massive centrifugal forces. The resulting flame is highly compact, considerably shortened and radially made wider.
- Nominal Split Ratio (SR = 0.6): This condition represents the optimal balance for the burner, producing the standard V- or M-shaped flame stabilized along the shear layers of a robust CRZ.
- Low Split Ratio (SR = 0.4 and 0.3): Diverting the majority of the air to the axial channels drastically reduces the global swirl number. At SR = 0.4, both configurations begin transitioning away from the aerodynamically stabilized V-shape. By SR = 0.3, the co-swirler flame structure becomes entirely jet-like, visually confirming the total collapse of the CRZ due to insufficient angular momentum.

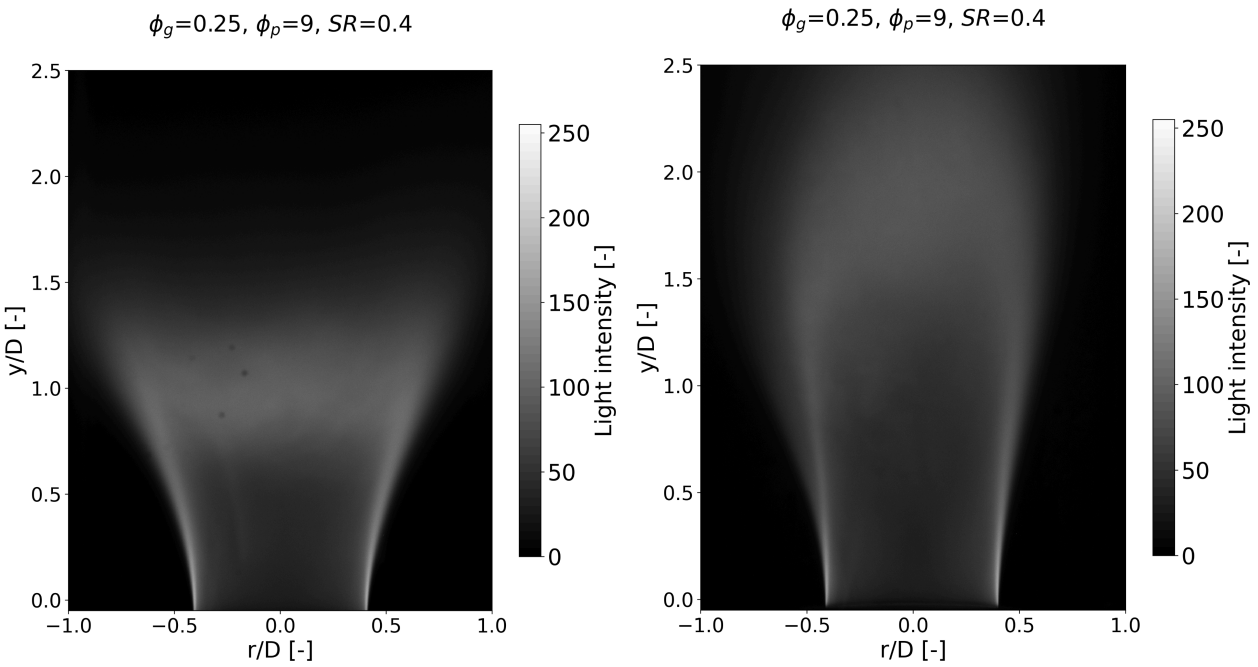


Fig 5.1 Comparison of averaged Flame topologies of Counter Swirl (left) and Co-swirl (right) configurations for identical operating conditions with premixing (inner swirl)- *this was the only case that satisfied the above conditions & for which data extraction occurred.*

5.1.2 Premixing Sensitivity and Flashback Limits- Test Observations

The most significant operational divergence between the co-rotating and counter-rotating injectors emerges when primary premixing is introduced.

In the co-swirl configuration at $\Phi_{global} = 0.45$, introducing a rich premixed stream ($\Phi_{premix} = 9 \,\&\, 7$) induces combustion oscillations consistent with possible thermoacoustic behavior, visually altering the flame morphology and generating a distinct humming noise. This instability only dampens when the premixing is weakened to $\Phi_{premix} = 5$. Flashback in the co-swirler is only triggered when pushed to very lean premixed $\Phi_{premix} = 3$ *(test 7 Feb-17'26)*

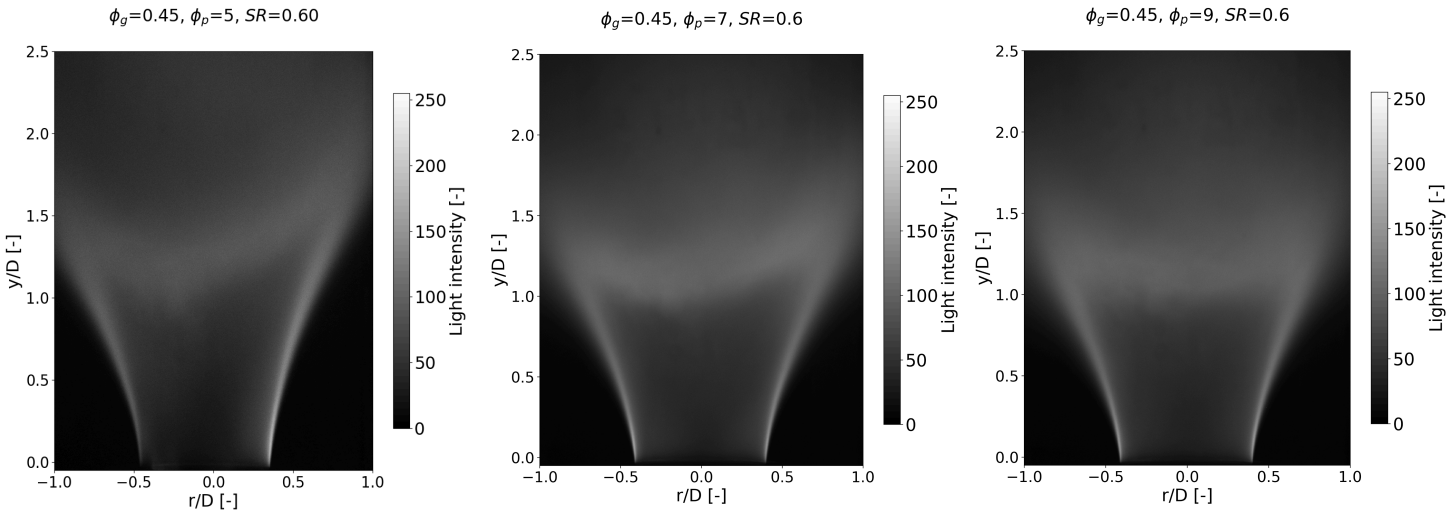


Fig 5.2 Different flame morphologies observed in Co-swirl Configuration while varying the $\Phi_{premix} = 5, 7 \,\&\, 9$ and fixing $\Phi_{global} = 0.45$ and SR = 0.6. *The damping is visible in the flame morphology.*

Conversely, the counter-swirl configuration demonstrates a severe vulnerability to flashback under premixed conditions, validating the CFD predictions of a weakened, highly sheared CRZ. At $\Phi_{global} = 0.45$ and $\Phi_{global} = 0.35$, the simple introduction of $\Phi_{premix} = 9$ causes immediate, violent flashback, resulting in the injector head becoming incandescent and precluding any stable data acquisition. The counter-swirl flame can only sustain premixing at the absolute leanest global condition $\Phi_{global} = 0.25$; however, even under these heavily diluted parameters at SR = 0.6, the flame only holds temporarily see **Fig 5.3** before flashing back after approximately four minutes of operation.

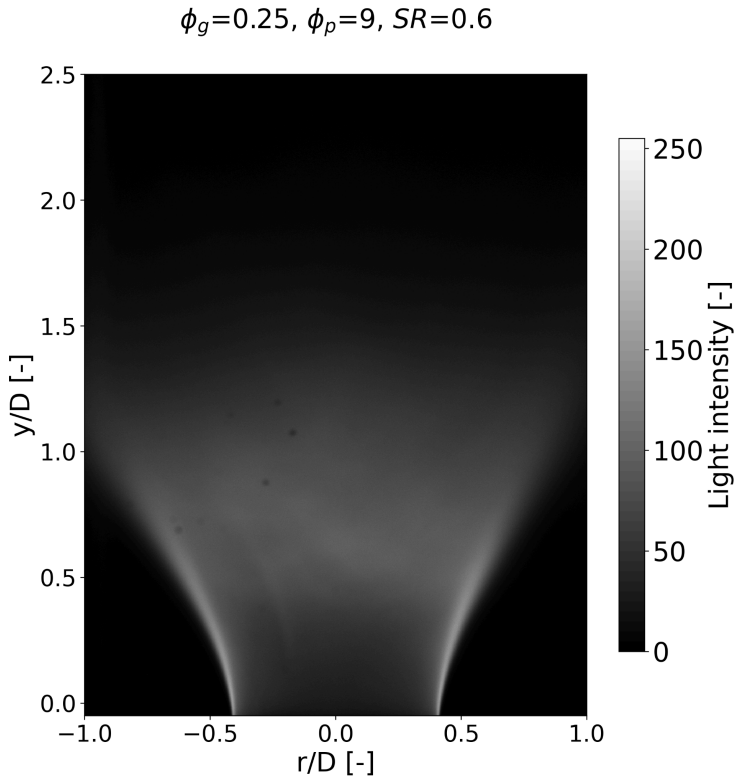


Fig 5.3 Counter swirl configuration with premixing $\Phi_{premix} = 9$ & SR = 0.6

5.1.3 Lean Blowout and Lift-off Dynamics

As the global equivalence ratio is reduced to the lean limit $\Phi_{global} = 0.25$ (non-premixed), the thermal power being fixed leads to an increase in excess air (dilution), which lowers the global flame temperature, stressing the aerodynamic anchoring. For the below statements refer **Fig 5.1**

- At SR = 0.6, the counter-swirler produces a completely lifted flame, indicating that the local flow velocity exceeds the turbulent flame speed at the injector exit.
- Under the same lean conditions but at SR = 0.4, the co-swirler initially produces a short flame constrained close to the injector, accompanied by a high-pitched acoustic howl. Eventually, the recirculation region abruptly disappears, and the flame transitions into a highly unstable, oscillating jet.

5.2 NOx Emission Results

The analysis of nitrogen oxide emissions focuses on comparing the standard Emission Index (EINOx) across the baseline co-swirl configuration and the modified counter-swirl configuration. The comparison utilizes corresponding operating points to isolate the aerodynamic effect of the inner swirler's rotation on pollutant formation.

Overall, the EINOx values across the tested matrices are comparable, though the co-swirl configuration generally produces slightly higher emissions than the counter-swirl setup under identical flow conditions. For example, at a global equivalence ratio of $\Phi_{global} = 0.25$, a premixed equivalence ratio of $\Phi_{premix} = 9$, and a split ratio of SR=0.40, the co-swirler generates an EINOx of 2.91 g/kg H_2, whereas the counter-swirler yields a reduced EINOx of 2.55 g/kg H_2. **Table 5.1** summarizes the experimental emission results for both configurations. The Combustion tests were carried out on **17-02-2026** for the Co-swirl (Model A) injector and 20-03-2026 for the Counter-swirl injector.

Table 5.1: Comparative NO_x Emission Index ($EINO_x$) for Co- & Counter-Swirl Injector configurations for 100% H_2 combustion

Φglobal	**SR**	**Φpremix**	**Co-Swirl EINOx (g/kg H_2)**	**Counter-Swirl EINOx (g/kg H_2)**
0.45	0.60	∞(Non-premixed)	2.44	2.11
0.45	0.40	∞	3.08	4.10
0.45	0.90	∞	2.29	2.17
0.45	0.60	9.00	2.40	*Flashback (No Data)*
0.35	0.60	∞	2.44	2.27

0.35	0.40	∞	2.76	2.47
0.35	0.90	∞	2.35	2.18
0.25	0.60	∞	2.54	2.15
0.25	0.40	∞	3.38	2.96
0.25	0.40	9.00	2.91	2.55

Observations on the Influence of Split Ratio (SR): The secondary split ratio heavily dictates the aerodynamic stretch and the intensity of the central recirculation zone, which directly impacts the flame topology and thermal NO_x formation.

- For the co-swirl configuration at (non-premixed), decreasing SR from 0.60 to 0.40 raises EINOx from 2.44 to 3.08 g/kg H_2. Further reduction to SR=0.30 causes emissions to spike to 4.61 g/kg H_2 as the flame transitions to a jet-like morphology and the recirculation region disappears.
- A similar trend is observed in the counter-swirl setup at Φ_{global}=0.45 (non-premixed): the nominal EINOx of 2.11 at SR=0.60 sharply increases to 4.10 g/kg H_2 when the SR is lowered to 0.4 . Higher split ratios (SR=0.90) in both configurations suppress emissions slightly compared to the nominal SR=0.60 baseline, aligning with the observed shortening and widening of the flame structures.

Observation on the Influence of Premixing and Flashback Limits: While the counter-swirler demonstrates a slight advantage in raw emission reductions, its operational envelope is significantly constrained by flashback limits when operating under premixed conditions.

- At leaner global equivalence ratios$\Phi_{global} = 0.25$, introducing partial premixing $\Phi_{premix} = 9$ at SR= 0.40 stably reduces the co-swirler $EINO_x$ from 3.38 to 2.91 g/kg H_2, and the counter-swirler EINOx from 2.96 to 2.55 g/kg H_2.
- However, the counter-swirl configuration cannot sustain stable combustion at higher thermal loads combined with premixing. Attempts to test the counter-swirler at $\Phi_{global} = 0.35$ andΦ_{global}=0.45 with $\Phi_{premix} = 9$ resulted in an immediate and strong flashback (with the injector head becoming incandescent due to heating) forcing us to stop data acquisition. In contrast, the co-swirler successfully maintained stability at Φ_{global}=0.45 across Φ_{premix} values of 9, 7 & 5, producing stable EINOx values of 2.40, 2.44, and 2.76 g/kg H_2 respectively, though notable combustion oscillations consistent with possible thermoacoustic behavior (humming) was recorded at the above premixed bounds.

6.Conclusions, Further developments & Declaration

The computational grid is designed with sufficient baseline resolution to accommodate various other operational flow rates. For moderate deviations in mass flow, the solver y^+ requirements remain satisfied with minimal to no near-wall refinement, while the CFL condition can be strictly maintained by simply adjusting the transient time step. Consequently, this mesh framework can efficiently simulate additional PIV test conditions. The resulting validated CFD datasets, coupled with sparse experimental PIV measurements, can serve as high-fidelity training data for Physics-Informed Neural Networks (PINNs) to reliably infer complex, unmeasured fluid dynamic parameters, such as the 3D pressure field.

The results of the PIV are averaged for time and for the over the thickness (1.2 mm appx Bezzi & Sala [35]) of the sheet, whereas in CFD while we have time averaging we have only the sheet that has one cell thickness, future improvements can be such that the volume averaging can be done by obtaining the data over a few planes in quick succession the slices mimicking the laser sheet thickness to get a more averaged data.

Another thing to consider when comparing the datasets is how the PIV cross-correlation actually works. The PIV pixel grid and interrogation windows essentially average out the velocities of the particles in that specific area, which naturally leads to a loss of sharpness in the velocity gradients. On the other hand, our high-resolution CFD mesh captures these sharp, localized shear gradients that the optical setup just can't see. For future validation work, a script could be used to depreciate or spatially average the CFD velocity fields to perfectly match the PIV interrogation window size. This would give a much fairer 1-to-1 comparison, ensuring we don't mistake an optical resolution limit for a CFD error.

Finally, the choice to follow an isothermal URANS methodology for this thesis was strictly dictated by the time, computational & storage limitations. Once more computational power and storage are made available, future campaigns should definitely pivot to modelling Large Eddy Simulation (LES). LES resolves the larger eddies directly instead of just modeling them, which would give a much more accurate picture of the Precessing Vortex Core (PVC) and the shear layer breakdown in the

counter-swirler. After that, the next major step is to simulate the Thermo-chemical Combustion, an actual hot run. By coupling the LES solver with detailed chemical kinetics (like Chemkin), a full 100% H_2 reacting flow can be modeled to investigate the flame structures and Aerothermics that influence the combustion. This can also help to understand how the heat release and thermal expansion alter the CRZ, and exactly where the thermal NO_x is forming.

Declaration:

I have made use of previous research & thesis data from Combustion lab, Open research literature; and AI to formulate my sentences, to code and to structure the contents of my thesis.

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A Appendix A

Derivation of Inlet Boundary Condition Velocities:
This appendix details the mathematical formulations and conversion procedures used to convert the experimental volumetric flow rate values measured in Normal Liters per minute (Nl/min) into inlet velocity boundary conditions (m/s) to be applied to the various inlet patches for the ANSYS Fluent- computational fluid dynamics (CFD) solver.

A.1 Governing Principles and Standard-to-Actual Conversion

Bronkhorst flowmeters in the experimental test rig record flow rates in Normal Liters per minute (Nl/Min). By international laboratory convention: "Normal" conditions are defined at standard reference values:

- Normal Pressure (P_n): 1.01325 bar (101,325 Pa)
- Normal Temperature (T_n)**:** 273.15 K (0 °C)

Because air density varies with temperature & pressure, the volumetric flow rate must be corrected to the true operating conditions of the laboratory environment (T_{op}) right before entering the combustor plenum or burner feed lines. Assuming the working fluid behaves as an ideal gas and the injection occurs under near-atmospheric static pressure $P_{op} \sim P_{n'}$ the real volumetric flow rate Q_{act} is derived from the Ideal Gas Law:

$$\frac{P_n \cdot Q_n}{T_n} = \frac{P_{op} \cdot Q_{act}}{T_{op}} \Rightarrow Q_{act} = Q_n \cdot \frac{T_{op}}{Tn} \cdot \frac{P_n}{P_{op}}$$

For an ambient operating temperature of T_{op} = 20 °C (293.15 K) and direct pipe feeds discharging near atmospheric pressure:

$$Q_{act} = Q_n \cdot \frac{293.15\ K}{273.15\ k} \sim 1.0732 \cdot Q_n$$

To convert this value into SI units of m^3/s we need to convert it from Minutes to Seconds.

$$Q_{act}\,[m^3/s] \;=\; \frac{Q_n[Nl/min] * 1.0732}{60000}$$

Once Q_{act} is established for each stream, the uniform inlet velocity (v) is calculated using the corresponding cross-sectional flow area (A), from the discharge formula:

$$v = \frac{Q_{act}}{A}$$

A.2 Detailed Calculations by Inlet Stream

A.2.1 Secondary Axial Air

- **E**xperimental Flow Rate (Q_n): 129.20 Nl/min
- Actual Volumetric Flow (Q_{act}):$Q_{act} = \frac{129.20*1.0732}{60000} = 2.311 * 10^{-3}\, m^3/s$

  Geometry: Concentric annulus with Outer Diameter Dout = 0.08 m (80mm) & Inner Diameter Din = 0.025m (25mm).
- Cross-Sectional Area ($A_{annulus}$):

  $A_{annulus} = \frac{\pi}{4}(D_{out}^{\,2} - D_{in}^{\,2}) = \frac{\pi}{4}\,(0.08^2 - 0.025^2) \sim 4.536 * 10^{-3}\, m^2$
- Inlet Velocity (V_{axial}):

  $V_{axial} = \frac{2.311*10^{-3} m^3/s}{4.536*10^{-3} m^2} \sim 0.510\, m/s$

A.2.2 Secondary Tangential Air

- Experimental Flow Rate (Q_n): 193.80 Nl/min
- Actual Volumetric Flow (Q_{act}):

  $Q_{act} = \frac{193.80*1.0732}{60000} = 3.467 * 10^{-3}\, m^3/s$
- Geometry: Distributed evenly across n = 8 parallel tangential injectors, each with diameter d = 0.011257m (11.257 mm).
- Total Cross-Sectional Area (A_{total}):

  $A_{total} = 8 * \frac{\pi}{4}d^2 = 8 * \frac{\pi}{4}(0.011257)^2 \sim 7.966 * 10^{-4}\, m^2$

- Inlet Velocity (V_{tan}):

$$V_{tangential} = \frac{3.467*10^{-3}\ m^3/s}{7.966*10^{-4}\ m^2} \sim 4.352\ m/s$$

A.2.3 Premix Air

- Experimental Flow Rate (Q_n): 31.95 Nl/min
- Actual Volumetric Flow (Q_{act}):

$$Q_{act} = \frac{31.95*1.0732}{60000} = 5.716\ *\ 10^{-4}\text{f}$$

- Geometry: Circular supply pipe with diameter d = 0.010m (10 mm).
- Cross-Sectional Area (A_{pipe}):

$$A_{pipe} = \frac{pi}{4}d^2 = \frac{\pi}{4}(0.010)^2 \sim 7.854\ *\ 10^{-5} m^2$$

- Inlet Velocity (V_{premix}):

$$v_{premix} = \frac{5.716*10^{-4} m^3/s}{7.854*10^{-5}} m^2 \sim 7.278 m/s$$

A.2.4 Air Substituting Hydrogen (H_2)

- Experimental Flow Rate (Q_n): 17.60 Nl/min
- Actual Volumetric Flow (Q_{act}):

$$Q_{act} = \frac{17.60*1.0732}{60000} = 3.149$$

- Geometry: Annular fuel/substitute injection line with Outer Diameter $D_{out} = 0.023m$ $(23mm)$ and Inner Diameter $D_{in} = 0.012$m (12mm).
- Cross-Sectional Area ($A_{annulus}$):

$$A_{annulus} = \frac{\pi}{4}(0.023^2 - 0.012^2) \sim 3.024\ *\ 10^{-4} m^2$$

- Inlet Velocity ($V_{sub\ H_2}$):

$$V_{sub\ H_2} = \frac{3.149*10^{-4} m^3/s}{3.024*10^{-4}\ m^2} \sim 1.041\ m/s$$

A.3 Summary of Final CFD Boundary Conditions & Convergence

The calculated operational velocities implemented as velocity-inlet boundary conditions in the numerical model are compiled below.

Inlet Stream Designation	Experimental Rate (Qn)	Effective Flow Area (A)	Computed Velocity (v)
Secondary Axial Air	129.30 Nl/min	$4.536 * 10^{-3}\, m^2$	**0.510 m/s**
Secondary Tangential Air	193.8 Nl/min	$7.966 * 10^{-4}\, m^2$	**4.352 m/s**
Premix Air	31.95 Nl/min	$7.854 * 10^{-5}\, m^2$	**7.278 m/s**
Air Substituting H_2	17.6 Nl/min	$3.024 * 10^{-4}\, m^2$	**1.041 m/s**

Final Residuals at the end of solution

- Counter- swirler:

continuity |X-velocity |Y-velocity |Z-velocity |k |ω

2.8110e-03 | 1.7880e-07| 1.4582e-06| 1.8488e-07 | 6.1369e-06| 3.2720e-05

- Co- swirler:

continuity |X-velocity |Y-velocity |Z-velocity |k |ω

2.5922e-03 | 1.5932e-07 | 3.4902e-07 | 1.9510e-07 |3.2629e-06 | 4.0278e-05

A.4 MESH information

A.4.1 Co-swirl Combustor model MESH data:

/mesh> size-info
Mesh Size

Level	Cells	Faces	Nodes	Partitions
0	**2968246**	19331716	15371985	12

5 cell zones, 22 face zones.

A.4.2 Counter-swirl Combustor model MESH data:

/mesh> size-info
Mesh Size

Level	Cells	Faces	Nodes	Partitions
0	**2184521**	14074799	11126440	12

4 cell zones*, 18 face zones. (* I Boolean united the first stage mixing zone and the inner swirler zone to aid a simpler share topology)

A.5 Cylindrical coordinate velocity Contours

When evaluating the flow on transverse cross-sections (the XZ planes), true cylindrical velocity components are computed to accurately depict the swirl and expansion mechanics. The radial position r and angular coordinate θ are defined as:

$$r = \sqrt{x^2 + z^2} \tag{4.1}$$

$$\theta = atan2(z, x) \tag{4.2}$$

The true radial velocity (V_r) and true tangential velocity (V_θ) are derived from the Cartesian components (V_x, V_z) using the following transformations:

$$V_r = \frac{x \cdot V_x + z \cdot V_z}{\sqrt{x^2 + z^2}} \tag{4.3}$$

$$V_\theta = \frac{x \cdot V_z - z \cdot V_x}{\sqrt{x^2 + z^2}} \tag{4.4}$$

Under this adopted convention, a positive V_r denotes radially outward flow (centrifugal expansion), while a negative V_r denotes radially inward flow. A positive V_θ indicates counter-clockwise rotation when viewed from above (looking down the negative Y-axis toward the origin).

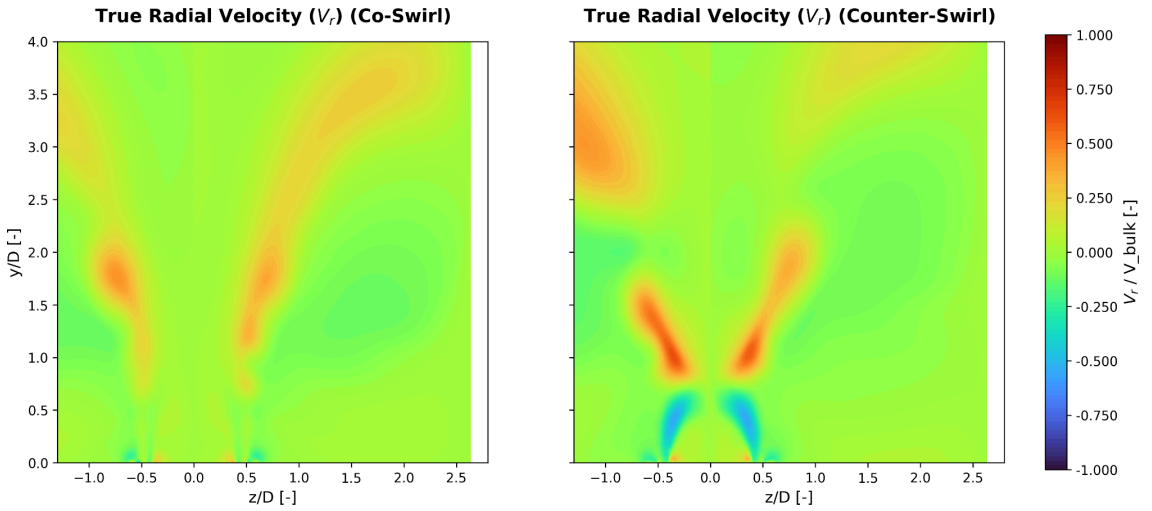

Fig A.1 Radial Velocity comparison YZ plane

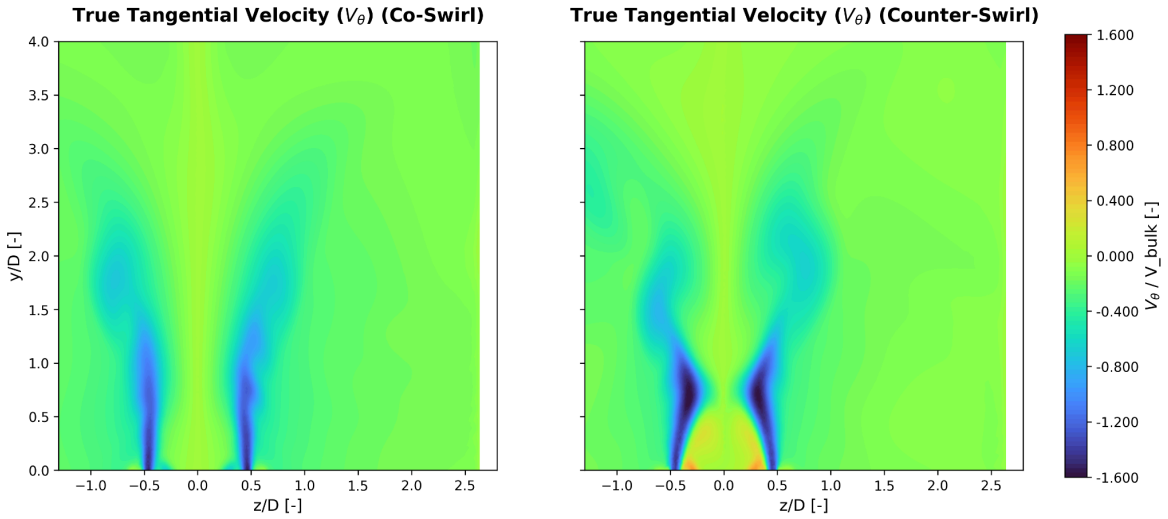

Fig A.2 Tangential Velocity comparison YZ plane

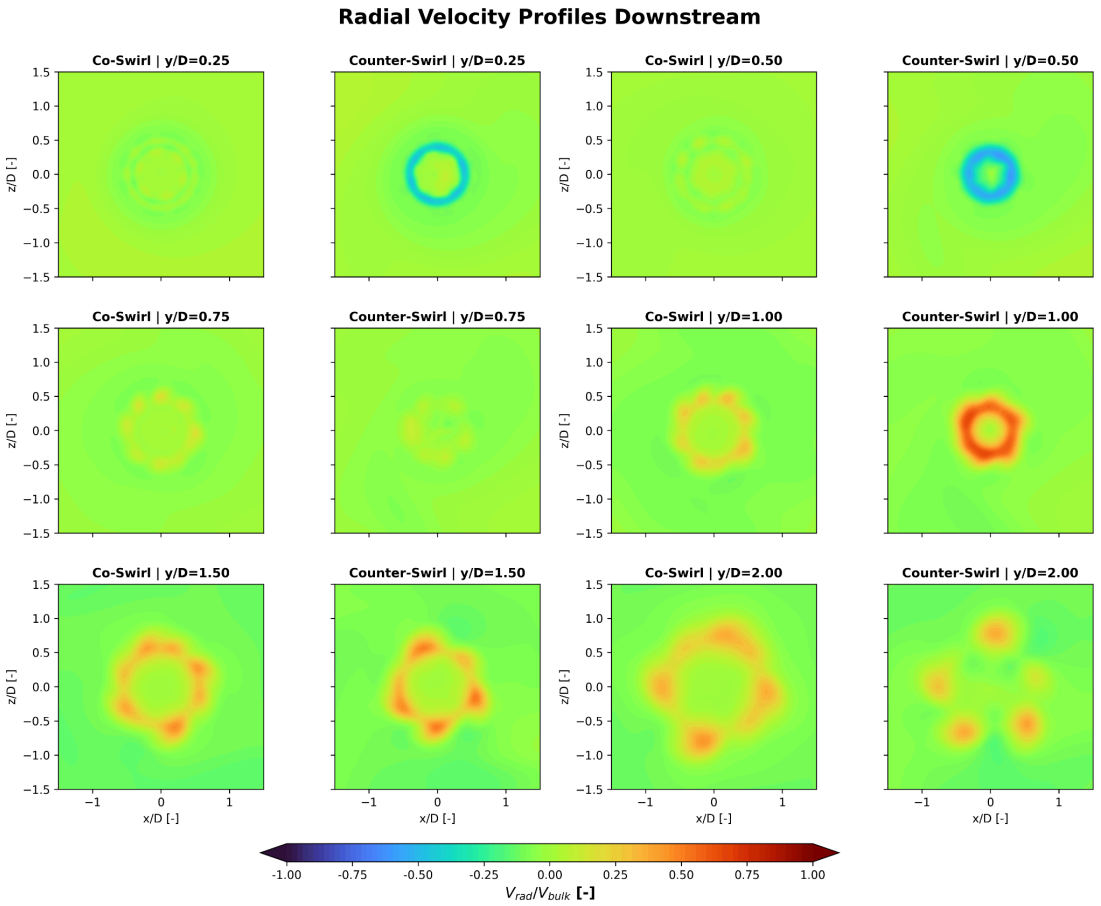

Fig A.3 Radial Velocity ZX plane downstream development comparison

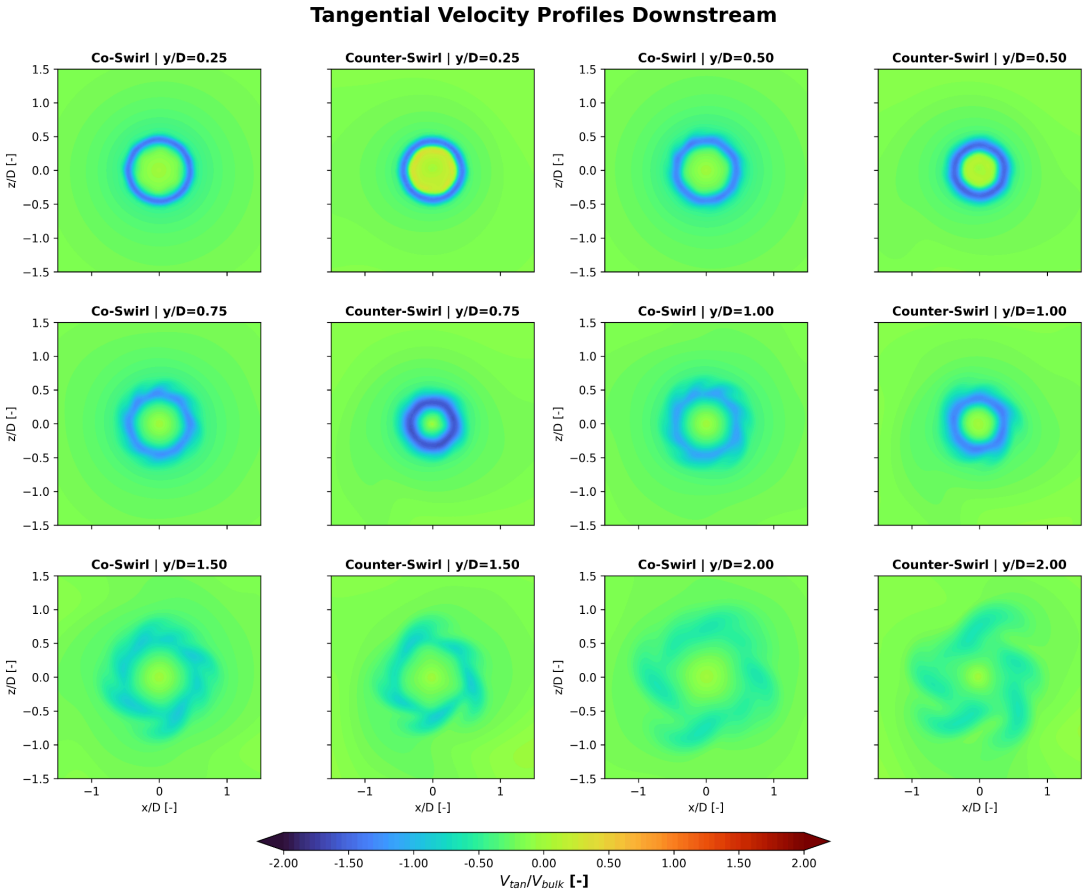

Fig A.4 Tangential Velocity ZX plane downstream development comparison

B Appendix B

Experimental Campaigns- thesis content relevant Observations

Test	Configuration	Φglobal	Φpremix	SR	Acoustic / Visual Observations	Data Acquisition
03	Co-Swirl	0.45	9.0	0.60	Strong instability (humming); altered flame morphology.	Acquired
07	Co-Swirl	0.45	3.0	0.60	Catastrophic flashback.	No Acquisition
09	Counter-Swirl	0.35	9.0	0.30	Strong flashback; injector head became incandescent.	No Acquisition
13	Counter-Swirl	0.25	∞	0.60	Flame lifted from the injector rim.	Acquired

List of Symbols

- A: Cross-sectional area [m^2]
- BOI: Body of Influence
- CFD: Computational Fluid Dynamics
- CFL: Cell Convective Courant Number
- CRZ: Central Recirculation Zone
- D, d: Diameter [m]
- $EINO_x$: Emission Index of nitrogen oxides [g/kg H_2]
- F/A: Fuel-to-Air ratio
- FVM: Finite Volume Method
- Gk, Pk: Turbulence production term
- G_x: Axial flux of axial momentum
- G_θ: Axial flux of angular momentum
- HPC: High-Performance Computing
- J: Momentum flux ratio
- k: Turbulent kinetic energy [m^2/s^2]
- LES: Large Eddy Simulation
- LHV: Lower Heating Value [kJ/Nl]
- M: Molar mass [g/mol]
- $\dot{m}$: Mass flow rate [kg/s]
- p: Static pressure [Pa]
- PIV / Stereo-PIV: (Stereoscopic) Particle Image Velocimetry
- P_{th}: Nominal thermal power [kW]
- PVC: Precessing Vortex Core
- Q, $\dot{V}$: Volumetric flow rate [Nl/min or m^3/s]
- R, r: Radius / Radial coordinate [m]
- S: Strain rate magnitude [s^{-1}]
- S_g: Geometrical Swirl Number
- S_n: Swirl Number
- SR: Secondary Split Ratio
- SST: Shear-Stress Transport
- t: Time [s]

- u_τ: Friction velocity [m/s]
- URANS: Unsteady Reynolds-Averaged Navier-Stokes
- $\vec{v}$, u: Velocity vector / velocity magnitude [m/s]
- X: Molar fraction
- y^+: Non-dimensional wall distance
- y, z: Spatial coordinates [m]
- κ: von Kármán constant
- μ: Dynamic viscosity [kg/($kg/m \cdot s$]
- μ_t: Turbulent (eddy) viscosity [$kg/m \cdot s$]
- ν: Kinematic viscosity [m^2/s]
- ω: Specific dissipation rate [s^{-1}]
- Ω: Vorticity magnitude [s^{-1}]
- Φ: Equivalence ratio
- ρ: Density [kg/m^3]
- τ_w: Wall shear stress [Pa]
- $\bar{\bar{\tau}}$: Mean viscous stress tensor [Pa]

Subscripts

- act: Actual (operating conditions)
- ax: Axial component
- bulk: Bulk (average) value
- e, s: External / Secondary
- eq: Equivalent
- g: Global
- i,p: Internal / Primary
- meas: measured
- n: Normal (standard reference conditions)
- rad: Radial component
- st: Stoichiometric
- tan,θ: Tangential component

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